

Prostate Cancer

Rapid Revision

Campbell-Walsh-Wein · Chapters 153–166

For MCh Urology theory & viva · NEET-SS / competitive MCQs

How to use (total under 7½ h even at a slow, careful pace)

- Each chapter reads top-to-bottom in the order you would write an essay answer: definition → pathology → diagnosis → management → complications → follow-up. The same material serves theory and viva.
- Boxed **ALGORITHM** and shaded **KEY POINT** panels carry the reasoning examiners ask about ("why do we...?").
- **MCQ one-liners** end each chapter: re-read these the day before the paper.
- The last section puts the big treatment choices side by side for essay questions.

Conventions: **underlined bold** = number to memorise · **MCQ** = exam trivia only · → *Ch 161* = cross-reference · † = not in the source chapter notes (added because it is essential in practice). Abbreviations are spelled out at first use in each chapter.

Chapter	Topic	Slow-reader time
153	Epidemiology, Aetiology & Prevention	29 min
154	Prostate Cancer Biomarkers	20 min
155	Prostate Biopsy	25 min
156	Pathology of Prostatic Neoplasia	29 min
157	Diagnosis & Staging	22 min
158	Active Management of Localised Disease	32 min
159	Active Surveillance	27 min
160	Open Radical Prostatectomy	27 min
161	Laparoscopic & Robotic Prostatectomy, PLND	31 min
162	Radiation Therapy	39 min
163	Focal Therapy	27 min
164	Locally Advanced Prostate Cancer	32 min
165	Recurrent & Newly Metastatic Disease	45 min
166	Castration-Resistant Prostate Cancer	38 min
★	Putting it together: the prostate cancer pathway	9 min
Total (30264 words, at 70 words/min)		7h 12min

153 Epidemiology, Aetiology and Prevention of Prostate Cancer

Reading time ≈ 29 min (slow pace)

1 · EPIDEMIOLOGY — THE HEADLINE

- Worldwide: **second commonest** non-skin cancer and **fifth** cause of cancer death in men; 2020 ≈ **1.4 million cases, 375,000 deaths**. First for incidence in 112/185 countries.
- Incidence varies ~13-fold (South Central Asia lowest, Northern Europe highest); United States (US) lifetime risk ≈ **1 in 8**; 5-year survival 97%, yet second cause of cancer death in US men.
- Incidence and mortality are decoupled**. Incidence tracks screening intensity and wealth (Northern/Western Europe, North America, Australia); mortality peaks where African ancestry meets poor access — **Caribbean has the highest mortality in the world** [MCQ]. Asia is lowest on both, but rising (Western diet, obesity, screening).
- Latent (autopsy) cancer** rises steeply with age: ~5% <30 y → 29% at 60-69 → **59% >79 y**; higher in Black men at every age.

KEY POINT Why is aetiology so hard to study? Natural history spans decades, and in a prostate-specific antigen (PSA)-screened population "a diagnosis of prostate cancer signifies, in most cases, that a man happened to have a PSA test at that time." Risk-factor studies may therefore identify determinants of *being tested*. The preferred end point is **aggressive disease** (advanced stage, metastasis, death). **Healthy-user bias**: health-conscious men seek PSA tests, so healthy habits spuriously appear to raise incidence while lowering mortality.

2 · ESTABLISHED RISK FACTORS

Only three are unequivocal: **age, race (ancestry) and family history**.

Age

- Mean age at diagnosis 66; ~60% diagnosed at ≥65; rare <40. Diagnosis at ≥70 carries a higher risk of dying of the disease.

Race

- Black men: highest incidence (172.6/100,000 in US) and **twice** the cancer-specific mortality of White men, with younger age and more metastatic disease at presentation.
- Drivers: structural inequity and unequal access (plus comorbidity), and genetics — **8q24** risk alleles (and 17q21) commoner in African ancestry.
- Equal access → equal outcome**: with similar treatment, Black men with non-metastatic disease have similar cancer-specific mortality.

Family history

~20% of patients have a family history; twin-study heritability up to **58%** — genes outweigh shared lifestyle.

Family history	Relative risk
Any first-degree relative (father/brother)	2-3×
Two first-degree relatives	~3.5×
Both father and brother	9×
Relative diagnosed <60 vs ≥60	~2.5× vs ~1.6×
Father/brother died of prostate cancer	2× risk of prostate cancer death

Risk scales with the number of relatives, closeness and youth at diagnosis; lethality itself is heritable.

3 · GERMLINE GENETICS

Most clinically important inherited genes are **DNA-repair genes**; they matter because they predict **aggressive** cancer, not just any cancer.

Gene	Pathway	Association	Clinical implication
BRCA2 , BRCA1	Homologous recombination (HR)	Higher incidence; BRCA2 familial, aggressive	Early targeted screening; poly(ADP-ribose) polymerase (PARP) inhibitor; caution in active surveillance (AS)
ATM	HR	Metastatic disease	As for BRCA
PALB2, CHEK2	HR / DNA repair	Familial; PALB2 lethal	CHEK2: targeted screening
MLH1, MSH2, MSH6, PMS2 (Lynch)	Mismatch repair (MMR)	2–5% of cases	Immune checkpoint inhibitor; very androgen deprivation therapy (ADT)-sensitive
HOXB13	Prostate-specific homeobox	Hereditary prostate cancer	Targeted screening
8q24, 17q21	Genome-wide association study (GWAS) loci	Commoner in African ancestry	Explain part of racial excess

- GWAS: >200 single-nucleotide polymorphisms (SNPs), each of small effect; steroid-pathway SNPs (HSD3B1, CYP19A1, HSD17B4) linked with progression; KLK3 (PSA gene) SNPs with risk.
- **Polygenic risk scores** combine SNPs to guide screening, but predict *any* cancer, not aggressive cancer — hence not yet practice-changing.
- † **Germline testing indications** (National Comprehensive Cancer Network, NCCN): metastatic, high/very-high-risk localised, or intraductal/cribriiform histology; strong family history of prostate, breast, ovarian, pancreatic or Lynch-type cancers; Ashkenazi Jewish ancestry. → *Ch 154*

4 · MOLECULAR AETIOLOGY AND INFLAMMATION

- **Chronic inflammation** is near-universal in biopsies; it raises PSA (detection bias), but genetic, molecular and pathological evidence supports a causal role → rationale for anti-inflammatory prevention.
- † **Proliferative inflammatory atrophy (PIA)** — proliferating atrophic epithelium beside inflammation — is the proposed precursor lesion, progressing via high-grade prostatic intraepithelial neoplasia (HGPIN) to cancer.
- **TMPRSS2-ERG** fusion: androgen-regulated TMPRSS2 promoter drives the ERG oncogene (ETS family); ~half of tumours in Western men, uncommon in East Asians (FOXA1 mutations commoner in Chinese men). → *Ch 154*

5 · POSSIBLE RISK FACTORS — BODY, METABOLISM, LIFESTYLE

The unifying rule: most lifestyle factors act on **progression** (aggressive/fatal disease) rather than initiation, and evidence is observational.

Factor	Total incidence	Aggressive / fatal
Height	~5% ↑ per 10 cm	↑ high-grade and death
Obesity	Mixed; ↓ low-grade	Consistently ↑
Smoking	Not ↑ (often ↓ — less screening)	↑ ~ 40% mortality ; no excess after quitting ≥10 y
Metabolic syndrome, hypertension, cardiovascular disease	Slight ↑	Metabolic syndrome: worse pathology
Diabetes	↓ (low testosterone, IGF-1, PSA; metformin)	—

Factor	Total incidence	Aggressive / fatal
Vigorous physical activity	No clear effect	↓ advanced disease
Alcohol	Unclear	↑ mortality ≥45 g/day; binge ↑

- **Mechanism** (Giovannucci): poor diet + inactivity → adiposity → ↑ insulin/insulin-like growth factor-1 (IGF-1), ↑ leptin, ↑ inflammation, ↑ oestrogen, ↓ sex hormone-binding globulin (SHBG) → ↑ proliferation, ↓ apoptosis.
- **Hormones**: adult serum testosterone does *not* predict risk **[MCQ]**; genetically higher free testosterone, IGF-1, insulin and early puberty modestly do.
- **Composite healthy lifestyle** (no smoking, activity, healthy weight, diet) → **30-50%** lower lethal cancer; even in the highest polygenic-risk quartile lifetime lethal risk falls from 5.3% to 1.6% — genes are not destiny.
- ADT is associated with cardiovascular disease in observational data (not in trial meta-analyses) → baseline cardiovascular assessment on ADT.

Diet

- International variation and migrant studies imply environmental causes, but World Cancer Research Fund (WCRF) 2018 found **no dietary factor with strong evidence**.
- **PSA-era downgrades**: dairy and calcium "probable ↑" → "limited"; tomatoes/lycopene "probably protective" → "no conclusion". Pre-PSA cancers were mostly aggressive, PSA-era ones mostly indolent, so associations changed.
- Lycopene works only from cooked tomato in oil (fat-soluble, bound in raw tomato); still linked with lower lethal cancer.

Assorted factors

- **Vasectomy**: not a risk factor (detection bias); American Urological Association: no need to discuss in counselling.
- Transgender women keep the prostate; on oestrogen median PSA ≈ **0.02 ng/mL** (50-fold lower) → normal-looking PSA is falsely reassuring; cut-points must be lowered.

6 · SUPPLEMENTS

No supplement benefits healthy men on a balanced diet; single high-dose agents have failed or harmed.

Agent	Evidence	Verdict
Selenium, vitamin E	Selenium and Vitamin E Cancer Prevention Trial (SELECT) — null; vitamin E alone ↑ risk	Avoid
Vitamin D	Vitamin D and Omega-3 Trial (VITAL) — no prostate effect	Not recommended
Calcium	↑ total; supplements ↑ fatal	Caution (relevant on ADT)
Zinc	≥75 mg/day or ≥15 y → ↑ lethal	Avoid prolonged high dose
High-dose antioxidants	↑ risk incl. high grade	Avoid
Soy, pomegranate, curcumin	Randomised trials null	Unproven
Green tea	↓ progression of HGPIN/atypical small acinar proliferation	Promising

KEY POINT Why did vitamin E work in the Alpha-Tocopherol, Beta-Carotene (ATBC) trial but fail in SELECT? ATBC used a tenfold lower dose in **heavy smokers** with minimal PSA screening (aggressive, clinically detected cancers); SELECT used 400 international units (IU) in non-smokers with widespread screening (indolent cancers). Different disease, different answer.

7 · CHEMOPREVENTION (PRIMARY PREVENTION)

No agent is approved by the US Food and Drug Administration (FDA) for prevention. The bar is now prevention of **aggressive** cancer, which no drug has met in phase III.

5 α -reductase inhibitors (5ARIs)

Finasteride and dutasteride block testosterone → dihydrotestosterone (DHT), shrinking the gland ~25% and halving PSA. Rationale: men with congenital 5 α -reductase deficiency have rudimentary prostates.

- **High-grade controversy:** both trials showed ~25% fewer cancers but slightly more high-grade (Gleason 7-10 / 8-10) cancers. Proposed artefacts — 5ARI makes PSA more cancer-specific and shrinks the gland so biopsy samples better — were **not accepted by FDA**, which added a high-grade warning to labels.
- Real-world data (>3 million men): fewer diagnoses, more Gleason 8-10, **no effect on mortality**. Conclusion: 5ARIs reduce detection of mostly indolent cancer.
- On a 5ARI, **double the PSA** for interpretation. → *Ch 154*

Other agents

- **Aspirin/non-steroidal anti-inflammatory drugs (NSAIDs):** small ↓ incidence, no mortality effect — not routine.
- **Statins:** observational ↓ advanced/fatal cancer; never tested in a prevention randomised controlled trial (RCT) — promising.
- **Metformin:** null.

Secondary prevention (after diagnosis)

- Prostate Cancer Lifestyle Trial (men on AS): plant-based diet, exercise, stress management → **5% vs 27%** needed treatment at 2 years.
- Activity ↓ cancer-specific mortality ~31%; aim ≥150 min/week moderate or ≥75 min/week vigorous. Western diet ↑ cancer death (hazard ratio 2.5); whole milk/high-fat dairy ↑ recurrence; continued smoking ↑ recurrence and death.
- Pomi-T capsule (pomegranate, green tea, turmeric, broccoli): smaller PSA rise in RCT.

KEY POINT Cardiovascular disease kills more men with non-metastatic prostate cancer than the cancer does — lifestyle advice benefits both.

8 · PSA SCREENING

PSA reaches blood by leakage through disrupted glandular architecture, so benign enlargement, inflammation and cancer all raise it. Screening is the strongest single predictor of detection.

- Screening reduces prostate cancer death by ~**20-30%**; the incidence excess in the screened arm shrinks with time while the mortality benefit builds then plateaus.
- Intention-to-screen underestimates benefit: 20% → 27% after correcting for non-attendance → 31% after also correcting for contamination.
- Harms: high negative-biopsy rate and overdiagnosis of Gleason score 6 / Grade Group 1 cancer → US Preventive Services Task Force (USPSTF) 2012: **Grade D** (against), revised in 2018 to **Grade C** (shared decision, age 55-69). Harms are now mitigated by AS for low-risk cancer and secondary testing (magnetic resonance imaging (MRI) or biomarkers) before biopsy. → *Ch 154*

9 · LANDMARK TRIALS

Trial	Design	Result	Take-home
ERSPC (European Randomized Study of Screening for Prostate Cancer)	>182,000 men; PSA every 2-4 y vs none	Cancer death ↓ 16% (9 y) → ~20% (16 y); excess incidence 85% → 36%	Screening saves lives, time-dependent; low contamination

Trial	Design	Result	Take-home
PLCO (Prostate, Lung, Colorectal, Ovarian) — US	Annual PSA vs usual care	Control arm heavily screened → contrast contaminated, effect diluted	Contamination dilutes effect — a trial of screening vs less screening
Göteborg-1	Swedish screening trial	Significant ↓ cancer-specific mortality	Supports ERSPC; basis of 4Kscore
† CAP (Cluster randomised trial of PSA testing, UK)	Single PSA invitation vs none	No benefit at 10 y; very small absolute ↓ death at 15 y	One-off screening is weak
PCPT (Prostate Cancer Prevention Trial)	18,882 average-risk men, PSA ≤3; finasteride 5 mg vs placebo 7 y; end-of-study biopsy	Cancer ↓ 24.8% (18.4 vs 24.4%); Gleason 7-10 ↑ 6.4 vs 5.1%; 18-y cancer death ↓ 25% (non-significant)	Fewer cancers, more high-grade; not approved
REDUCE (REduction by DUtasteride of prostate Cancer Events)	High risk: PSA 2.5-10, prior negative biopsy; dutasteride 0.5 mg vs placebo 4 y	Cancer ↓ 22.8% ; Gleason 8-10 ↑ (0.9 vs 0.6%; years 3-4: 12 vs 1)	Same answer as PCPT; FDA warning
SELECT	35,533 men; selenium 200 µg ± vitamin E 400 IU vs placebo	Stopped early for futility; vitamin E ↑ risk (P = 0.06)	No supplements for prevention
VITAL	>25,000; vitamin D3 2000 IU vs placebo 5 y	No prostate effect	Vitamin D not preventive

10 • MCQ ONE-LINERS

- Highest prostate cancer mortality region: Caribbean.
- Latent cancer at autopsy >79 years ≈ 59%.
- Three established risk factors: age, race, family history.
- Father and brother both affected: ~9-fold risk.
- Twin-study heritability: up to 58%.
- Black men: twice the cancer-specific mortality of White men.
- Ancestry-related risk locus: 8q24.
- HOXB13 mutation: hereditary prostate cancer, prostate-specific homeobox gene.
- BRCA2 carriers: aggressive disease; caution with active surveillance.
- MMR-deficient (Lynch) disease: responds to immune checkpoint inhibitors.
- Polygenic risk scores predict any cancer, not aggressive cancer.
- Smoking: no increase in incidence but ~40% higher mortality.
- Vasectomy is not a prostate cancer risk factor.
- Transgender women on oestrogen: median PSA 0.02 ng/mL.
- Adult serum testosterone does not predict prostate cancer risk.
- High-dose zinc supplements: increased lethal cancer.
- SELECT: selenium/vitamin E no benefit; vitamin E possibly harmful.
- PCPT: finasteride reduced cancer by 24.8%.
- PCPT and REDUCE: more high-grade cancer on 5ARI.
- No drug is FDA-approved for prostate cancer prevention.
- 5ARIs lower PSA ~50% — double measured value.
- ERSPC: ~20% reduction in prostate cancer death.
- PLCO negative because the control arm was contaminated by screening.
- USPSTF: Grade D (2012) → Grade C for age 55-69 (2018).
- Composite healthy lifestyle: 30-50% fewer lethal cancers.

Reading time ≈ 20 min (slow pace)

1 • WHY BIOMARKERS BEYOND PSA

- Prostate-specific antigen (PSA) is **organ-specific, not cancer-specific**: benign prostatic hyperplasia (BPH) and inflammation also raise it → negative biopsies and overdiagnosis of Grade Group (GG) 1 cancer. Screening evidence → *Ch 153*.
- Guidelines now advise **secondary testing** (imaging or biomarkers) before biopsy when PSA is raised; the target is **clinically significant cancer (GG ≥ 2)**.
- Three settings: (1) before initial or repeat biopsy; (2) newly diagnosed favourable-risk cancer — active surveillance (AS) vs radical treatment; (3) after radical prostatectomy (RP) — adjuvant/salvage decisions.
- Generations: first = organ (PSA); second = cancer (PCA3, TMPRSS2:ERG, kallikreins); third = high-grade cancer (SChLAP1). Second-generation markers track grade only indirectly, via tumour volume.

2 • PSA BIOLOGY

- PSA = human kallikrein 3 (**hK3**), gene KLK3 on **chromosome 19q13**; androgen-regulated serine protease from luminal epithelium that **liquefies semen** (cleaves semenogelin). Half-life **2-3 days**.
- Processing: pre-pro-PSA → **proPSA** (244 amino acids (aa), 7-aa leader) → **hK2 cleaves the leader** → active PSA (237 aa).
- Why serum PSA rises**: loss of basal-cell layer and basement membrane lets PSA leak out. Per cell, cancer makes *less* PSA than benign tissue, but Gleason pattern 4 contributes several-fold more serum PSA per cm³.

Serum form	% of PSA	Measured?	Note
PSA-α1-antichymotrypsin (ACT)	60-90	Yes	Predominant complex; inactive
PSA-α2-macroglobulin (A2M)	10-20	No — all epitopes masked	Retains some activity
PSA-α1-protease inhibitor	1-5	Yes	—
Free PSA (fPSA)	5-45	Yes	proPSA, benign PSA (BPSA), intact PSA

Confounders of PSA

- Biopsy: rise lasts **≥ 4 weeks**; cycling >55 km ~10% rise; ejaculation transient in men ≥50 (back to baseline in 24 h — ask, and repeat after abstinence).
- 5α-reductase inhibitors (5ARIs)**: PSA falls ~**50%** by 12 months → **double** it (overestimates in first 6 months, underestimates after years). 5ARI improves PSA's screening performance (Prostate Cancer Prevention Trial, PCPT).
- Age, race and prostate volume alter PSA without cancer. † Age-specific upper limits (ng/mL): 40-49 → 2.5; 50-59 → 3.5; 60-69 → 4.5; 70-79 → 6.5.

3 • PSA DERIVATIVES

Derivative	Definition / values	Clinical use
PSA density (PSAD)	PSA ÷ prostate volume; cut-off 0.15 (range 0.08-0.15)	Better than PSA; used after negative/equivocal magnetic resonance imaging (MRI) (data for 0.15 there are lacking); ≤0.15 = Johns Hopkins AS entry; predicts AS reclassification

Derivative	Definition / values	Clinical use
PSA velocity	Change per year	Minimal value; never an isolated biopsy trigger ; unclear in AS
% free PSA	Cancer PSA escapes processing and binds ACT → low %fPSA = cancer ; thresholds 15–25%	Food and Drug Administration (FDA): PSA 4–10 with benign digital rectal examination (DRE); mainly after negative biopsy ; unaffected by 5ARI (a ratio)

KEY POINT Limit of %fPSA: validated against *any* cancer, not GG ≥2, so it overestimates clinically significant risk. Risk calculators (PCPT, Prostate Biopsy Collaborative Group) incorporate it for shared decisions.

Free PSA isoforms

- **[-2]proPSA (p2PSA)**: incompletely cleaved, peripheral zone, rises with Gleason score → basis of PHI.
- **BPSA**: internally clipped, transition zone, tracks BPH/volume — no prognostic value.
- **Intact PSA (iPSA)**: component of 4Kscore.

4 · BLOOD AND URINE TESTS (NCCN 2023 PRE-BIOPSY BIOMARKERS)

The National Comprehensive Cancer Network (NCCN) lists six: three blood, three urine. All estimate risk of GG ≥2 to avoid unnecessary biopsies; **thresholds are setting-specific** (higher for repeat biopsy).

Test	Sample / components	Threshold / key fact
PHI (Prostate Health Index)	Blood: (p2PSA ÷ fPSA) × √tPSA (tPSA = total PSA)	24–28.6 proposed (none validated); ~30% biopsies avoided; reported against any-grade cancer
4Kscore	Blood: total PSA, fPSA, iPSA, hK2 + age, DRE, prior biopsy	7.5% : sensitivity 94%, negative predictive value (NPV) 95%; ≤7.5% = low long-term metastasis/death risk
IsoPSA	Blood: PSA isoform <i>structure</i> (two-phase partition)	Unaffected by 5ARI or α-blockers
SelectMDx	Post-DRE urine: distal-less homeobox 1 (DLX1), homeobox C6 (HOXC6) messenger RNA (mRNA) + age, DRE, PSA, volume	NPV ~95%; needs prostate volume (ultrasound/MRI)
ExoDx Prostate IntelliScore (EPI)	Urine exosomes: PCA3, ERG, SPDEF	No DRE , no clinical variables; 15.6 initial, 20 repeat
MyProstateScore (MPS)	Post-DRE urine PCA3 + TMPRSS2:ERG + serum PSA	10 initial (sensitivity 97%), 15 repeat; validated in Black men

PCA3 and TMPRSS2:ERG

- **Prostate cancer antigen 3 (PCA3)**: long non-coding RNA on 9q21–22, overexpressed in cancer. Score = (PCA3 mRNA ÷ KLK3 mRNA) × 1000 in urine collected after **attentive DRE**; thresholds 25/35. **Independent of prostate size**. FDA-approved for men with a **previous negative biopsy**; now used only within EPI and MPS.
- **TMPRSS2:ERG**: fusion of androgen-regulated TMPRSS2 with the E26 transformation-specific (ETS) transcription factor ERG; **~100% specific** but in only ~half of tumours → combined with PCA3 (doubles specificity at 95% sensitivity).
- **hK2** (~80% homologous to PSA) is expressed more intensely in cancer and with grade, independently of PSA → complementary.

KEY POINT Why must urine tests follow a DRE? Urine carries cancer RNA only from shed prostate cells, and an attentive DRE massages cells into the urethra. ExoDx avoids this because exosomes (small vesicles carrying RNA) are shed into urine spontaneously. Urinary markers are highly cancer-specific; their weakness is coverage, so they are combined into panels.

5 • TISSUE-BASED BIOMARKERS

Methylation — after a negative biopsy

ConfirmMDx: methylation of GSTP1, APC, RASSF1 in negative cores; NPV ~88-90%. Rationale: the **field ("halo") effect** — methylation extends into benign tissue beside a tumour, so a core that missed cancer still carries its signature.

Genomic classifiers — prognosis, not detection

	Prolaris	Oncotype DX Prostate	Decipher
Genes	31 cell-cycle progression + 15 housekeeping	12 cancer genes (4 pathways) + 5 housekeeping	22-marker classifier
Predicts	Progression, cancer death; recurrence after RP/radiotherapy	Adverse pathology at RP (primary pattern 4 or pT3)	Metastasis and cancer death
Main use	Biopsy: AS vs treatment	Biopsy: AS vs treatment	Post-RP (score ≥ 0.4 may benefit from radiotherapy); now also biopsy

- Post-Operative Radiation Therapy Outcomes Score (PORTOS): 24-gene Decipher-derived signature predicting radiotherapy response.
- **Limitations:** multifocality — a low-grade focus does not predict an unsampled high-grade one; evidence mostly retrospective. G-MAJOR is a prospective randomised trial of Decipher in favourable-risk disease.

6 • GERMLINE TESTING

- Up to 40-50% of risk is familial/hereditary; one affected relative → 2-4-fold risk. Most heritable genes are **DNA-damage-repair** genes.
- STHLM3: 4 kallikreins + 232 single-nucleotide polymorphisms + clinical variables for Gleason score ≥ 7 .

Gene	Risk	Phenotype
BRCA2	4.5× <65 y; ~8.6× overall	Younger, higher grade, shorter survival
BRCA1	1.8× <65 y	—
ATM, CHEK2	↑ (CHEK2 odds ratio ~2-3.4)	ATM aggressive
Lynch (MSH2 etc.)	MSH2 ~10×	Also colorectal, upper tract cancers
HOXB13 G84E	4.5×	Not more aggressive

Testing by next-generation sequencing: BRCA1/2 alone, or multigene panels (BRCA1/2, ATM, CHEK2, MLH1, MSH2, TP53, EPCAM) when no familial mutation is known. NCCN focuses on BRCA; carriers need earlier, tailored detection. Indications → *Ch 153*.

7 • CLINICAL ALGORITHM

ALGORITHM

Raised PSA → repeat (exclude infection, instrumentation, ejaculation, 5ARI) → MRI and/or biomarker (PHI, 4Kscore, IsoPSA, SelectMDx, EPI, MPS; PSA density) → biopsy if risk high

Negative biopsy, persistent suspicion → MRI, %fPSA, PCA3/EPI/MPS (repeat thresholds), ConfirmMDx → targeted re-biopsy

Newly diagnosed favourable-risk → PSA density, Prolaris/Oncotype/Decipher → AS vs treatment

After RP → † PSA should be undetectable; ≥ 0.2 ng/mL confirmed = biochemical recurrence → Decipher/PORTOS for adjuvant/salvage radiotherapy

8 • MCQ ONE-LINERS

1. PSA is human kallikrein 3; gene on chromosome 19q13.
2. PSA liquefies semen by cleaving semenogelin.
3. Serum PSA half-life: 2–3 days.
4. hK2 activates proPSA by cleaving its 7-aa leader.
5. Major PSA complex: $\alpha 1$ -antichymotrypsin.
6. PSA- $\alpha 2$ -macroglobulin is not detected by standard assays.
7. Cancer: lower % free PSA.
8. Per cell, cancer produces less PSA than benign tissue.
9. 5ARI halves PSA; % free PSA unchanged.
10. PSA rise after biopsy persists ≥ 4 weeks.
1. PSA density cut-off: 0.15.
2. PSA velocity should never alone trigger biopsy.
3. [-2]proPSA: peripheral zone; BPSA: transition zone.
4. $PHI = (p2PSA \div fPSA) \times \sqrt{tPSA}$.
5. 4Kscore unique component: hK2; cut-off 7.5%.
16. IsoPSA unaffected by 5ARI.
17. PCA3: long non-coding RNA, independent of prostate volume.
18. PCA3 requires post-DRE urine; FDA-approved after negative biopsy.
19. ExoDx: urine exosomes, no DRE needed.
20. TMPRSS2:ERG: $\sim 100\%$ specific, present in half of tumours.
21. SelectMDx genes: DLX1 and HOXC6; needs prostate volume.
22. ConfirmMDx: methylation of GSTP1, APC, RASSF1 — field effect.
23. Prolaris: cell-cycle progression genes.
24. Oncotype DX predicts adverse pathology at prostatectomy.
25. Decipher ≥ 0.4 post-RP may favour radiotherapy.
26. BRCA2: highest-risk, most aggressive germline mutation.
27. HOXB13 G84E raises risk but not aggressiveness.

155 Prostate Biopsy: Techniques and Imaging

Reading time ≈ 25 min (slow pace)

1 · INDICATIONS AND CONTRAINDICATIONS

Transrectal ultrasonography (TRUS) shows posterior peripheral zone (PZ) and central zone as homogeneous echogenic tissue and the anterior transition zone (TZ) as heterogeneous; zonal anatomy itself is pathological, not sonographic. Most cancers arise in the posterolateral PZ, so biopsy geometry follows the PZ. Grey scale is a poor cancer test: **39%** of cancers are isoechoic; sample hypoechoic PZ lesions, but their absence never precludes biopsy. Imaging alone never diagnoses cancer: tissue is mandatory.

- **Initial biopsy:** raised prostate-specific antigen (PSA) plus risk factors after informed choice; suspicious digital rectal examination (DRE) or nodule; symptoms or signs of metastasis; cancer at transurethral resection for presumed benign disease.
- **Repeat biopsy:** rising/persistent PSA; atypical small acinar proliferation (ASAP; **≈50%** cancer risk); atypical intraductal proliferation; positive urinary prostate cancer gene 3 (PCA3)/ConfirmMDx with persistent suspicion; suspicious magnetic resonance imaging (MRI) lesion; recurrence after ablation. **Follow-up:** active surveillance (AS).
- National Comprehensive Cancer Network (NCCN): biopsy for abnormal DRE with raised PSA, or repeat PSA >3 ng/mL with Prostate Imaging Reporting and Data System (PI-RADS) ≥3; PSA velocity alone, or abnormal DRE with normal PSA, is insufficient. Biomarkers → *Ch 154*; screening → *Ch 157*.
- PSA density (PSAD) = PSA ÷ volume; **>0.15 ng/mL/cc** suggests cancer. Volume (ellipsoid) = $\pi/6 \times \text{transverse} \times \text{anteroposterior} \times \text{length}$; 1 cm³ ≈ 1 g.
- **Contraindications:** significant coagulopathy, severe immunosuppression, acute urinary tract infection (UTI)/prostatitis. Painful anorectal disease → general/regional anaesthesia. No anus (after abdominoperineal resection) → in-bore MRI or transperineal biopsy.

2 · PRE-BIOPSY MULTIPARAMETRIC MRI (MPMRI)

mpMRI = T2-weighted imaging (T2WI) + diffusion-weighted imaging (DWI) + dynamic contrast enhancement (DCE). It acts as a **triage test**: no lesion → biopsy may be avoided; lesion → targeted cores. Targeting finds more high-grade and fewer low-grade cancers, which curbs overdiagnosis.

PI-RADS scores 1-5 the likelihood of clinically significant cancer (Gleason ≥7 and/or volume >0.5 cm³ and/or extraprostatic extension [EPE]). Lesion **≥1.5 cm** or definite EPE → PI-RADS 5. Validation: cancer in ≈1/3 of PI-RADS 3, >1/2 of 4, ≈90% of 5. AUA/Society of Abdominal Radiology 2016: mpMRI and targeting of PI-RADS ≥3 after a negative biopsy with persistent PSA rise.

Feature	Peripheral zone	Transition zone
Dominant sequence	DWI	T2WI
Dominant score 1, 2, 4, 5	PI-RADS = DWI score	PI-RADS = T2WI score
Dominant score 3	DCE positive → 4; negative → 3	DWI 5 → 4; else 3
Role of DCE	Upgrades 3 → 4 only	None
† v2.1 change (2019)	—	T2WI 2 with DWI ≥4 → PI-RADS 3

KEY POINT Trap: DCE never creates or downgrades a PI-RADS 4-5, and never upgrades a TZ lesion. mpMRI still misses a minority of significant cancers (mostly 3+4; a third primary pattern 4): MRI-invisible lesions, reader discordance, registration or needle error, small lesion in a large gland. Hence systematic cores remain relevant.

3 · TARGETING TECHNIQUES

	Cognitive	MRI/ultrasound software fusion	In-bore MRI
Method	Operator reviews MRI and aims mentally during TRUS	Software coregisters marked MRI with live TRUS (3D model, elastic deformation corrects probe distortion)	Prone in gantry; needle guide in rectum; repeated short sequences confirm needle
Equipment	None extra	Fusion platform (UroNav freehand electromagnetic; Artemis mechanical arm)	MRI-compatible kit
Drawback	Needle position cannot be recorded; operator-dependent; poor for small lesion in large gland	Cost; coregistration error	Time, cost, logistics

Fusion: PI-RADS 5 targets are nearly always positive; targeted cores are $\approx 3\times$ likelier than systematic cores to find intermediate/high-risk disease. Meta-analysis (2022): **no significant difference** among the three in detection. Take **2-4 cores per target** (American Urological Association [AUA]/Society of Urologic Oncology [SUO] 2023: 2-3, systematic biopsy may be added). Targeted biopsy indicated whenever biopsy is indicated and a PI-RADS ≥ 3 lesion exists; also persistent PSA after negative biopsy, AS, focal-therapy planning.

4 · SYSTEMATIC TEMPLATES AND TECHNIQUE

- **Sextant** (Hodge 1989): base, mid, apex parasagittal; sampled too much TZ, missed lateral PZ → inadequate today.
- **Extended 12-core** (double sextant, adds lateral and apical cores): AUA standard; ↑ detection, fewer repeat biopsies. 6 → 12 raises detection, 12 → 18-21 does not: **geometry, not core number**.
- TZ/anterior cores: repeat setting or gland >50 mL. Seminal vesicle cores: palpable abnormality.
- **Transrectal technique**: left lateral decubitus, DRE → volume → survey base to apex → 18-gauge spring-loaded gun (throw 15-25 mm; tip runs 0.5 cm past the core). Indent capsule, withdraw a few mm, then fire, else the core samples anterior tissue and misses the PZ. Never reposition with the needle on the rectal wall. ≤ 2 cores per formalin jar.

5 · TRANSRECTAL VS TRANSPERINEAL (TP)

	Transrectal	Transperineal
Position	Left lateral	Lithotomy, scrotum taped up
Needle passes	Rectal wall (Class III, contaminated)	Sterilisable perineal skin
Prophylaxis	Mandatory for all	Skin flora only, or none
Infection	Higher, driven by fluoroquinolone resistance	Lower (meta-analysis of randomised trials)
Rectal bleeding	Common	Eliminated
Apex/anterior access	Needs probe torque	Ready (anterior-only cancers 6-20%)
Template	Single 12-core	Several (Ginsburg; Michigan Urological Surgery Improvement Collaborative [MUSIC] 12-core, which omits midline cores → less pain/haematuria); none universal

- European Association of Urology (EAU) 2023: TP **preferred** (strong). Detection at least non-inferior. Slogan "TREXIT".

- TP prophylaxis: skin prepared with povidone-iodine or chlorhexidine, so no rectal flora is inoculated; skin-flora cover or none, as omitting antibiotics did not change the very low sepsis rate.
- Access: brachytherapy grid (mapping, 5-mm), freehand needle-guide kit (PrecisionPoint), or two preplaced 14-gauge cannulas (cheapest, most versatile; learning curve 4–6 weeks).
- † **PREVENT** (randomised, 2024): TP without antibiotics vs transrectal with targeted prophylaxis: no significant difference in infection; detection comparable.

6 • ANTIBIOTIC PROPHYLAXIS

- Transrectal: every patient, regardless of risk. AUA 2019: fluoroquinolone; 1st/2nd-generation cephalosporin ± aminoglycoside; or 3rd-generation cephalosporin. Give ≥1 h before; duration **≤24 h**. Oral route only for quinolones.
- Single-dose agents: levofloxacin 500 mg, gentamicin 5 mg/kg, amikacin 15 mg/kg, ceftriaxone 1–2 g. Renal impairment: aztreonam replaces aminoglycoside.
- Food and Drug Administration (FDA) 2016 fluoroquinolone warning: tendon rupture, neuropathy, central nervous system effects, myasthenia.
- Infection risk factors: antibiotics within **6 months** (most consistent), diabetes, immunosuppression, prior UTI/prostatitis/biopsy infection, many prior biopsies (AS), travel (India, South-East Asia), health-care worker, rectal fluoroquinolone-resistant *E. coli*. Response: augmented (>1 agent) or rectal-swab-targeted prophylaxis, or go transperineal.
- Endocarditis/prosthesis risk: IV ampicillin (vancomycin if allergic) + gentamicin → oral quinolone 2–3 days.
- Enema: better acoustic window; no proven infection benefit.

7 • ANTICOAGULATION AND ANAESTHESIA

Agent	Action
Low-dose aspirin	Continue
Warfarin, clopidogrel	Stop 5–7 days; biopsy when international normalised ratio (INR) <1.5
Direct oral anticoagulants	Stop 2–5 days
High thromboembolic risk (mechanical valve)	Bridge with unfractionated or low-molecular-weight heparin

Also stop herbal supplements. Biopsy is an intermediate bleeding-risk procedure.

- **Transrectal**: periprostatic block, 1–2% lidocaine 5 mL each side at the seminal vesicle-prostate junction fat pad via a 22-gauge spinal needle; beats topical intrarectal gel.
- **Transperineal**: skin/tract 1–2 mL → **periapical triangle** 5 mL (bounded by levator ani laterally, rhabdosphincter medially, external anal sphincter posteriorly, apex superiorly) → 3 mL along capsule, all under ultrasound. Pain worst at injection, then biopsy, then probe.

8 • COMPLICATIONS AND MANAGEMENT

Complication	Rate	Management
Haematuria	23–63%	Self-limiting; clot retention (0.7%) → catheter, washout, irrigation
Haematospermia	10–50%	Counsel: lasts ≈1 month
Rectal bleeding	2–22%	Escalation ladder (below)
Infection / UTI	0.1–7% / 2–6%	Oral or IV antibiotics
Sepsis	0.3–3.1%	Carbapenem until cultures; up to 2 weeks if admitted
Vasovagal	1.4–5.3%	Trendelenburg, IV fluids

Complication	Rate	Management
Retention	0.2-0.4%	Temporary catheter (large gland, high International Prostate Symptom Score)
Erectile dysfunction	—	Transient; anxiety contributes

ALGORITHM

Post-biopsy fever → assess for sepsis (fever/hypothermia, tachycardia, tachypnoea, confusion, hypotension) → cultures → **no empirical fluoroquinolone** (resistant *E. coli*; also resistant to co-trimoxazole, ampicillin) → carbapenem (or amikacin/3rd-generation cephalosporin per sensitivity) → resuscitate.

ALGORITHM

Persistent rectal bleeding → pressure (probe/finger) → tamponade (inflated sheath or Foley balloon) → endoscopy with adrenaline/sclerosant → angioembolisation → transrectal suture.

KEY POINT "Commonest complication": bleeding overall; infection among serious complications/admissions. Admissions for infection rose from 1.0% to 4.1% in the resistance era. The asymmetry (bleeding frequent but trivial, infection rare but dangerous) drives universal prophylaxis and the move to the perineum.

9 · REPEAT AND SATURATION BIOPSY

- Yield falls each round: European Randomized Study of Screening for Prostate Cancer (ERSPC) **22% → 10% → 5% → 4%** on biopsies 2-4.
- Second biopsy if suspicion persists; third/fourth only for high suspicion. Strategy now: **mpMRI ± targeted biopsy**, tissue methylation (ConfirmMDx), anterior/apical and TZ cores, transperineal route.
- Saturation** (>12, usually 18-21+ cores; Barzell transperineal template mapping): mainly after prior negatives; needs anaesthesia; a large prospective study showed no significant gain. Transurethral biopsy is obsolete.
- Micro-ultrasound (29 MHz; PRI-MUS score) approaches mpMRI without contrast: emerging.

10 · LANDMARK TRIALS

Trial	Design	Result	Take-home
PROMIS (2017)	576 men: mpMRI, 12-core TRUS and 5-mm template mapping (reference)	Sensitivity mpMRI 93% vs TRUS 48%; negative predictive value (NPV) 89% vs 74%; 27% could avoid biopsy	mpMRI is a better triage test than TRUS biopsy
PRECISION (2018)	Randomised, 500 biopsy-naive men: MRI → targeted (PI-RADS ≥3) vs 12-core TRUS	Significant cancer 38% vs 26%; 28% avoided biopsy	MRI pathway superior; fewer insignificant cancers
† MRI-FIRST (2019)	Prospective paired: systematic + targeted in each man	Each method alone missed significant cancers the other found	Combine targeted + systematic for maximal detection
PREVENT (2024)†	Randomised TP (no antibiotics) vs transrectal	Infection not significantly different	Both acceptable; TP avoids antibiotics

11 · MCQ ONE-LINERS

- Percentage of cancers isoechoic on TRUS: 39%.
- Commonest false-positive hypoechoic lesion: benign prostatic hyperplasia foci and prostatitis.
- PSAD suggesting malignancy: >0.15 ng/mL/cc.
- Sextant biopsy described by Hodge, 1989.
- Current standard systematic biopsy: 12 cores (AUA).
- Needle gauge for core biopsy: 18 G; throw 15-25 mm.

7. Maximum cores per formalin jar: 2.
8. Transrectal biopsy wound class: III (contaminated).
9. AUA prophylaxis duration: ≤ 24 h.
10. Most consistent infection risk factor: antibiotic exposure within 6 months.
11. Dominant organism in post-biopsy sepsis: fluoroquinolone-resistant *E. coli*.
12. Empirical drug for post-biopsy sepsis: carbapenem.
13. Aminoglycoside substitute in renal insufficiency: aztreonam.
14. Aspirin before biopsy: continue low dose.
15. INR threshold for biopsy: <1.5 .
16. Commonest complication overall: bleeding (haematuria).
17. Haemospermia lasts ≈ 1 month.
18. Dominant PI-RADS sequence in PZ: DWI; in TZ: T2WI.
19. DCE role: upgrades PZ score 3 to 4 only.
20. Lesion size giving automatic PI-RADS 5: >1.5 cm.
21. Significant cancer definition in PROMIS: Grade Group ≥ 3 or core length ≥ 6 mm.
22. PROMIS mpMRI sensitivity 93% vs TRUS 48%.
23. PRECISION: targeted 38% vs systematic 26% significant cancer.
24. Cores per MRI target: 2–4.
25. Cognitive vs fusion vs in-bore: no significant detection difference.
26. EAU 2023 preferred route: transperineal.
27. Commonest transperineal block: periapical triangle.
28. Repeat-biopsy yield ERSPC biopsy 2: 22%.
29. ASAP on biopsy: $\approx 50\%$ cancer on repeat.
30. Absent seminal vesicle: ipsilateral renal agenesis in 79%.
31. Müllerian cysts midline; wolffian cysts lateral.

156 Pathology of Prostatic Neoplasia

Reading time ≈ 29 min (slow pace)

1 · PRECURSOR AND INTRADUCTAL LESIONS

Prostatic intraepithelial neoplasia (PIN) = architecturally benign acini/ducts lined by cytologically atypical cells; low-grade (LGPIN) vs high-grade (HGPIN).

- **LGPIN is not reported:** irreproducible vs benign, and no excess cancer risk on repeat biopsy.
- **HGPIN is a precursor:** carcinoma seen budding off it; ~**20%** of HGPIN carries the transmembrane protease serine 2 : ETS-related gene (**TMPRSS2:ERG**) fusion; further hits needed for invasion.
- **No repeat biopsy for isolated HGPIN, unifocal or multifocal:** once multiparametric MRI (mpMRI), digital rectal examination (DRE) and prostate-specific antigen (PSA) are considered, unsampled significant cancer is very unlikely. Same for HGPIN on transurethral resection (TUR).
- HGPIN is now less worrisome because **cribriform HGPIN is no longer diagnosed** — it is reclassified as intraductal carcinoma or atypical intraductal proliferation (AIP).

Intraductal carcinoma of the prostate (IDC-P): atypia clearly beyond HGPIN (e.g. dense cribriform) with **basal cells retained**. Usually **retrograde spread of high-grade invasive cancer** into ducts (clonally related); rarely a high-grade precursor. Prevalence rises with risk (~2% low risk → >50% metastatic). Independently predicts early biochemical recurrence (BCR). With grade group 1 + IDC-P, radical prostatectomy (RP) upgrades in **≥80%**. Strict (Guo-Epstein) criteria license definitive therapy without proven invasion.

Lesion	Defining feature	Meaning	Action on isolated biopsy finding
LGPIN	Mild atypia; irreproducible	No excess risk	Not reported
HGPIN	Benign architecture, atypical cells; non-cribriform	Weak precursor	No repeat biopsy
PINATYP	HGPIN + adjacent small atypical glands	As "atypical"	Manage as atypical, suspicious for carcinoma
AIP	Atypia beyond HGPIN, below IDC-P criteria	May herald IDC-P/significant cancer	Consider re-biopsy (if benign or grade group 1 elsewhere)
IDC-P	Dense cribriform/marked atypia, basal cells retained	Usually high-grade cancer already present	Definitive therapy (or re-biopsy); excludes active surveillance (AS)

KEY POINT Intraductal lesions are graded by what they signify: HGPIN is a weak risk marker, IDC-P usually marks high-grade cancer already present. Traps: re-biopsy for HGPIN; surveillance for grade group 1 with IDC-P.

2 · ADENOCARCINOMA: LOCATION, SPREAD, STAGING

Zonal anatomy (McNeal) and tumour location

Zone	% of glandular tissue	% of cancers	Typical pathology / clinical relevance
Peripheral zone (PZ)	†~70%	†~70%	Main mass in clinical T2 and 85% of T1c; †posterolateral → palpable on DRE, well sampled by systematic biopsy
Transition zone (TZ)	†~5% (young adult)	†~20%	†Origin of benign prostatic hyperplasia (BPH); anterior TZ tumours are impalpable and under-sampled by transrectal ultrasound (TRUS)-guided systematic biopsy; usually lower grade

Zone	% of glandular tissue	% of cancers	Typical pathology / clinical relevance
Central zone (CZ)	†~25%	†~5%	†Cone around the ejaculatory ducts at the base; cancers here rare but more aggressive (higher grade, seminal vesicle invasion)
Anterior fibromuscular stroma	†Non-glandular	†Rarely primary site	†Invaded by anterior PZ/TZ tumours
Periurethral glands	†<1%	—	†Contribute to BPH (median lobe)

- **Anterior tumours** (TZ or anterior horn of PZ) ~15% of RPs: same BCR risk, but extraprostatic extension (EPE) more often has positive margins.
- **Multifocal in >85%**; contralateral tumour usually small, but ~20% show adverse size, EPE, grade or margins.

Spread

- No discrete histological capsule → say "**extraprostatic extension**", not capsular penetration. Peripheral tumours exit via **perineural spaces**, posterolaterally.
- **Seminal vesicle invasion (SVI)** = tumour in the SV *muscular wall*. Routes: commonest — out at the base → periseminal soft tissue → SV; less common — along ejaculatory ducts or direct from base; almost never discontinuous.
- Metastases: nodes and bone commonest; may present in left supraclavicular nodes. Lung common at autopsy (with bone).
- Volume vs stage: EPE uncommon <**0.5 cm³**; nodes/SVI uncommon <**4 cm³**.

Pathological stage

Stage	Definition
T1a / T1b	Incidental on TUR of prostate (TURP)/enucleation; ≤5% / >5% of tissue
T1c	Non-palpable, needle biopsy (usually raised PSA)
pT2	Organ-confined at RP (T1a-c are restaged pT2/pT3)
pT2+ (pT2x)	Positive margin at intraprostatic incision or ill-defined boundary (apex) — stage there unassessable
pT3a	EPE; includes microscopic bladder neck invasion
pT3b	Seminal vesicle invasion
pT4	†Invasion of adjacent structures other than SV (rectum, levators, external sphincter, pelvic wall)

Trap: microscopic bladder neck invasion is pT3a, not pT4; T1a/T1b hinge on the 5% cut-off, not grade.

3 · GRADING: GLEASON AND GRADE GROUPS

- **Gleason**: architecture at low power, patterns 1-5; modified 2005 and later. Original = commonest + second commonest; **2005 biopsy rule = commonest + highest**. Single pattern doubled (3+3).
- Lowest score in practice = **6**. Expanded pattern 4 moved many old 6s to 7: modern 6 is homogeneous and favourable — virtually no progression if organ-confined, **no nodal metastatic capacity**, EPE uncommon.

Pattern	Morphology
3	Discrete well-formed glands, whatever size or packing
4	Cribriiform (incl. glomeruloid) , poorly formed, fused glands
5	Sheets, cords, single cells, solid nests, comedonecrosis

Grade group (GG) system = International Society of Urological Pathology (ISUP) grade; endorsed by World Health Organization (WHO) 2016; better reflects biology than Gleason score.

GG	Gleason	Nodes at RP	5-yr BCR-free
1	3+3=6	0%	<u>96%</u>
2	3+4=7	0.6%	<u>88%</u>
3	4+3=7	4.3%	<u>63%</u>
4	4+4, 3+5, 5+3 = 8	—	<u>48%</u>
5	4+5, 5+4, 5+5	—	<u>26%</u>

KEY POINT The steep drop is between GG2 and GG3, so 3+4 and 4+3 are never lumped as "7". Trap: 3+5 and 5+3 are GG4; GG5 starts at 9. Because pure GG1 cannot metastasise, whether to call it "cancer" is debated.

- **Percent pattern 4** in score 7 correlates with adverse pathology and BCR → report it for GG2–3† (ISUP 2014 recommendation). GG2 with limited pattern 4 may suit AS; GG3 with 90% pattern 4 (near GG4) → neoadjuvant hormones with radiotherapy recommended, at 60% possibly avoidable.
- **Tertiary pattern 5** at RP (<5% of a 3+4/4+3 nodule): raises BCR risk (between GG2 and GG3, or equal to GG3) but **stays in its GG** because nodal risk is not raised.

ISUP 2019 vs Genitourinary Pathology Society (GUPS) 2019

Issue	ISUP 2019	GUPS 2019
% pattern 4 in GG2–3	†Report	†Report
IDC-P with invasive cancer	† Incorporated into Gleason grade	† Not graded ; reported separately (comment)
Cribriform pattern	†Report presence	†Report presence
MRI-targeted cores of one lesion	† Aggregate grade for the lesion (plus per-core)	†Grade each core; aggregate optional

- †**Minor high-grade pattern**: on biopsy, any higher-grade pattern is included as the secondary pattern, however small; at RP, a higher-grade pattern <5% is reported as a tertiary (minor) pattern and does not change the GG.
- †Both bodies treat cribriform glands as pattern 4 regardless of extent.

KEY POINT †Why it matters: cribriform pattern and IDC-P mark aggressive disease and argue against active surveillance even in GG2, so they must appear in the report. Aggregate grading of a targeted lesion avoids both under-grading (one core alone) and over-grading (highest core alone).

Upgrading (sampling error)

Setting	Risk of higher grade
Biopsy GG1 → RP	~ $\frac{1}{3}$ upgraded
Biopsy GG2 → RP	~20% become GG3
GG1 with negative mpMRI (Prostate Imaging–Reporting and Data System [PI-RADS] 1–2)	Still ~20% at 6 yr

Predictors: greater biopsy extent, higher PSA, smaller prostate, fewer cores. Early (<3 yr) reclassification = missed component, not progression. Trap: negative mpMRI does not remove GG1 upgrading risk.

4 · NEEDLE BIOPSY ASSESSMENT

- **Separate jars per site:** per-core grade and extent; targeted re-biopsy; $\leq 2-3$ cores/block avoids buried foci and fragmentation; avoids pitfalls (**base: SV/central zone mimic HGPIN; apex: Cowper gland mimics cancer**).
- Report **number of positive cores + one other extent measure** (mm, % per core, total %): none superior.
- **Perineural invasion:** report on biopsy ($\uparrow$ EPE; adverse on AS; prognostic for external-beam radiotherapy); at RP very common and not prognostic — omit.

Diagnosis and mimics

- **Basal cell markers — high-molecular-weight cytokeratin (HMWCK), p63:** stain benign glands; **absence supports cancer**. Pitfall: partial atrophy, adenosis and HGPIN may also lack basal cells (patchy).
- **α -methylacyl-CoA racemase (AMACR):** labels carcinoma and HGPIN — adjunct only (false positives and negatives).
- **Adenosis** — chief mimic of low-grade cancer: transition zone, multifocal, incidental in TURs ($\sim 1.6\%$); **no increased cancer risk**.
- **"Atypical, suspicious for carcinoma" (ASAP):** $\sim 5\%$ of biopsies; cancer on follow-up $\sim 40\%$, PSA does not stratify; $\sim 80\%$ of these GG1, $\sim 5\%$ GG4-5. **Seek expert review before re-biopsy** (most often changed diagnosis); then markers, imaging or re-biopsy sampling the atypical site.

KEY POINT Sampling error is one-directional: a bad biopsy is believed, a good biopsy is provisional, an atypical biopsy is re-examined before it is repeated.

5 · TUR AND RADICAL PROSTATECTOMY SPECIMENS

- TUR: 6-8 cassettes detect all T1b. RP: whole-mount sections give no extra information.
- **Grade each major nodule;** dominant (largest) usually highest grade/stage — if not, label #1, #2.
- **Nodes:** uncommon with Gleason < 8 ; predicted by stage, PSA, grade. Node-positive almost always progresses: at 15 yr BCR-free $\sim 7\%$, cancer-specific survival $\sim 58\%$. Intraductal/cribriform at RP $\rightarrow \uparrow$ nodal and distant metastasis.
- **EPE:** focal (few glands) vs nonfocal — extent predicts progression. **SVI far more ominous: 65%** 5-yr progression.
- **Margins:** $\sim 15\%$ positive in robotic series; only $\sim 50\%$ progress. Distance to ink irrelevant. Report **extent (focal vs nonfocal, cut-off 3 mm) and grade at margin** — guide adjuvant radiotherapy.
- **Intraprostatic incision** (not "capsular incision"): risk equals focal EPE with positive margin.
- **Independent predictors of biochemical progression:** grade, EPE extent, margins (extent, grade), SVI, nodes. **Tumour volume is not independent** $\rightarrow$ rough estimate suffices.

6 · TREATMENT EFFECT

Treatment	Histological effect	Gleason score?
Hormone therapy	Grade appears artefactually higher	No where effect present
Finasteride	No consistent change	Yes
Radiotherapy	Radiation atypia (up to 72 months); benign glands mimic cancer	Only if no treatment effect
Cryotherapy / high-intensity focused ultrasound (HIFU)	Infarction; residual cancer unaltered	Yes

Post-radiotherapy biopsy: benign with radiation effect / cancer without effect / cancer with effect (prognosis $\approx$ no cancer).

7 · SUBTYPES AND OTHER TUMOURS

Subtype	Grading	Behaviour
Ductal (pure 0.4–0.8%; mixed ~5%)	Pattern 4 by default (PIN-like 3; comedonecrosis 5)	Periurethral → exophytic lesion at verumontanum mimicking papillary urothelial cancer; DRE/PSA may be normal; prognosis ≈ Gleason 8; visceral recurrence at low PSA
Mucinous	As usual	As acinar (incl. bone metastases)
Neuroendocrine differentiation (immunohistochemical; ~half of acinar)	As usual	No prognostic significance; risk of misdiagnosis as small cell
Small cell (~50% mixed)	Not graded	Like lung small cell; survival <1 yr
Squamous (pure)	—	Osteolytic metastases; hormone-unresponsive
Sarcomatoid (= carcinosarcoma)	—	Dismal

- **Urothelial carcinoma in prostate** (35–45% of cystoprostatectomies): intraductal only → bladder stage decides. From prostatic urethral carcinoma in situ (CIS): suburethral invasion pT1, stromal invasion pT2; **transmural from bladder into stroma = pT4**.
- Immunohistochemistry: prostate — PSA; if weak, P501S (prostein), NKX3.1. Urothelial — **GATA3** (most sensitive/specific).
- Sarcomas 0.1–0.2%: **rhabdomyosarcoma in children, leiomyosarcoma in adults**. Stromal tumour of uncertain malignant potential (STUMP): may recur/progress → close follow-up.

8 · MCQ ONE-LINERS

- | | |
|---|--|
| <ol style="list-style-type: none"> 1. LGPIN — not reported. 2. Isolated HGPIN, even multifocal — no repeat biopsy. 3. IDC-P — basal cells retained; usually spread of high-grade invasive cancer. 4. GG1 + IDC-P on biopsy — upgraded at RP in >80%; not for AS. 5. Commonest tumour site — posterior peripheral zone; multifocal >85%. 6. Microscopic bladder neck invasion — pT3a. 7. 2005 biopsy Gleason — commonest + highest pattern. 8. Cribriform and glomeruloid glands — pattern 4. 9. 3+5=8 — GG4. 10. 5-yr BCR-free GG1→5 — 96, 88, 63, 48, 26%. 11. Biopsy GG1 upgraded at RP — about one-third. 12. Tertiary pattern 5 <5% — stays in its GG. 13. Basal cell markers — p63, HMWCK; absence supports cancer. 14. ASAP — ~40% cancer on follow-up; PSA does not stratify. | <ol style="list-style-type: none"> 15. Cowper gland mimics cancer at apex; SV/central zone mimic HGPIN at base. 16. Perineural invasion — report on biopsy, omit at RP. 17. Focal vs nonfocal positive margin — 3 mm. 18. SVI — 65% 5-yr progression. 19. Hormone-treated cancer — not graded; post-ablation residual cancer — graded. 20. Ductal adenocarcinoma — graded pattern 4; mimics papillary urothelial cancer at verumontanum. 21. Urothelial vs prostatic — GATA3 vs PSA/NKX3.1/prostein. 22. †Largest zone and commonest site of cancer — peripheral zone (~70%). 23. †BPH arises from — transition zone; central zone cancers are rare but aggressive. 24. †IDC-P in Gleason grade — included (ISUP 2019); excluded, reported separately (GUPS 2019). 25. †Several MRI-targeted cores from one lesion — aggregate grade (ISUP 2019). |
|---|--|

157 Diagnosis and Staging of Prostate Cancer

Reading time ≈ 22 min (slow pace)

1 • CLINICAL PRESENTATION

- Early cancer is **asymptomatic**; symptoms imply advanced disease. Most cancers are now found by prostate-specific antigen (PSA) and digital rectal examination (DRE), then transrectal ultrasound (TRUS)-guided biopsy.
- Locally advanced**: obstructive voiding, ureteric obstruction (loin pain, renal failure), haematuria, reduced ejaculate, rarely erectile dysfunction.
- Metastatic**: bone pain, pathological fracture, leg oedema, spinal cord compression; rarely retroperitoneal fibrosis, paraneoplastic syndromes, disseminated intravascular coagulation (DIC).

2 • SCREENING IN BRIEF

Feature	ERSPC	PLCO
Trial	European Randomised Study of Screening for Prostate Cancer	Prostate, Lung, Colorectal and Ovarian (United States)
Cancer-specific mortality	20% relative reduction (16 years)	No difference (17 years)
Number needed	570 invited, 18 extra diagnoses per death prevented	—
Why	—	Heavy PSA contamination of control arm; ≈3× fewer events: really screening vs less screening

- US Preventive Services Task Force (USPSTF): Grade D (2012) → **Grade C (2018)**, age 55–69, shared decision; not ≥70.
- American Urological Association (AUA): shared decision 55–69, 2-yearly. National Comprehensive Cancer Network (NCCN): baseline PSA at 45. No body advises unselected population screening; harm = overdiagnosis and overtreatment, worst >65 years.

3 • DIGITAL RECTAL EXAMINATION AND PSA

- DRE**: fair reproducibility; low screening value at PSA <3 ng/mL; but an abnormal DRE raises risk of Grade Group (GG) 4–5 and multiplies cancer risk at every PSA level, so most urologists combine both. DRE barely changes PSA.
- PSA**: kallikrein serine protease; **organ-specific, not cancer-specific**. Disrupted gland architecture (cancer, benign prostatic hyperplasia [BPH], prostatitis, biopsy, massage, resection) leaks PSA into blood. Rises ≈4% per mL of volume.
- Lower PSA: hypogonadism, obesity, statins, 5α-reductase inhibitors (5-ARIs), BPH surgery. On 5-ARI (finasteride 1 or 5 mg, dutasteride): **double PSA** after ≥12 months, or treat the nadir as new baseline and investigate any rise.
- Risk is **continuous**: no safe cut-off. Prostate Cancer Prevention Trial (PCPT): cancer in 15% with PSA ≤4, ≈15% of these Gleason ≥7. Thresholds used: 4 (original), 2.5–3 (ERSPC 3). A higher threshold misses curable significant cancers; a lower one adds unnecessary biopsies and overdiagnosis; 10 ng/mL has been suggested above 70 years. Repeat a raised PSA before biopsy; no empirical antibiotics without urinary symptoms.

PSA (ng/mL)	Biopsy yield, normal DRE	Biopsy yield, suspicious DRE	High-grade (normal / suspicious DRE)
0–1	8.8%	—	0.9%
4–10	17–32%	45–51%	4.1% / 11.7%
>10	43–65%	70–90%	19.4% / 50.5%

About 1 in 11 men biopsied at PSA 0–1 has cancer; a suspicious DRE multiplies risk in every band. Baseline PSA in the 40s predicts metastasis and death >25 years later (Malmö Preventive Project).

- Derivatives: PSA density ≥ 0.15 ; velocity > 0.75 ng/mL/yr; free PSA $\leq 25\%$ (PSA 4–10); Prostate Health Index, 4Kscore, urine and tissue tests → *Ch 154*. Pre-biopsy multiparametric magnetic resonance imaging (mpMRI), Prostate Imaging Reporting and Data System (PI-RADS) and biopsy technique → *Ch 155*.

4 • STAGING: PRINCIPLES

Clinical stage uses pre-treatment DRE, PSA, biopsy and imaging; **pathological stage** uses the radical prostatectomy (RP) specimen and nodes, is more accurate and predicts outcome better. Biochemical recurrence-free and cancer-specific survival fall with pathological stage; post-RP predictors are grade, margins, extracapsular extension, seminal vesicle invasion (SVI) and nodes.

5 • TNM (AJCC 8TH EDITION, 2017)

Category	Definition
T1a / T1b	Incidental at transurethral resection: $\leq 5\%$ / $> 5\%$ of tissue
T1c	Impalpable, found on needle biopsy (one or both sides)
T2a / T2b / T2c	Palpable, confined: $\leq \frac{1}{2}$ one side / $> \frac{1}{2}$ one side / both sides
T3a	Extraprostatic extension (uni- or bilateral)
T3b	Seminal vesicle invasion
T4	Fixed or invading other structures: external sphincter, rectum, bladder, levators, pelvic wall
N1	Regional (pelvic) node(s)
M1a / M1b / M1c	Non-regional nodes / bone / other sites $\pm$ bone

The American Joint Committee on Cancer (AJCC) 8th edition combines tumour, node, metastasis (TNM) with PSA and GG into prognostic stage groups. Whitmore–Jewett is historical.

KEY POINT T1 is defined by how the cancer was **found**, T2 by what is **felt**. Traps: SVI is T3b, not T4; non-regional nodes are M1a, not N1; a biopsy-found impalpable cancer is T1c even if bilateral; clinical T is assigned by DRE, not by MRI.

6 • PREDICTING TUMOUR EXTENT

- PSA** correlates with stage but cannot stage an individual (overlap, BPH contribution, high-grade tumours make less PSA per gram). Organ-confined: $\approx 80\%$ at PSA < 4 , 66% at 4–10, $< 50\%$ at > 10 . Nodes: $\approx 20\%$ at PSA > 20 , **75% at > 50** .
- DRE**: both over- and under-stages; useful only in combination.
- Biopsy grade** is the most important biopsy finding → *Ch 156*. GG1 ≤ 6 ; GG2 3+4; GG3 4+3; GG4 8; GG5 9–10. Gleason 7 spans two groups: primary pattern matters. Tertiary pattern 5 on biopsy is recorded as the secondary pattern (3+4 with tertiary 5 → 3+5). Number/percentage of positive cores, cribriform and intraductal patterns add information; tumour in seminal vesicle or periprostatic fat on biopsy worsens prognosis.
- Prostatic acid phosphatase: abandoned.

7 • RISK STRATIFICATION

Combining T stage, PSA and grade predicts extent and outcome better than any single parameter.

D'Amico	Stage	PSA (ng/mL)	Gleason	10-year disease-free after RP
Low (all criteria)	T1–T2a	≤ 10	≤ 6	83%
Intermediate (any)	T2b	> 10 –20	7	46%

D'Amico	Stage	PSA (ng/mL)	Gleason	10-year disease-free after RP
High (any)	T2c	>20	8-10	29%

Low risk needs **every** criterion; the single worst feature assigns intermediate or high.

† NCCN group	Criteria
Very low	T1c, GG1, PSA <10, PSA density <0.15, <3 positive cores, ≤50% of any core
Low	T1-T2a, GG1, PSA <10
Favourable intermediate	One intermediate factor (T2b-c, GG2-3, or PSA 10-20), GG1-2, <50% cores positive
Unfavourable intermediate	≥2 intermediate factors, or GG3, or ≥50% cores positive
High	T3a, or GG4-5, or PSA >20
Very high	T3b-T4, primary pattern 5, >4 cores GG4-5, or ≥2 high-risk features

- **Cancer of the Prostate Risk Assessment (CAPRA)**: validated multiparameter score for counselling.
- **Nomograms** outperform groups for individual prediction. † Partin tables (PSA, clinical T, grade → probability of organ-confined disease, extracapsular extension, SVI, nodes) and Memorial Sloan Kettering Cancer Center (MSKCC) nomograms are the classic tools; nodal-risk nomograms decide pelvic lymph node dissection (PLND).
- **Tissue genomic tests** (Prolaris: cell-cycle progression; Oncotype DX: 17 genes; Decipher: 22 RNA markers) predict adverse pathology, recurrence and death; may refine risk groups; added value still awaits prospective validation.

8 • STAGING IMAGING

Imaging yield follows **pre-test probability**: the risk group chooses the scan. Nodal and bone disease are rare in screened men, so staging scans are overused.

Risk	Imaging
Low / intermediate, asymptomatic	No routine abdominopelvic computed tomography (CT) or bone scan
High	Bone scan + CT or mpMRI; consider prostate-specific membrane antigen (PSMA) positron emission tomography (PET)

- **mpMRI** (T2-weighted + diffusion + dynamic contrast): locoregional staging (extraprostatic extension, SVI), treatment planning, active surveillance selection; recommended before biopsy.
- **PSMA PET**: ⁶⁸Ga-PSMA-11 and piflufolastat F-18 (¹⁸F-DCFPyL) are US Food and Drug Administration (FDA)-approved for initial staging of high-risk disease; more sensitive and specific than conventional imaging for nodes and distant disease; proof of clinical benefit when conventional imaging is negative remains limited.
- Ultrasound advances (Doppler, contrast, HistoScanning): not routine.

KEY POINT Why not image low-risk men? With a low pre-test probability, most positive scans are false positives, which cause further tests, anxiety and wrong denial of curative treatment.

9 • LYMPH NODE STAGING

- PSMA PET is more sensitive and specific than CT/MRI for nodal disease.
- **PLND**: no consistent oncological benefit; its value is **staging** (refines risk, counselling, guides adjuvant/secondary therapy).
- Perform at RP when a nomogram-predicted nodal risk exceeds the guideline threshold (thresholds vary), balanced against PLND morbidity.

10 • LANDMARK TRIALS

Trial	Design	Result	Take-home
ERSPC	European randomised screening, from 1993	20% relative reduction in cancer mortality at 16 years	Screening saves lives at cost of overdiagnosis
PLCO	US randomised screening, from 1993	No mortality difference	Contaminated control arm; not proof of futility
PCPT	Placebo arm biopsied regardless of PSA	15% cancer at PSA ≤ 4	No PSA level is safe; risk is continuous
† proPSMA (2020)	Randomised: PSMA PET-CT vs CT + bone scan, high-risk	Accuracy $\approx 92\%$ vs 65% ; fewer equivocal results, less radiation	PSMA PET is the better first-line staging test

11 • MCQ ONE-LINERS

1. Early prostate cancer: usually asymptomatic.
2. ERSPC mortality reduction: 20% relative.
3. PLCO null result explained by control-arm contamination.
4. Current USPSTF grade (55-69): C.
5. PSA is organ-specific, not cancer-specific.
6. PSA rises $\approx 4\%$ per mL of prostate volume.
7. Correction for 5-ARI after 12 months: double PSA.
8. Finasteride 1 mg lowers PSA as much as 5 mg.
9. Main source of serum PSA without cancer: transition zone.
0. Free PSA cut-off suggesting cancer: $\leq 25\%$.
1. PSA velocity threshold (PSA 4-10): >0.75 ng/mL/year.
2. Incidental cancer in $>5\%$ of resected tissue: T1b.
3. Impalpable cancer found on needle biopsy: T1c.
4. Tumour $>$ half of one lobe, not both: T2b.
5. Seminal vesicle invasion: T3b.
6. Rectal or bladder invasion: T4.
7. Non-regional node metastasis: M1a.
8. Bone metastasis: M1b.
9. Gleason 4+3: Grade Group 3.
0. Most important biopsy prognostic finding: grade.
1. D'Amico high risk: PSA >20 , Gleason ≥ 8 or T2c.
2. D'Amico low risk needs all three criteria.
3. Nodal involvement at PSA >50 : $\approx 75\%$.
4. Organ-confined at PSA <4 : $\approx 80\%$.
5. No routine bone scan in asymptomatic low/intermediate risk.
6. FDA-approved PSMA agents: ^{68}Ga -PSMA-11, piflufolastat F-18.
7. Decipher: 22-RNA tissue classifier.
8. Oncotype DX Prostate: 17 genes.
9. Prolaris measures cell-cycle progression genes.
0. PLND value: staging, not proven therapy.
1. Pathological stage beats clinical stage for prognosis.

158 Active Management Strategies for Localised Prostate Cancer

Reading time ≈ 32 min (slow pace)

1 · THE MODERN QUESTION: TIMING AS WELL AS MODALITY

- Prostate-specific antigen (PSA) era → radical treatment for almost everyone. Evidence against this: low metastasis/death rates in observed low-risk disease; randomised treatment-vs-observation trials diluted by early other-cause death; screening trials without robust mortality benefit.
- 15-year randomised data in favourable risk: surveillance → **more metastasis but no difference in cancer mortality**, because of crossover to treatment within 5–10 years, competing mortality and better treatment of metastatic disease.
- Management spectrum**: likely to die of competing risks → **observation (watchful waiting, WW)**; good longevity but benefit unclear → **active surveillance (AS)**; needs therapy → radical prostatectomy (RP), external-beam radiation therapy (EBRT), brachytherapy, or whole/partial-gland ablation.

KEY POINT Cancer mortality risk is always weighed against competing mortality. A competing risk argues against treatment only when it will act *before* the cancer does.

2 · HOST EVALUATION

Longevity and competing risks

- National Comprehensive Cancer Network (NCCN): life-expectancy estimation is critical; use Social Security Administration or World Health Organization (WHO) life tables.
- Prostate Cancer Outcomes Study (PCOS): 10-year prostate cancer mortality **3–18%** by risk group vs **≥33%** other-cause mortality with any comorbidity.
- The blunt "**10-year life expectancy**" rule undertreats older and overtreats younger men → individualise using age + comorbidity + cancer risk.
- Charlson Comorbidity Index (CCI)**: most used; designed for 1-year inpatient mortality. **Prostate cancer-specific CCI (PCCI)** reweights for 10-year mortality. **Care Assessment Needs (CAN) score** (automated, Veterans Health Administration): best predictor of 5-year survival **MCQ**.
- Age bias**: clinicians judge longevity by age, not comorbidity → comorbid young men overtreated, healthy old high-risk men undertreated.
- Aggressive (high-grade, advanced) cancer may merit treatment despite shortened longevity, but absolute benefit falls and number needed to treat (NNT) rises steeply with Charlson score.

Baseline urinary, sexual and bowel function

Baseline function sets the size of the quality-of-life (QoL) risk; measure with the International Prostate Symptom Score (IPSS) and the University of California, Los Angeles (UCLA) Prostate Cancer Index.

Baseline state	Favours	Reason
Obstructive lower urinary tract symptoms (LUTS), large gland	Surgery	RP removes obstruction durably; brachytherapy ↑ retention
Severe irritative LUTS	Surgery	Highest risk of worsening after radiation
Gland >60 cm ³	Surgery (or neoadjuvant androgen-deprivation therapy, ADT)	Pubic arch interference for brachytherapy; >80 g → ADT (shrinks 25–40%) before radiation
Chronic retention / poor compliance / neurogenic bladder	Surgery	Radiation causes progressive bladder fibrosis; if radiation chosen, switch catheter to intermittent self-catheterisation

Baseline state	Favours	Reason
Prior transurethral resection (TUR) defect	Surgery	Brachytherapy dosimetry difficult; ↑ incontinence. Do transurethral resection of prostate (TURP) <i>before</i> radiation, delay EBRT 6-8 weeks
Inflammatory bowel disease, bowel dysfunction, prior rectal surgery	Surgery	Radiation causes more bowel toxicity

- **Sexual:** recovery depends on age, baseline erectile function, phosphodiesterase type 5 (PDE5) inhibitor use, vascular risk factors. Poor baseline → more non-nerve-sparing surgery.

Contraindications by modality

Modality	Absolute	Relative
Radiotherapy	Previous radical pelvic irradiation	Bowel trapped/in field (prior pelvic surgery, hernia); poor healing (uncontrolled diabetes); inflammatory bowel disease; bowel dysfunction
Surgery	—	Previous pelvic radiation; chronic steroids; severe cardiac/pulmonary/haematological comorbidity

3 · RISK STRATIFICATION

NCCN group	Criteria	Staging imaging
Very low	All: T1c; Gleason ≤6/grade group (GG) 1; PSA <10 ng/mL; <3 positive cores, ≤50% each; PSA density < 0.15	Not indicated
Low	All: T1-T2a; GG1; PSA <10	
Favourable intermediate	Intermediate-risk factors (IRF): T2b-c, GG2, PSA 10-20. All: 1 IRF; GG1-2; <50% cores positive	
Unfavourable intermediate	Any: 2-3 IRFs; GG3; ≥50% cores positive	Bone scan + computed tomography (CT) or magnetic resonance imaging (MRI) of pelvis/abdomen, chest imaging; prostate-specific membrane antigen (PSMA) positron emission tomography (PET) may replace (NCCN 2023)
High	Any: T3a; GG4-5; PSA >20	
Very high	Any: T3b-T4; primary pattern 5; >4 cores GG4-5	

KEY POINT Favourable groups need ALL criteria; adverse groups ANY one. Traps: PSA density appears only in very low risk; GG3 alone makes intermediate risk unfavourable. American Urological Association (AUA)/American Society for Radiation Oncology (ASTRO)/Society of Urologic Oncology (SUO) use very low/low/intermediate/high without substratification.

- Gleason → GG: 3+3 = 1; 3+4 = 2; 4+3 = 3; 4+4 = 4; 9-10 = 5. International Society of Urological Pathology (ISUP) 2005 moved cribriform patterns to pattern 4 → today's Gleason 6 is "purer" (grade migration).
- **Epstein criteria** (insignificant cancer): PSA density <0.15, Gleason ≤6, <3 positive cores, ≤50% any core.
- Nomograms: **Partin tables** (pathological stage, accurate 72.4%); **Kattan** (relapse after RP); **University of California, San Francisco-Cancer of the Prostate Risk Assessment (UCSF-CAPRA)** (recurrence and cancer mortality; risk jumps at score ≥6).
- **proPSMA** (randomised, 302 high-risk): gallium-68 PSMA PET-CT vs conventional imaging — accuracy **92% vs 65%**, sensitivity 85% vs 38%. PSMA PET's advantage is specificity/positive predictive value; a negative scan cannot exclude small nodal disease.
- Genomic tests (Decipher, Prolaris, Oncotype DX): AUA 2022 — not routine; use only where it changes an AS decision (high-volume 3+3 considering treatment, 3+4 considering AS).

Intraprostatic heterogeneity limits them.

4 • OBSERVATION VS ACTIVE SURVEILLANCE

	Active surveillance	Watchful waiting
Intent	Curative — defer treatment until needed	Palliative — avoid treatment
Candidate	Good longevity, low risk	Limited longevity (competing risks)
Monitoring	Scheduled PSA, MRI, biopsy	Periodic PSA, examination
Trigger	Progression → curative therapy	Symptoms/metastasis → ADT; palliative radiotherapy

- AS is **preferred for very low and most low risk** (AUA, ASTRO, SUO, NCCN). Selected favourable intermediate (low-volume GG2) is acceptable; most upgrading is to GG2, of low metastatic potential in the first decade.
- Limitations: **25-45%** occult high grade on systematic biopsy; declining adherence; no agreed triggers. Reclassification predictors: PSA density >0.15, core cancer density.
- Inclusive (Toronto) vs restrictive (Johns Hopkins) programmes, entry criteria, MRI and biopsy schedules → *Ch 159*.

5 • COMPARATIVE ONCOLOGICAL EVIDENCE

- Biochemical-free survival is a flawed end point for comparing RP and radiation: definitions differ, radiation kills slowly and leaves PSA-producing tissue, ADT delays PSA rise. Valid end points (cancer-specific, overall mortality) need long follow-up.
- Population studies** (Swedish, Cancer of the Prostate Strategic Urologic Research Endeavor (CaPSURE), Surveillance, Epidemiology, and End Results (SEER)): higher cancer mortality after radiation than RP (Swedish: subdistribution hazard ratio ≈ 1.8 in non-metastatic) — largely **selection bias** (older, sicker, higher-risk men got radiation).
- PCOS** (cohort, RP vs EBRT): RP overall hazard ratio (HR) 0.60, disease-specific HR 0.35 — but curves separated by 3 years, implausible for early-stage disease → confounding.

6 • LANDMARK TRIALS

Trial	Design	Result	Take-home
SPCG-4 (Scandinavian Prostate Cancer Group-4)	RP vs WW; 695 men; pre-PSA era, ~90% palpable	23.2 yr: cancer deaths 63 vs 99, relative risk (RR) 0.56 ; absolute difference 11%; NNT 8 . Greatest benefit age <65 (RR 0.45) and intermediate risk	RP saves lives in palpable, intermediate-risk disease in younger men; not representative of screen-detected cancer
PIVOT (Prostate Cancer Intervention Versus Observation Trial)	RP vs observation; 731 men (planned 2000); PSA era; older, comorbid	10 yr: no difference in all-cause or cancer mortality; benefit if PSA >10. 19.5 yr: all-cause absolute risk reduction (ARR) 5.5% (not significant, NS), cancer ARR 4.0% (NS); significant all-cause benefit only in intermediate risk (ARR 14.5%). Bone metastases 4.7% vs 10.6%	No benefit of RP in low risk; underpowered; ~40% dead at 10 yr → poor longevity selection
ProtecT (Prostate Testing for Cancer and Treatment)	1643 screen-detected men aged 50-69; active monitoring vs RP vs radiotherapy + 3-6 months ADT; 77% low risk	15 yr: cancer mortality 2.7% vs 2.2% vs 2.9% (P = 0.53); metastasis higher with monitoring (half of all progression); 60% of monitored men treated by 15 yr; RP = radiotherapy (RT) for mortality and metastasis	Initial AS is safe in favourable risk (incl. much favourable intermediate); surgery and radiation are oncologically equivalent; age <65 → lower cancer mortality with surgery than RT

KEY POINT Why do randomised trials show so little benefit? Competing mortality and low-risk enrolment dilute the effect, so benefit appears only in younger, intermediate-risk men with long follow-up. The consistent observational advantage of RP over radiation is mostly selection bias. ProtecT expected 15% mortality and observed ~1% at 10 years — hence low power.

7 • FUNCTIONAL OUTCOMES COMPARED

Domain	RP	EBRT	Brachytherapy
Sexual	Worst ; persists at 24 months	Mild early decline; near AS by 24 months	Mild early decline
Incontinence	Worst ; persists at 24 months	No change	No change
Obstructive/irritative	Slightly worse at 3 months, then better than AS	Worse at 3 months; resolves by 12	Worst acute ; resolves by 24 months
Bowel	No change	Worse , persists at 24 months	No change

Propensity-weighted, vs AS (Chen et al., JAMA 2017).

- **PCOS**: RP → more incontinence and erectile dysfunction (ED), radiation → more bowel urgency at 2 and 5 years; **no difference at 15 years** (both groups decline with age).
- **ProtecT**: RP worst for sexual function and continence throughout; radiation → transient bowel, voiding and sexual effects, rising bloody stools; monitoring → gradual decline. No difference in anxiety, depression or general QoL.
- **CEASAR** (Comparative Effectiveness Analysis of Surgery and Radiation): in low risk, 3-year sexual score better after EBRT than RP; RP's erectile disadvantage confined to the top quartile of baseline function.
- **Open vs robotic RP**: one randomised trial — similar function at 12 weeks; outcomes depend on the surgeon, not the approach → *Ch 160/161*.

KEY POINT Surgery front-loads urinary and sexual harm that partly recovers; radiation spares early function but adds bowel toxicity and slow cumulative decline. Curves converge by 15 years — the choice is *when and where* the patient accepts harm.

8 • SHARED DECISION-MAKING AND COUNSELLING

- No treatment is clearly superior for an individual, so **patient preference carries unusual weight**. Shared decision-making is endorsed by AUA, ASTRO and SUO.
- Patients who perceive active involvement → better informed, less decisional regret. Decision aids: little effect on health outcomes; newer ones improve decision quality.
- Surgery choosers fear cancer death; non-surgical choosers prioritise QoL; obstructive LUTS → surgery. The **urologist's recommendation is the most influential factor** — beware financial and specialty bias.
- Give specific, quantitative risks; **longevity assessment may be the most important component** of counselling.

9 • STEPWISE DECISION ALGORITHM

ALGORITHM

Decision 1 — treat or not? Disease (grade, stage, genomics, family history) + host (age, comorbidity, longevity, preference) → no treatment = WW (short longevity) or AS (very low/low, selected favourable intermediate).

Decision 2 — partial or whole gland? Focality, size, grade, potency, LUTS → partial = focal ablation (limited-focality intermediate risk, ideally in a trial) → *Ch 163*.

Decision 3 — surgery vs radiation vs whole-gland ablation → expected outcomes, toxicities, baseline function, risk of secondary treatment.

Decision 4 — delivery: open vs robotic RP, lymph node dissection, surgeon; EBRT vs brachytherapy, fractionation, ADT → *Ch 162*.

Risk / longevity	Usual recommendation
Any risk, limited longevity, low lethal potential	Watchful waiting; ADT at symptoms
Very low	AS (best option)
Low	AS preferred for most; all therapies are options
Favourable intermediate	RP or RT; AS for select (low-volume GG2) — NCCN: AS and all therapies are options; focal therapy in trials
Unfavourable intermediate	RP or RT + ADT; AS strongly discouraged (unless age/comorbidity)
High / very high	RP or RT + ADT; consider even with competing risks if cancer threatens within 10 yr; no focal therapy

Surgery vs radiation — counselling points

Surgery	Radiation
Early incontinence and ED are the norm → set a realistic recovery timeline to avoid regret	Avoids short-term urinary and sexual effects
Quote surgeon-specific outcomes	Incremental yearly decline in urinary/sexual function
Obstructive LUTS may improve	Secondary rectal/bladder cancer at 15-20 years (controversial)
Age <65: lower cancer mortality than RT in ProtecT	Intermediate/high risk → ADT, which drives much early toxicity

- **Whole-gland ablation:** little long-term data, frequent retreatment, salvage surgery hampered by fibrosis. Cryoablation → ED almost always; high-intensity focused ultrasound (HIFU) → better function but retreatment **29-38%**.
- Radiation forms (EBRT, brachytherapy, stereotactic body radiotherapy, protons) have comparable 10-year oncology; choose by tolerable side effects.

10 • MCQ ONE-LINERS

1. SPCG-4: RP vs watchful waiting, pre-PSA era, ~90% palpable disease.
2. SPCG-4 at 23 years: cancer-death RR 0.56, NNT 8.
3. SPCG-4 benefit greatest in men <65 years and intermediate risk.
4. PIVOT: RP no mortality benefit in low risk; underpowered (731 of 2000 planned).
5. PIVOT 19.5 years: significant all-cause benefit only in intermediate risk.
6. ProtecT 15-year cancer mortality: ~2-3% in all three arms (P = 0.53).
7. ProtecT: monitoring → more metastasis but no mortality difference.
8. ProtecT radiotherapy arm received 3-6 months neoadjuvant ADT.
9. ProtecT: 60% of active monitoring arm treated by 15 years.

- | | |
|--|--|
| 0. Early divergence of survival curves in observational RP vs RT studies indicates selection bias. | 20. Charlson index originally predicted 1-year inpatient mortality. |
| 1. Brachytherapy, not EBRT, causes the worst acute obstructive/irritative symptoms. | 21. CAN score outperforms CCI and PCCI for 5-year survival. |
| 2. EBRT signature toxicity: bowel problems persisting at 24 months. | 22. PSA density criterion appears only in NCCN very-low-risk group. |
| 3. RP improves obstructive/irritative scores beyond AS2 by 1-2 years. | 23. GG3 alone makes intermediate risk unfavourable. |
| 4. PCOS: functional differences between RP and RT disappear by 15 years. | 24. Primary Gleason pattern 5 = very high risk. |
| 5. CEASAR: RP erectile disadvantage seen only in the highest baseline-function quartile. | 25. Staging imaging begins at unfavourable intermediate risk. |
| 6. Previous pelvic radiation is the absolute contraindication to radiotherapy. | 26. Partin tables predict pathological stage; accurate 72.4%. |
| 7. Brachytherapy technically difficult with gland >60 cm ³ (pubic arch interference). | 27. Epstein: PSA density <0.15, Gleason ≤6, <3 cores, ≤50% any core. |
| 8. Delay EBRT 6-8 weeks after TURP. | 28. Most important counselling element: longevity assessment. |
| 9. Radiation planned in retention: change to intermittent catheterisation to cycle the bladder. | 29. Most influential factor in patient choice: urologist's recommendation. |
| | 30. Whole-gland HIFU retreatment 29-38%; cryoablation causes ED almost always. |

159 Active Surveillance of Prostate Cancer

Reading time ≈ 27 min (slow pace)

1 • RATIONALE: OVERDIAGNOSIS

- Active surveillance (AS) is now the initial strategy for most favourable-risk cancer (in some regions >90%), reversing near-universal radical treatment 10–15 years ago.
- Latent reservoir:** autopsy microfocal cancer ≈ age as % (30% in 30s, ≥80% over 80). Prostate-specific antigen (PSA) + biopsy detects cancer in 15–25%, but lifetime risk of prostate cancer death is only **3%**. Harbours : death ≈ 20:1; diagnosis : death ≈ 6:1. Estimated prostate overdiagnosis 87–94%.
- United States Preventive Services Task Force (USPSTF) on PSA screening: 2012 Grade D → 2018 Grade C, largely because AS uptake reduced overtreatment.
- Two premises of AS:** a long lead time to clinical progression, and treatment still curable at the first sign of higher risk.

2 • AS VERSUS WATCHFUL WAITING

	Active surveillance	Watchful waiting (WW)
Candidate	Fit for radical therapy (life expectancy usually ≥10 yr)	Limited life expectancy (age, comorbidity)
Intent	Curative, deferred	Palliative
Monitoring	Close: PSA, digital rectal examination (DRE), MRI, biopsy	Infrequent PSA and clinical review; no biopsy or imaging
Action	Definitive therapy on reclassification	Palliation of symptomatic/advanced disease

- Switch AS → WW when life expectancy falls to <5–10 years (typically ~80 years; age-comorbidity nomograms). No hard rule: preference is central.
- Trap: older observation trials and registries reflect WW, not modern AS.
- Pitfall — neuroendocrine dedifferentiation:** progression without PSA rise, seen in elderly men with unrecognised hypogonadism. Measure testosterone if suspected and obtain interval imaging.

KEY POINT AS defers curative intent; WW abandons it. Life expectancy, not grade, chooses between them.

3 • BIOLOGY OF GRADE GROUP 1

Grade group (GG) 1 = Gleason 3+3=6; GG2/3 = 3+4/4+3; GG4/5 = 8/9–10 → *Ch 156*. Grade is by far the most powerful predictor of significance.

- Pattern 3 is molecularly near-normal;** patterns 4–5 show the hallmarks of cancer: epidermal growth factor receptor (EGFR) overexpression, HER2 amplification, BCL2 strong, increased angiogenesis, **PTEN deleted in 70–90%**; cyclin D2 is the one abnormality of *loss* in pattern 4.
- PTEN loss alone is not metastatic; **MYC amplification + PTEN deletion** drives genomic instability and metastasis.
- Metastatic potential of pure GG1 is effectively zero:** every nodal metastasis in surgical Gleason 6 series showed higher grade on re-review. Local invasion (occasional EPE — extraprostatic extension) does occur — one argument for keeping the word "cancer".
- True grade progression ~**1–2% per year**, mostly high-volume Gleason 6; follow-up must be long term.
- Deaths in low-risk series = **occult higher grade missed at biopsy**, present in ~**30%** after 10–12-core systematic biopsy (85% of it 3+4).

Hazard	Misattribution of grade	True biological progression
Nature	Occult higher grade missed by systematic biopsy	Pattern 3 → 4/5
Frequency	Commonest (25-30%)	1-2.3%/year
Usual result	Mostly GG2	Many 3+4 remain indolent
Remedy	MRI, targeted biopsy, clinical predictors	Long-term surveillance

KEY POINT The main hazard of AS is **sampling error at entry**, not biology over time ("wolf in sheep's clothing"). The protocol's first job is to get the entry grade right.

4 • PATIENT SELECTION

Most GG1 should be on AS: consensus of American Urological Association (AUA), European Association of Urology (EAU), American Society of Clinical Oncology (ASCO), National Institute for Health and Care Excellence (NICE), Cancer Care Ontario.

	Very low risk (Epstein criteria)	Low risk
Definition	GG1; ≤2 positive cores; no core >50%; PSA density <0.15	Any GG1, PSA <10
Progression to treatment	Lower	Higher
Metastasis	Exceptionally low	Exceptionally low

The split reflects the likelihood of **occult higher grade**, not volume-dependent metastatic potential.

High-volume GG1 remains a candidate but needs scrutiny: multiparametric MRI (mpMRI) + targeted biopsy, extended re-biopsy and/or tissue genomic assay.

Programmes: inclusive vs restrictive

	Toronto (inclusive)	Johns Hopkins (restrictive)
Entry	All Gleason 6 + selected intermediate (Gleason 7 or PSA >10)	Epstein very low risk
Eligible at diagnosis	~50%	~20%
15-yr cancer mortality	5% actuarial (1.5% actual)	0.5%
Intervention at 5 yr	~25%	~25%

- **Inclusion criteria set mortality; intervention is constant (~25% at 5 yr).** In Toronto, baseline Gleason 7 had ≥20% metastasis at 15 yr (hazard ratio [HR] 3.75 vs low risk); PSA >10 barely predictive.
- Prostate Cancer Research International: Active Surveillance (PRIAS; n=5302): cancer mortality <1% at 10 yr. Other-cause mortality far exceeds cancer mortality in every cohort; cardiovascular disease is the commonest cause of death.
- **Favourable intermediate risk:** AS offered selectively — small-volume GG2 (limited pattern 4). Only NICE lists AS and radical treatment as equal options for intermediate risk.
- Exclude: GG ≥3; intraductal/cribriform histology raises concern → *Ch 156*.

Other risk factors

- **Black men:** data inconsistent (more upgrading in some series; no difference in protocol-driven Canary Prostate Active Surveillance Study [PASS]). Unequivocally AS candidates — consider more intense surveillance; ensure social factors do not impede compliance.
- **Family history** rarely justifies treatment ("family history inflation" from PSA-era screening of relatives). Exception: metastatic disease <60 yr in a first-degree relative → germline testing.
- **BRCA2:** highest upgrading risk on AS (especially to GG ≥3, within 10 yr); BRCA1, ATM less. Consider definitive therapy even for low grade; AS only with very close follow-up and excellent

compliance.

- National Comprehensive Cancer Network (NCCN) germline testing: metastatic/high risk; strong family history; consider in intermediate risk with intraductal/criform histology.
- Higher-grade risk factors: **PSA density >0.15**, Black race, biopsy volume; restricted diffusion on MRI is a powerful predictor.

5 • MONITORING PROTOCOL

ALGORITHM

Diagnosis → mpMRI (if not already done) → **confirmation biopsy** (systematic + targeted) within 1-3 yr → PSA 6-monthly, DRE every 1-2 yr, MRI ≤ every 18-24 months → repeat biopsy every 3-5 yr while stable → trigger → definitive therapy, or switch to WW when life expectancy falls

- Confirmation biopsy targets undersampled areas; may be omitted if the initial biopsy was MRI-targeted. Modelling allows delay to year 3 without harm.
- Tailor biopsy frequency to risk. **Compliance is paramount**: compliers with confirmation biopsy had better cancer mortality and metastasis-free survival.

Guideline	Intermediate risk	Schedule / MRI
Cancer Care Ontario	Treat; AS if selected	PSA 3-6-monthly; DRE yearly; biopsy within 6-12 mo then 3-5-yearly; MRI if discordant
AUA/ASTRO (American Society for Radiation Oncology) 2022	AS if selected	PSA 6-12-monthly; DRE 1-2-yearly; biopsy 1-4-yearly; MRI before initial or confirmation biopsy
NICE 2019	Radical treatment or AS	PSA 3-4-monthly year 1, then 6-monthly; MRI at enrolment

Triggers for intervention

Prompts definitive therapy	Prompts evaluation only
GG ≥3 (unless life expectancy significantly shortened)	Increased Gleason 6 volume → MRI or molecular biomarker
Traditionally any Gleason ≥7 (now relaxed)	Small-volume GG2 → may continue AS
	PSA rise alone → repeat MRI or biopsy

KEY POINT Grade triggers treatment; volume and PSA trigger investigation. 10-15% leave AS by choice (anxiety) without a trigger.

6 • MRI IN AS

- **Why MRI sees what matters**: higher grade → denser, disorganised architecture → impeded water diffusion, so diffusion-weighted imaging (DWI) signal correlates inversely with Gleason score. MRI tends to miss low grade but locates and sizes higher grade.
- **Entry stratification — strong evidence**: mpMRI before confirmation biopsy, with **systematic + targeted cores** (~1/3 upgraded, found by both routes). Recommended by NICE, option in NCCN.
- **Serial monitoring alone — weak**: only moderate sensitivity/specificity for progression to GG ≥2 (negative predictive value [NPV] ~0.8-0.9) → MRI must not be the sole means of excluding progression.
- MRI-invisible cancers have a more favourable natural history. A visible lesion predicts progression to treatment.
- MRI NPV depends on underlying risk: higher-risk men still need systematic biopsy.

7 · BIOMARKERS AND GENOMICS

Assay	Basis	AS evidence
Decipher (Genomic Classifier)	22-marker RNA expression	Predicts upgrading in GG1, not GG2; small gain
Oncotype DX (Genomic Prostate Score)	12 cancer genes, 4 pathways	Mixed; no association in Canary PASS
Prolaris (Cell Cycle Progression)	31 cell-cycle + 15 housekeeping genes	None specific

- None was built for AS. Clinical parameters already stratify well, so incremental benefit is modest.
- **Bayesian problem:** with 3% prior risk of metastasis, a 90%-specific test labels 10% "significant" — false positives $\approx 3\times$ true risk. Hence test only higher-risk candidates (high volume, PSA density, GG2).
- Heterogeneity between foci; repeat testing unhelpful.
- Use to set surveillance intensity, **never as sole basis for AS**; a favourable score does not override adverse clinicopathology.

8 · LIFESTYLE, DRUGS AND PSYCHOLOGY

- No significant psychological harm: anxiety follows diagnosis, not management.
- Counsel: **stop smoking**, vigorous exercise (ERASE trial: interval training lowered PSA and PSA velocity), healthy weight (obesity $\rightarrow$ more high-grade cancer and mortality), vegetable-based diet. Prostate-healthy = heart-healthy.
- Micronutrients: only vitamin D 1000–2000 international units (IU)/day reasonable; **selenium and vitamin E contraindicated** (SELECT: vitamin E $\rightarrow$ more prostate cancer).
- **5-alpha-reductase inhibitors:** REDEEM (dutasteride) cut "progression" by 38% entirely through less GG1 volume (cytoreduction), no change in upgrading $\rightarrow$ not used to prevent progression. Statins reasonable for cardiovascular benefit; metformin, apalutamide, enzalutamide investigational.

9 · LANDMARK TRIALS

Trial	Design	Result	Take-home
SPCG-4 (Scandinavian Prostate Cancer Group 4)	n=695, pre-PSA, mostly palpable; prostatectomy vs WW; 23 yr	Prostate cancer death 19.6% vs 31.3% (relative risk 0.55)	Benefit in age <65 and GG ≥ 3 ; none in GG1–2
PIVOT (Prostate Cancer Intervention versus Observation Trial)	n=731, US veterans, PSA era; prostatectomy vs observation; 18.6 yr	All-cause death 68% vs 73% (not significant); low-risk cancer death 2.7%	Modest benefit, mainly intermediate risk
ProtecT (Prostate Testing for Cancer and Treatment)	n=1643, PSA-detected; active monitoring vs prostatectomy vs radiotherapy; 15 yr	Cancer mortality <3%, no difference; metastases 9.4% vs 4.5% vs 5.0% ; 61% of monitoring arm treated	Mortality-safe; active monitoring $\neq$ AS (no scheduled biopsy, $>1/3$ intermediate/high risk); quality of life same or better
ASIST	GG1 on AS; MRI-targeted + systematic vs systematic confirmation biopsy	Upgrading 33% vs 27% (not significant); $\sim 50\%$ less AS failure at 2 yr	MRI benefit shows over follow-up; quality matters
REDEEM	Dutasteride vs placebo on AS	38% less "progression", no upgrading difference	Cytoreduction, not disease modification

10 • MCQ ONE-LINERS

- | | |
|---|---|
| 1. Lifetime risk of prostate cancer death (North America) — 3%. | 15. Family history inflation — screening artefact, not biology. |
| 2. AS intent — curative, deferred; WW — palliative. | 16. Confirmation biopsy — within 1-3 years; systematic + targeted. |
| 3. Switch AS to WW — life expectancy <5-10 years. | 17. PSA on AS — 6-monthly. |
| 4. Epstein very low risk — ≤ 2 cores, no core >50%, PSA density <0.15. | 18. Repeat biopsy while stable — every 3-5 years. |
| 5. Pure Gleason 6 metastatic potential — effectively zero. | 19. Definite trigger — GG ≥ 3 . |
| 6. Occult higher grade after systematic biopsy — ~30%. | 20. PSA rise alone — prompts MRI/biopsy, not treatment. |
| 7. True grade progression on AS — ~1-2% per year. | 21. DWI signal — inversely related to Gleason score. |
| 8. Main hazard of AS — misattribution of grade at entry. | 22. Serial MRI alone — insufficient to exclude progression. |
| 9. Pattern 3 vs 4 — PTEN deleted in 70-90% of pattern 4. | 23. Decipher — 22-marker RNA classifier. |
| 10. Cyclin D2 — lost in pattern 4. | 24. Genomic test on AS — only higher-risk candidates; never sole basis. |
| 1. Intervention on AS at 5 years — ~25%, regardless of cohort. | 25. ProtecT active monitoring — more metastases, no mortality difference at 15 years. |
| 2. 15-yr cancer mortality — Toronto 5% actuarial vs Hopkins 0.5%. | 26. SPCG-4 benefit — age <65 and GG ≥ 3 . |
| 3. Commonest cause of death on AS — cardiovascular disease. | 27. REDEEM — dutasteride reduced volume progression, not upgrading. |
| 4. Highest upgrading risk germline mutation on AS — BRCA2. | 28. Selenium and vitamin E — contraindicated on AS. |
| | 29. USPSTF PSA screening — Grade D (2012) → Grade C (2018). |
| | 30. Neuroendocrine progression without PSA rise — check testosterone, image. |

160 Open Radical Prostatectomy

Reading time ≈ 27 min (slow pace)

1 • PRINCIPLES

- Radical prostatectomy (RP) is the best cure for clinically localised cancer and the only treatment shown in a randomised controlled trial (RCT) to reduce metastasis and cancer death (Scandinavian Prostate Cancer Group-4, **SPCG-4**, Bill-Axelsson).
- Most Western RPs are now robot-assisted, without proof of better cancer control or function. Open lessons transfer to the robot; open is the fallback on conversion → *Ch 161*.
- Goals in order:** cancer control → avoid technical complications → continence → potency.

KEY POINT The order of goals settles every intraoperative trade-off: when nerve sparing and a clear margin conflict, the margin wins.

2 • SURGICAL ANATOMY

Dorsal venous complex (DVC) and arteries

- Deep dorsal vein** → pierces urogenital diaphragm → **superficial branch** (between puboprostatic ligaments) + right and left **lateral plexuses** (Santorini) → inferior vesical → internal iliac vein. Free communication with pudendal, obturator and vesical plexuses → any tear bleeds heavily.
- The **common trunk lies directly over the urethra** — where the DVC is transected.
- Inferior vesical artery** ends as **urethral** branches (bladder neck, **5 and 7 o'clock** pedicles) and **capsular** branches (wrapped in nerves → visible marker of the cavernous nerves).
- Accessory pudendal arteries** cross the anterolateral prostate; dividing them reduces penile inflow.

Pelvic plexus and neurovascular bundle (NVB)

- Pelvic plexus = parasympathetic **S2-S4** + sympathetic (hypogastric nerve); beside the rectum, **5-11 cm** from the anal verge, midpoint at the seminal vesicle tip. Ligating the "lateral pedicle" mid-way cuts both vessels and nerves.
- NVB of Walsh** = capsular vessels + cavernous nerves in the lateral pelvic fascia, posterolateral to the prostate. Nerves are invisible; the vessels show where they are.
- At the apex branches spray anteriorly and posteriorly → basis of **high anterior release**.

Striated urethral sphincter

- A **vertical tubular sheath** (Oelrich), not a flat diaphragm: the growing prostate invades and thins it, so it attaches broadly over the apex — apical dissection decides continence.
- Passive continence:** sphincter, slow-twitch fibres. **Active:** levator ani, voluntary. Both mainly **pudendal nerve**, so pelvic floor exercises train both.

Fascial layers

Layer	Position	Surgical rule
Denonvilliers fascia	Between rectum and prostate, over seminal vesicles	Excise completely
Prostatic fascia	On gland; carries DVC tributaries	Leave on prostate in nerve sparing
Levator fascia	Over pelvic muscles; fuses laterally as lateral pelvic fascia	Incise to reach NVB

KEY POINT Nerves and vessels run between levator and prostatic fascia. Leaving prostatic fascia on the gland and Denonvilliers fascia on its back lets nerve sparing and clear margins coexist.

3 · PATIENT SELECTION AND PREOPERATIVE CARE

- Ideal: curable man who will live long enough to benefit (at 65–70, 50% chance of 15 more years).
- **Harder but not contraindications:** prior pelvic surgery or radiotherapy, transurethral resection of the prostate (TURP), hernia mesh, large gland.
- **Defer 6 weeks after biopsy, 12 weeks after TURP** so haematoma and adhesions settle — crucial for NVB preservation and avoiding rectal injury.
- Inguinal hernia: ~15% preop, up to 20% after RP (unmasked) → examine standing; repair preperitoneally at RP.
- No autologous donation (only deepens anaemia); type and screen all. Erythropoiesis-stimulating agents raise haematocrit. Cephalosporin prophylaxis. Low-dose aspirin often continued. Bowel prep: clear fluids, magnesium citrate, morning enema.
- Magnetic resonance imaging (MRI) guides how much NVB and urethra to preserve. **Fibreoptic headlight** = most useful instrument (operation is beneath the pubis).

4 · RETROPUBIC RP — OPERATIVE STEPS

ALGORITHM

Lower midline extraperitoneal incision → pelvic lymph node dissection (PLND) → endopelvic fascia → puboprostatic ligaments → DVC → urethra → posterior sphincter → NVB release → pedicles → bladder neck and seminal vesicles → reconstruction → anastomosis

- Relative hypotension (systolic ≤ 100 mm Hg), crystalloid ≤ 1500 mL until prostate out; 16-Fr silicone catheter. Leave tissue over the external iliac artery (leg lymphatics → oedema, lymphocele).
- **PLND:** for intermediate/high risk; omitted for Grade Group 1–2 low-volume disease. External iliac vein → femoral canal (node of Cloquet spared) → iliac bifurcation → obturator fossa, protecting the obturator nerve → *Ch 161*.
- **Endopelvic fascia:** open at sidewall reflection; too medial → Santorini plexus bleeding. Pudendal perforators: clip, don't coagulate.
- **Puboprostatic ligaments:** divide, but spare the **pubourethral** part (anchors sphincter).

DVC control

Feature	Stapler (preferred)	Suture ligation
Method	Vascular load; fire only if catheter moves freely	3-0 Monocryl figure-of-eight to pubic perichondrium + back-bleeding stitch
Advantage	~200–400 mL less blood loss; less traction	Recreates puboprostatic support

- **Anterior apex = commonest positive margin site** (surface hard to see). Divide lateral sphincter edges midway between apex and pelvic floor — an apical vessel tethers the NVB near the apex.
- Venous control must be absolute before touching the apex: oversee the distal DVC as a "hood".
- **Urethra:** transect close to apex; six 3-0 sutures, **mucosa and submucosa only** (muscle bites delay continence). Posterior sphincter: left → right → centre, never all from one side (endangers contralateral NVB).

Nerve sparing

Plane	Where	Use
Interfascial	Between levator and prostatic fascia	Correct nerve-sparing plane
Intrafascial	Beneath prostatic fascia	High margin risk — not used
Wide excision (extrafascial)	NVB + lateral pelvic fascia removed	Induration, fixed bundle, doubtful margin

- Both NVBs usually preserved: extracapsular extension rarely exceeds 1–2 mm, and margin rates are the same with or without nerve sparing (most margins are apical → posterior → posterolateral).

- Decision is never final before surgery: weigh erectile status, apical lesion, Partin tables/MRI. Intraoperative **induration or a fixed bundle** → **excise that side**. Touch is open surgery's advantage — absent robotically.
- **Technique**: roll the prostate (release prostate from bundle, not bundle from prostate) → find posterolateral groove → start **at the apex** (branches seen before cut) → apex → base → ligate posterior branch at seminal vesicle. Small clips; **never thermal energy**.
- **High anterior release** (before urethral division) → less traction, faster recovery. Needs complete haemostasis: incise levator fascia along the lateral DVC edge, preserving the glistening prostatic fascia; clip, never cauterise. Unilateral release works as well as bilateral, so it is done on the favourable side only — the gain is lower traction, not the anterior branches. Avoid if ≥ 2 **Tsuzuki criteria**: prostate-specific antigen (PSA) >10, Gleason >6, >20% core involvement, >33% cores positive, perineural invasion, abnormal digital rectal examination.
- **Wide excision**: clamp **medial** → **lateral** on rectum (reverse risks rectal injury).

Pedicles, bladder neck, reconstruction

- Lateral pedicles in 3 layers (all at once cuts into prostate). Sponge stick on catheter, not prostate (false-positive margin).
- **Bladder neck**: incise anteriorly → divide 5 and 7 o'clock vessels → posterior neck hugging the seminal vesicles, watching ureteric orifices → clip vasa.
- **Complete seminal vesicle excision is essential**; pelvic plexus lies on their lateral surface. Doubtful posterolateral margin on specimen → excise that NVB.
- **Tennis-racquet** closure + mucosal eversion; **intussusception** sutures → "**turtle sign**". Remove nearby clips (contracture).
- Tension-free anastomosis; Babcock on bladder while tying, no catheter traction. Peritoneal window lets haematoma decompress.

KEY POINT Every step does one of three jobs: absolute venous control before the apex; the right fascia on the specimen; no traction or heat on the NVB.

5 • PERINEAL RP (BRIEF)

- Reflects lateral and anterior fasciae off the prostate, sparing the DVC → **less bleeding**; retropubic RP works outside the fasciae, so the DVC must be ligated.
- † PLND is not possible through the perineal incision; it needs a separate approach.

6 • INTRAOPERATIVE COMPLICATIONS

Complication	Setting	Management
Haemorrhage (commonest, venous)	Endopelvic fascia, DVC	Divide DVC completely and oversee (best measure)
Obturator nerve	PLND	Reanastomose, fine non-absorbable
Rectal injury	Small prostate; apical/posterior plane	Two-layer repair + omentum; colostomy if prior radiotherapy
Ureteric injury	Dissecting in trigone	Reimplantation

- Partial DVC division bleeds more: retraction pulls cut veins open. Typical loss 300–1000 mL; homologous transfusion rarely needed.
- Rectal injury risk is highest with small prostates during apical dissection.
- **Rectal repair**: finish prostatectomy → anal dilatation shows injury → 2-layer closure → omental pedicle between rectum and anastomosis (prevents rectourethral fistula) → broad-spectrum antibiotics.

7 • POSTOPERATIVE CARE

- Mobilise same evening; discharge day 1-2. Ketorolac preferred to spare opioids (if creatinine <1.2 mg/dL, no haematuria).
- **Catheter ~1 week**; day-4 removal causes frequent retention. Never remove with significant extravasation. Delayed healing = perianastomotic haematoma.
- Before removal at 1 week: routine cystogram, transperineal saline sonocystography, or no imaging if haematocrit falls <6%. Bladder spasm → reassurance, oxybutynin 5 mg.

8 • POSTOPERATIVE COMPLICATIONS

- Mortality **0.2%**.
- **Delayed haemorrhage** (hypotension, tachycardia, oliguria) → **early exploration and evacuation**; a retained haematoma drains through the anastomosis → contracture and incontinence. Exploration uses similar blood but shortens stay; most still go home day 1-2.
- **Thromboembolism** = major cause of death; highest with PLND, peaks **14-28 days**. Compression, early ambulation, patient education; treat with low-molecular-weight heparin.
- **Bladder neck contracture** (~3 months, poor stream): dilatation → cold-knife incision 3, 6, 9 o'clock + self-catheterisation → triamcinolone injection → rarely reconstruction + artificial sphincter.
- **Lymphocele** → *Ch 161*.
- **Incontinence**: intrinsic sphincter deficiency, mainly sphincter injury at DVC division; also deep bites, NVB injury, poor bladder neck. Prevention: preserve apical sphincter; precise tension-free mucosal anastomosis; small supple neck. Pre-existing outlet obstruction slows recovery. Treat: less fluid, no caffeine, stop α-blockers, pelvic floor exercises; imipramine or adrenergic agonists by cause. Continence = **0-1 pad/24 h**.
- **Erectile dysfunction (ED)**: recovery predicted by **age <65, preoperative potency, no diabetes, bilateral nerve sparing**; continues up to 4 years. Phosphodiesterase type 5 (PDE5) inhibitor rehabilitation helps; nightly not better than on-demand. Also penile shortening, climacturia.

KEY POINT Most late complications trace to the pelvic haematoma: it distracts the anastomosis (contracture, incontinence) and inflames the NVBs — hence early re-exploration, clips near the bundles and a peritoneal window.

9 • OUTCOMES

Time	Continent	Potent (Walsh)
3 months	80%	38%
12 months	93%	73%
18-24 months	97%	86%

- Continence returns faster than potency. Single-expert figures; another centre found only a third at baseline potency at 2 years.
- RP improves lower urinary tract symptoms for up to 15 years.

10 • LANDMARK TRIALS

Trial	Design	Result	Take-home
SPCG-4	RCT, RP vs observation	Less metastasis and cancer death	RP cuts cancer death
Padma-Nathan	RCT, nightly sildenafil vs placebo	Better erections at 1 year	Rehabilitation works

Trial	Design	Result	Take-home
REINVENT	RCT, vardenafil on-demand vs nightly	On-demand better	Daily dosing unproven

11 • MCQ ONE-LINERS

1. RP goals in order: cancer, complications, continence, potency.
2. Defer RP 6 weeks after biopsy, 12 weeks after TURP
3. Most useful open RP instrument: fiberoptic headlight.
4. DVC common trunk lies directly over the urethra.
5. Pelvic plexus: 5–11 cm from anal verge, centred on seminal vesicle tip.
6. Striated sphincter: vertical tubular sheath, not flat diaphragm.
7. Correct nerve-sparing plane: interfascial.
8. Denonvilliers fascia is excised completely with the specimen.
9. Commonest positive margin site: apex.
10. Pubourethral ligament is spared to support the sphincter.
11. Stapled DVC control saves 200–400 mL blood.
12. Urethral sutures: mucosa and submucosa only.
13. Near the NVB: clips, never thermal energy.
14. Induration or fixed bundle → excise that NVB.
15. Tsuzuki ≥ 2 criteria → no high anterior release on that side.
16. Complete seminal vesicle excision is essential.
17. "Turtle sign": intussuscepted bladder neck.
18. Rectal injury: two-layer repair plus omental interposition.
19. Delayed haemorrhage needing transfusion → explore and evacuate.
20. Major cause of death after RP: thromboembolism, peak 14–28 days.
21. Recalcitrant contracture: incision plus triamcinolone.
22. Post-RP incontinence: intrinsic sphincter deficiency.
23. Perineal RP spares the DVC → less bleeding.
24. Pelvic plexus: parasympathetic S2–S4 plus hypogastric sympathetic fibres.
25. Hernia after open RP: up to 20% — repair preperitoneally at RP.
26. Wide excision clamp passes medial → lateral to protect the rectum.
27. Continence definition: 0–1 pad per 24 hours.
28. ED recovery predictors: age < 65 , potent, no diabetes, bilateral sparing.

161 Laparoscopic and Robotic-Assisted Laparoscopic Radical Prostatectomy and Pelvic Lymphadenectomy

Reading time ≈ 31 min (slow pace)

1 · BACKGROUND

- Laparoscopic radical prostatectomy (LRP; Schuessler 1997, standardised in France 2000) was limited by laparoscopic suturing of the anastomosis.
- Robotic-assisted laparoscopic radical prostatectomy (RALRP) removed that barrier: wristed (EndoWrist) instruments + 10× three-dimensional view. Now the dominant approach in the United States; LRP persists where robots are unaffordable.

KEY POINT The robot did not change the operation; it removed the suturing barrier. Surgical anatomy (dorsal venous complex [DVC], neurovascular bundle [NVB], fasciae, sphincter) is as in open surgery → Ch 160.

2 · PATIENT SELECTION

- Indications = open surgery: clinically localised cancer (cT1-T2, or cT3 resectable with wide excision).
- Staging: high risk → computed tomography (CT) or magnetic resonance imaging (MRI) + bone scan; consider in unfavourable intermediate risk; prostate-specific membrane antigen positron emission tomography (PSMA PET)-CT if high risk with negative conventional imaging.

Absolute contraindications	Relative (difficulty only)
Uncorrectable bleeding diathesis; unfit for general anaesthesia	Neoadjuvant hormones, complex prior pelvic surgery, morbid obesity, prostate >80 g, median lobe, prior transurethral resection of the prostate (TURP), salvage after radiotherapy or ablation

- Avoid relative contraindications early in the learning curve. Prior extraperitoneal mesh repair → **transperitoneal** route (Retzius obliterated).
- Obesity**: long trocars placed more cephalad, lower pneumoperitoneum (10 mm Hg); watch compression neuropathy and ventilation.
- Large gland/median lobe**: longer surgery, more blood loss, but continence comparable. **Prior TURP**: ureteric orifices close to bladder neck; outcomes may be worse.
- Multiparametric MRI (mpMRI): specificity **0.88**, sensitivity **0.61** for extracapsular extension (ECE); guides extent of nerve sparing and wider resection.

3 · PREOPERATIVE PREPARATION AND ANAESTHESIA

- No routine bowel preparation (optional enema). Cefazolin before incision; treat positive urine culture. Low-dose aspirin may continue.
- Consent specific to minimally invasive surgery (MIS): **conversion to open; general anaesthesia mandatory**. Baseline Sexual Health Inventory for Men (SHIM) and International Prostate Symptom Score (IPSS).
- Positioning**: arms tucked and padded (median, ulnar nerves); **no fixed shoulder rests** (brachial plexus injury); avoid exaggerated hip flexion (femoral neuropraxia).
- Carbon dioxide (CO₂) pneumoperitoneum**: vagal response, oliguria, hypercarbia → acidosis. Anaesthetist raises minute ventilation; surgeon lowers pressure.
- Fluids restricted (1.5–2 L, mostly after anastomosis) — urine output is unreliable and excess urine obscures the field.
- Steep Trendelenburg**: facial, laryngeal and corneal oedema, raised intraocular pressure → protect eyes; marked facial oedema → consider delaying extubation.

4 · ACCESS

Transperitoneal vs extraperitoneal

Feature	Extraperitoneal	Transperitoneal
Working space	Smaller; collapses with suction	Larger, familiar landmarks
Bowel	Protected; less ileus	Exposed
Tilt	Less steep — obesity, lung disease	Steep
CO ₂ absorption	Higher (more hypercarbia)	Lower
Prior extraperitoneal mesh	Difficult	Preferred

- Verdict: surgeon preference; no consistent advantage. Extraperitoneal: infraumbilical incision → balloon dilator develops Retzius.
- Transperitoneal **anterior** (commonest; Retzius first) vs **retrovesical** (seminal vesicles first in the cul-de-sac — useful with a median lobe).

Port placement (RALRP)

- Veress or Hasson entry; insufflation 12–15 mm Hg. Camera port at umbilicus, **15–17 cm** from pubic symphysis; three 8-mm robotic ports; assistant 12-mm and 5-mm ports on the patient's right.
- Instruments: left Maryland bipolar, right monopolar scissors, fourth arm ProGrasp. 0° lens; 30° down helps at bladder neck and apex.

5 · OPERATIVE STEPS (TRANSPERITONEAL ANTERIOR RALRP)

ALGORITHM

Drop bladder (Retzius) → endopelvic fascia → DVC ligation → bladder neck → seminal vesicles and vasa → posterior plane (Denonvilliers) → pedicles and NVB release → apex: DVC and urethra → inspect → pelvic lymph node dissection (PLND) → entrapment → posterior reconstruction → anastomosis → extraction

- **Retzius**: divide urachus high, incise peritoneum lateral to medial umbilical ligaments, stay inside pelvic brim (epigastric and iliac vessels); mobilise bladder laterally to the vas for a tension-free anastomosis.
- Divide endopelvic fascia sharply, sweep levator off — **no cautery** (sphincter, NVB). Preserve accessory pudendal arteries (sometimes the dominant penile supply).
- **DVC**: pneumoperitoneum tamponades venous bleeding. Suture (0 polydioxanone) close to pubis, away from apex; or stapler; or delayed division and oversewing at the end.

Bladder neck — no tactile feedback

- Cues: end of prevesical fat line; catheter balloon traction; bladder dome traction; instrument "pinch".
- Divide anteriorly in the midline → deflate balloon, lift catheter → inspect for median lobe and orifices → divide posterior neck angled **45°** downward.
- Prior TURP: balloon falls into defect → division too distal. Balloon off midline → suspect median lobe. When in doubt, **divide more proximally** (avoids bladder neck margin).

Seminal vesicles and posterior plane

- Incise posterior Denonvilliers fascia ~0.5 cm below vesicle bases; blunt dissection to apex. Low ECE risk → plane between Denonvilliers and prostatic fascia; posterolateral ECE risk → **excise Denonvilliers** (plane on perirectal fat). Bleeding = too close to prostate. Mobilise rectum widely now to protect it later.

Pedicles and nerve sparing

Plane	Boundaries	Use / cost
Extrafascial	Outside levator and Denonvilliers fascia	Widest margin; erectile dysfunction (ED) if bilateral
Interfascial	Between prostatic and levator fascia	Default for organ-confined disease
Intrafascial	Between capsule and prostatic fascia	Maximal nerve preservation; more positive margins; select carefully

- Moving inward trades margin for nerves. Periprostatic vessels are a visual surrogate for nerves.
- NVB injury mechanisms: incision, entrapment, **thermal**, **traction**.
- **Athermal ("clipless"/veil) release**: incise levator fascia on the anteromedial mid-prostate → interfascial plane → sweep NVB posterolaterally off the white prostatic fascia → groove forms → Hem-o-lok clips on pedicle → sharp division. Release the NVB **before** mobilising the specimen (less traction). Bleeding from NVB → fine sutures, not cautery.
- Pedicle clips near the anastomosis can migrate into bladder → haematuria, infection, stones.

Apex

- **Apex = commonest positive margin site**; its dissection governs continence and potency. Do not enter anterior prostate dividing the DVC.
- Preserve maximal urethral length but watch for a protruding apical lip; a small urethral rim on the apex lowers margins without harming continence. Sharp division, minimal cautery.
- Inspect whole specimen; excise extra tissue at a close margin. Entrap prostate and nodes in a sack.

Reconstruction

- Bladder neck reconstruction usually unnecessary (neck "parachutes" onto urethra). Needed for large gland, median lobe, prior TURP → tennis-racquet; identify orifices (methylene blue) and imbricate at 5 and 7 o'clock or stent.
- **Rocco stitch (posterior reconstruction)**: remnant Denonvilliers + posterior bladder → posterior rhabdosphincter, running 2-0, before the anastomosis. Early continence benefit inconsistent; agreed benefit = **tension-free anastomosis**. Anterior resuspension to arcus tendineus restores anterior support.
- **Van Velthoven running anastomosis**: start 6 o'clock, two arms (now barbed, double-armed) run anteriorly; no eversion sutures needed (running suture gives mucosal apposition). Pass catheter under vision; irrigate to test. Avoid NVB in urethral bites.

6 · CONTINENCE-PRESERVING VARIANTS

Feature	Standard anterior	Retzius-sparing (Bocciardi)	Hood (Tewari)
Route	Retzius	Pouch of Douglas only	Anterior
Detrusor apron, puboprostatic ligaments, DVC	Divided	Untouched	Preserved as a "hood"
Anastomosis	Posterior → anterior	Anterior first ("upside-down")	Posterior → anterior
Continence	Reference	Faster to 3 months; equal at 12	Promising, small cohort
Margins	Reference	Higher — avoid anterior or locally advanced tumours	Unknown

Rationale: anterior compartment structures stabilise the urethra and are thought to be part of the continence mechanism, so preserving them beats reconstructing them.

7 · PELVIC LYMPH NODE DISSECTION

Template	Nodes	Drainage captured
Limited	Obturator	—
Standard	Obturator + external iliac	~38%
Extended (ePLND)	+ internal iliac	~ 63%
+ Common iliac	ePLND + common iliac	~75%
Super-extended	+ presacral	—

- Standard template misses most drainage → **if PLND is done, use the extended template.** Justified as the best **staging** tool; no proven cancer-specific or overall survival benefit; morbidity rises with extent.
- Indications:** low risk (nodes <5%) → may omit. American Urological Association (AUA) 2018: unfavourable intermediate or high risk. European Association of Urology (EAU) 2021: nomogram nodal risk **≥5%**. AUA/American Society for Radiation Oncology (ASTRO) 2022: risk-stratified by nomogram.
- Caveat: **Briganti** and Memorial Sloan Kettering Cancer Center (MSKCC) nomograms predate mpMRI and PSMA PET and assume systematic biopsy.
- Extended boundaries:** proximal 2 cm above common iliac bifurcation (ureter crossing); distal node of Cloquet; lateral genitofemoral nerve; medial bladder wall; all obturator fossa tissue and around internal iliac artery.
- Technique: peritoneal incision lateral to medial umbilical ligament → packet off external iliac vein → protect obturator nerve and vessels, accessory obturator vein → clip and extract. **Clip lymphatics** → fewer lymphoceles. Open and laparoscopic PLND equivalent; node count is a poor quality surrogate.

PLND complications (~1 in 7)

- Vascular: small tear → fine polypropylene; major → open conversion.
- Obturator nerve** (medial thigh numbness, weak adduction): clip on nerve → remove; transection → primary repair.
- Ureter: at risk proximally over the iliac vessels — minimise heat there.
- Lymphocele — commonest:** radiological up to 60%, clinically significant **2-5%**. **Transperitoneal access is not protective** (bladder adheres and walls off the fossa) → **peritoneal interposition flap** reduces it. Presents with irritative symptoms and pelvic pressure, classically ipsilateral deep vein thrombosis (DVT); may get infected. Asymptomatic → leave. Symptomatic → percutaneous drain → recurrence: sclerosant or laparoscopic marsupialisation.

8 · POSTOPERATIVE CARE

- Enhanced Recovery After Surgery (ERAS): local anaesthetic to ports, ketorolac, regular paracetamol + non-steroidal anti-inflammatory drug, opioids only if needed. Diet within 4-8 h. Same-day discharge feasible in selected patients (lower pneumoperitoneum, extraperitoneal route).
- Drain not routine after a watertight running anastomosis.
- Normal activity by 4 weeks, strenuous at 6.

Scenario	Catheter	Cystogram
Routine	~1 week, not before day 5	No
Extensive neck reconstruction / doubtful	≥10 days	Yes
Salvage	2-3 weeks	Strongly consider

- Removal at day 3-4 → more retention. Extravasation delays but does not reduce long-term continence.

- Pelvic floor training: start preoperatively; physiotherapist-guided after; begin ≥ 3 days after catheter removal.
- **Penile rehabilitation:** ED from neural, vascular and smooth muscle injury. Phosphodiesterase type 5 inhibitors (PDE5i) help on-demand erections, but early rehabilitative use does not improve later response (3 randomised controlled trials [RCTs]). Intracavernosal injection for non-responders; vacuum device preserves length.

9 • MIS-SPECIFIC COMPLICATIONS

Complication	Rate	Management
Transfusion	1-2%	Pneumoperitoneum compresses veins
Bladder neck stricture	<2% (open up to 16%)	Dilatation/incision → rarely urethroplasty
Thromboembolism	~0.5%	Pneumatic compression routine; drugs only with multiple risk factors
Robotic failure	<0.5%	Cancel or convert
Recognised rectal injury	Rare; higher in salvage	Multilayer repair + omental/peritoneal flap, no colostomy
Unrecognised rectal injury	Rare	Rectovesical fistula → colostomy, later repair
Urine leak	Uncommon	Percutaneous drain, longer catheter, cystogram

- **Small bowel injury** is usually off-screen (instrument exchange, adhesiolysis) → delayed peritonitis → urgent repair.
- **Ureteric injury:** behind vas (posterior approach); at posterior bladder neck after TURP/median lobe (caught in anastomosis — imbricate or stent); at proximal PLND.
- DVT → image pelvis for lymphocele/haematoma compressing iliac vein; significant lymphocele → duplex for occult DVT.

KEY POINT Robotic injuries are fewer but less visible: bleeding and strictures fall with magnified vision, while bowel injury and positioning neuropathy happen outside the camera's field and must be anticipated.

10 • OUTCOMES

- **Oncologic:** no difference in positive surgical margin (PSM) rate open vs robotic; RALRP PSM ~15% overall, ~9% in organ-confined. PSM driven by tumour (Gleason ≥ 7 , perineural invasion, stage) and surgeon volume — **not** obesity, gland size or prior surgery. Biochemical recurrence (BCR) = prostate-specific antigen (PSA) >0.2 ng/mL.
- **Continence:** passive = smooth muscle and suspensory ligaments; active = striated sphincter (pudendal nerve branches enter at the apex). Experienced surgeons: no-pad rates mid-90s% at 1 year. Persistent mild → sling or adjustable balloons (ProACT); severe (<2%) → artificial urinary sphincter.
- **Potency:** cavernous nerves run 3-5 mm from the capsule and swing to 3 and 9 o'clock at the apex — where most damage occurs. Up to 90% potent at 1 year in expert hands without prior ED; recovery continues to 5 years.
- **Salvage RALRP** (after radiotherapy, brachytherapy, high-intensity focused ultrasound or cryotherapy): exclude metastases (PSMA PET); full bowel prep; rectal injury behind Denonvilliers (air-bubble test); catheter 2-3 weeks; continence only 33-80%; usually impotent.

11 • LANDMARK TRIALS

Trial	Design	Result	Take-home
Yaxley 2016	RCT, one open vs one robotic surgeon	Equal continence, potency, early oncology at 3 months; blood loss ~1400 vs 300 mL; shorter stay	Robot wins on recovery, not cancer or function
Retzius-sparing RCTs (Cochrane)	Single-centre RCTs	Earlier continence; equal at 12 months; more PSM	Use selectively

12 • MCQ ONE-LINERS

- Only absolute contraindications: uncorrectable bleeding and unfit for general anaesthesia.
- Prior extraperitoneal mesh hernia repair → transperitoneal approach.
- mpMRI for ECE: specificity 0.88, sensitivity 0.61.
- Retrovesical approach suits a median lobe.
- Posterior bladder neck divided angled 45° downward.
- When in doubt, divide bladder neck more proximally.
- Default nerve-sparing plane: interfascial.
- Posterolateral ECE risk → excise Denonvilliers fascia.
- Cavernous nerves are highly heat-sensitive — athermal release.
- Commonest positive margin site: apex.
- Agreed benefit of Rocco stitch: tension-free anastomosis.
- Retzius-sparing: faster early continence, higher positive margins.
- Extended PLND captures ~63% of prostatic drainage.
- EAU threshold for PLND: nomogram nodal risk >5%.
- Extended PLND lateral limit: genitofemoral nerve.
- Commonest PLND complication: lymphocele.
- Transperitoneal approach does not prevent lymphocele; peritoneal flap does reduce it.
- Do not remove catheter before day 5 (retention).
- Bladder neck stricture: <2% robotic vs up to 16% open.
- Critical robotic failure: <0.5%.
- Yaxley RCT: equal function, less blood loss robotic.
- Obesity and gland size do not raise PSM.
- Cavernous nerves lie at 3 and 9 o'clock at mid-urethra.
- Salvage RALRP continence: 33-80%.

162 Radiation Therapy for Prostate Cancer

Reading time ≈ 39 min (slow pace)

1 • RADIOBIOLOGY AND FRACTIONATION

- Every advance in radiation therapy (RT) answers three problems of a deep pelvic target: **penetration** (megavoltage linear accelerator, 6–20 MV, maximum dose several cm deep, skin sparing), **sparing** of rectum, bladder and femoral heads, and **targeting accuracy** despite organ motion.
- Conventional fractionation = **1.8–2 gray (Gy)** daily (1 Gy = 1 J/kg). The **linear-quadratic model** predicts a tissue's response to fraction size from its **α/β ratio**.

Feature	Low α/β (2–4 Gy)	High α/β (≥ 8 Gy)
Tissue	Late-responding normal tissue (rectum, bladder)	Most tumours (many cycling cells)
Sensitivity to fraction size	High	Low

- Prostate cancer has few cycling cells, so it behaves like a late-responding tissue: $\alpha/\beta \approx \mathbf{1.5\text{ Gy}}$ (external beam radiation therapy [EBRT] cohorts 1.0–1.7), **lower than the rectum ($\sim 3\text{ Gy}$)**.
- Rationale for hypofractionation:** because the tumour, not the rectum, is the tissue most sensitive to fraction size, fractions $>2\text{ Gy}$ favour tumour kill over late toxicity. A lower total dose in fewer fractions gives equal or better control with less late injury.

Category	Dose/fraction	Typical course
Conventional	1.8–2 Gy	~8 wk (78 Gy/39)
Moderate hypofractionation	2.25–4 Gy	~4 wk (60 Gy/20)
Ultrahypofractionation = stereotactic body RT (SBRT)	$\geq 5\text{ Gy}$	$\leq 5\text{--}7$ fractions (36.25 Gy/5)

2 • EBRT TECHNIQUES

Planning and delivery

- Simulation:** computed tomography (CT) in the treatment immobilisation device → contour target and organs → prescribe dose and normal-tissue constraints → optimise → **dose-volume histograms (DVHs)** record the dose to each structure.
- Multiparametric magnetic resonance imaging (mpMRI) is fused with CT to delineate prostate and dominant lesion; MRI alone has no electron-density data, so it cannot compute dose. Prostate-specific membrane antigen positron emission tomography (PSMA PET) is appropriate before RT for unfavourable intermediate- and high-risk disease and changes management in up to half → Ch 152.

	3D conformal RT (3D-CRT)	Intensity-modulated RT (IMRT) / volumetric modulated arc therapy (VMAT)
Planning	Forward: beams shaped to target outline	Inverse: constraints entered, optimiser computes fluence
Dose	Conforms to target shape	Steep gradients; concave dose around rectum; VMAT halves treatment time
Toxicity	IMRT ↓ GI (gastrointestinal) toxicity, not GU (genitourinary) — rectum lies outside the target, the urethra inside it. No randomised trial, yet IMRT is now standard.	

Volume	Prostate application
Gross tumour volume (GTV)	Dominant intraprostatic lesion (DIL) on MRI

Volume	Prostate application
Clinical target volume (CTV)	Whole prostate ± proximal seminal vesicles ± pelvic nodes (common, external and internal iliac, obturator, presacral)
Planning target volume (PTV)	CTV + margin for motion and set-up: 3-7 mm with image guidance

Image-guided RT (IGRT), fiducials and spacers

- **Interfraction** motion (prostate position differs daily) → daily online imaging: implanted **gold fiducial markers**, cone-beam CT, on-rail CT or megavoltage CT. Benefit: better biochemical control, less GU bother and proctitis; cost ~2-3 cGy per scan. Transabdominal ultrasound abandoned (interuser variability).
- **Intrafraction** motion → real-time tracking, radiofrequency transponders (MRI artefact) or endorectal balloon; bladder-filling and rectal-emptying instructions help both.
- IGRT is the standard (National Comprehensive Cancer Network, NCCN); smaller margins are what make large fractions safe.
- **Rectal spacer**: biodegradable hydrogel or hyaluronic acid injected transperineally into Denonvilliers' fascia separates prostate from anterior rectal wall. Randomised: late rectal toxicity **2% vs 7%** (hydrogel, 79.2 Gy); grade ≥2 rectal 2.9% vs 13.8% (hyaluronic acid, 60 Gy/20). Safe with EBRT, SBRT and brachytherapy.
- **MRI-linear accelerator**: no fiducials, daily plan adaptation; slower. MIRAGE: 2 mm margins ↓ acute GU toxicity vs CT-guided SBRT.
- **Protons**: Bragg peak, no exit dose, but they shrink only the low-dose bath, not the high-dose rectal volume that drives late toxicity. No randomised benefit over IMRT.

3 · DOSE ESCALATION

- All six randomised trials (MD Anderson 70 vs 78 Gy; PROG 95-09; Dutch 68 vs 78; MRC RT01 64 vs 74; GETUG 06; Radiation Therapy Oncology Group (RTOG) 0126 70.2 vs 79.2) improved **biochemical control in all risk groups**, with or without androgen deprivation therapy (ADT); **none improved overall survival (OS)** → dose escalation (≥~78 Gy conventional) is standard.
- Why no OS gain: effective salvage therapy; distant metastasis fell only borderline (RTOG 0126: 6% → 4%).
- Cost: MD Anderson grade ≥2 late toxicity doubled at 78 Gy; rectal risk rises when >25% of rectum gets 70 Gy.
- **Focal boost** — FLAME: boost to MRI-defined DIL improved biochemical progression-free survival (hazard ratio [HR] 0.45) without more toxicity.

4 · HYPOFRACTIONATION

Moderate (2.25-4 Gy)

- **CHHiP**: 74 Gy/37 vs **60 Gy/20** vs 57 Gy/19 (all 3-6 mo ADT). 60/20 noninferior; 57/19 not shown noninferior; no toxicity difference.
- **PROFIT** (intermediate risk) 60/20 noninferior; **RTOG 0415** (low risk) 70 Gy/28 noninferior but more late GI/GU toxicity; **HYPPO** 64.6 Gy/19 **not superior** (a superiority design).
- Postoperative: HYPPO 62.5 Gy/25 noninferior to 66.6 Gy/37 for patient-reported late toxicity.

Ultrahypofractionation / SBRT

- **HYPO-RT-PC**: intermediate-high risk, no ADT, **42.7 Gy/7** vs 78 Gy/39 → noninferior failure-free survival (~84% at 5 yr); more early acute toxicity, equal late toxicity.
- **PACE-B**: low-intermediate risk, **36.25 Gy/5** vs 78/39 or 62/20 → similar 2-yr late grade ≥2 GU and GI toxicity.
- **Dose ceiling**: no oncological gain above **40 Gy/5**; higher doses buy toxicity, not control.

- Delivery: alternate days (normal-tissue repair), margins 3–5 mm, fiducial tracking or on-board CT. Rectal risk factor: >35% of circumference >39 Gy; spacer markedly ↓ rectal ulcers.

5 · BRACHYTHERAPY (BT)

	Low-dose-rate (LDR)	High-dose-rate (HDR)
Source	Permanent seeds (^{125}I , ^{103}Pd , ^{131}Cs)	Temporary ^{192}Ir (380 keV) in remote afterloader via transperineal catheters
Technique	Transrectal ultrasound (TRUS)-guided transperineal template, lithotomy, general anaesthesia; peripheral loading spares urethra	Dose shaped by dwell times/positions; no post-implant dosimetry; shielded room
Uses	Monotherapy low/intermediate; boost	Boost (e.g. 15 Gy × 1 after 46 Gy); monotherapy 27 Gy/2, 34–38 Gy/4
Pitfall	Post-implant oedema, seed migration	19 Gy × 1 fails: 5-yr local failure 29% vs 3% (27 Gy/2)

Isotope	Half-life	Monotherapy	Boost (with 45 Gy EBRT)
^{125}I	59.6 d	140–160 Gy	108–110 Gy
^{103}Pd	17 d	110–125 Gy	90–100 Gy
^{131}Cs	9.7 d	100–115 Gy	80–85 Gy

- Shorter half-life → faster delivery → lower prescription. Low energy (21–30 keV) → dose falls off within mm: spares tissue but punishes misplacement.
- **Post-implant dosimetry** (CT ± MRI at 14–30 days): **D90** (dose covering 90% of gland) >90% of prescription; **V100** (% of gland receiving prescription) >85%; urethral V150 <0.05 cm³; rectal V100 <1 cm³. Low D90 predicts biochemical failure. **[MCQ]** D90 is a dose, V100 a volume.

Selection and contraindications

The main acute toxicity is obstructive/irritative uropathy (temporary catheter in a minority), so urinary function drives eligibility. Patient must tolerate anaesthesia, lithotomy and TRUS.

- Relative contraindications: **International Prostate Symptom Score (IPSS) >20; maximum flow rate (Qmax) <10 mL/s**; large median lobe; large transurethral resection of prostate (TURP) defect or TURP within 3–6 mo (incontinence); prior pelvic RT; inflammatory bowel or connective tissue disease; **pubic arch interference** (large gland); rectal fistula; cannot assume lithotomy.
- Large gland: short ADT shrinks ~30% (cardiovascular toxicity); 5α-reductase inhibitor ~20%, less morbid.
- Retention risk: IPSS >20 ≈ 29% vs IPSS <10 ≈ 2%. α-blockers started before implant ease symptoms.

BT with EBRT

- EBRT 45 Gy + implant at 60–70% of monotherapy dose; EBRT covers periprostatic tissue, seminal vesicles ± nodes.
- **RTOG 0232** (intermediate risk): BT alone = EBRT + BT for biochemical control, more late toxicity with combination → **BT alone is standard for intermediate risk**.
- **ASCENDE-RT** (intermediate-high risk; 12 mo ADT + pelvic EBRT 46 Gy): **LDR boost** vs EBRT boost to 78 Gy → EBRT boost had **twice the biochemical failure** (HR 2.04); no OS difference; price = 5-yr grade 3 GU toxicity **18.4% vs 5.2%**.
- UK HDR boost trial: 55 Gy ± HDR 17 Gy/2 → 21% absolute relapse-free gain at 12 yr. Gleason 9–10 pooled analysis: EBRT + BT lowest prostate cancer mortality.

KEY POINT Why BT works: the steep fall-off lets intraprostatic dose far exceed EBRT. Hence outcome depends on implant accuracy (D90), eligibility on whether the urethra tolerates swelling, and EBRT is added only where disease may lie beyond the implant.

6 • ADT WITH RT

- **Synergy:** ADT induces apoptosis, shrinks the target and ↓ Ku70 → impaired non-homologous end-joining repair (radiosensitisation); RT up-regulates androgen receptor signalling, which adjuvant ADT blocks.
- ADT added to RT, and RT added to ADT, each improve survival in intermediate- to high-risk disease (trials in §12; locally advanced detail → *Ch 164*).

Risk group (NCCN)	With EBRT	With EBRT + BT	Key evidence
Low / favourable intermediate	None	—	RTOG 94-08: no benefit in low risk
Unfavourable intermediate	4-6 mo	—	RTOG 94-08, DFCI 95-096, TROG 96.01 (6 > 3 mo); RTOG 99-10: 9 mo no better than 4
High / very high	24-36 mo (≥18)	12-36 mo	European Organisation for Research and Treatment of Cancer (EORTC) 22863, RTOG 92-02, EORTC 22961 (36 > 6 mo); PCS IV: 18 = 36 mo

- Required duration rises with risk and falls as RT dose rises: in **TROG 03.04 RADAR** the benefit of 18 vs 6 mo ADT disappeared after an HDR boost.
- **STAMPEDE:** adding 2 yr abiraterone + prednisolone to ADT + RT in high-risk N0/N1 M0 improved metastasis-free and OS → *Ch 164*.

7 • PELVIC NODAL RT

- **RTOG 94-13** (node risk >15%): early progression-free survival (PFS) advantage for whole-pelvis RT lost on update. **GETUG-01:** no benefit. Intact prostate: no proven benefit from older trials.
- Why negative: fields ending at L5/S1 miss the **common iliac nodes**, the site of over half of first nodal recurrences. RTOG 09-24 uses an **L4/L5** superior border.
- † **POP-RT** (high/very high risk, node risk >20%, long-term ADT): whole-pelvic RT to the L4/L5 border improved 5-yr biochemical failure-free survival (~95% vs ~81%) and disease-free survival; no OS difference.
- Toxicity: more acute GI; with VMAT, 24-month quality of life unchanged.

8 • POST-PROSTATECTOMY RT

- **Adjuvant vs early salvage:** RADICALS-RT, GETUG-AFU 17 and RAVES (64-66 Gy) and their meta-analysis → **observation with early salvage is preferred** for most; adjuvant RT may still suit very high-risk (Gleason 8-10, pT3b) men → *Ch 164*.
- **SPPORT (RTOG 0534)**, salvage: prostate bed RT vs + short-term ADT vs + pelvic nodes + short-term ADT → **pelvic RT + short-term ADT** gave best freedom from progression and fewer distant metastases.
- Dose ≥64 Gy; acute GU toxicity lower than for intact prostate (RTOG 96-01: grade ≥2 14%). PSMA PET, fluciclovine or choline PET localise recurrence better than conventional imaging.

9 • FOLLOW-UP AND BIOCHEMICAL FAILURE AFTER RT

- Benign glands persist, so prostate-specific antigen (PSA) does not become undetectable as after radical prostatectomy (RP). Nadir falls with dose: ~0.6 ng/mL (70 Gy), ~0.3 (79 Gy), ≤0.1 (EBRT + BT). Time to nadir ~1.5 yr EBRT, ~2.5 yr BT.

	American Society for Therapeutic Radiology and Oncology (ASTRO) 1996	Phoenix 2005 (current)
Failure	3 consecutive rises	PSA $\geq$ nadir + 2 ng/mL
Date	Backdated midway nadir-first rise	When reached (or salvage started)
Notes	Delays $\geq$ 18 mo; biased; invalid with ADT/BT	Best correlates with clinical relapse; valid with short ADT; fewer false calls from bounces

- Not failure: laboratory error, treated prostatitis, benign bounce. Meeting a definition does not by itself mandate intervention.
- Poor prognosis: biochemical failure <18 mo; nadir <12 mo; nadir >2 ng/mL at 1 yr; **PSA doubling time (PSADT) <3 mo** (cancer death; <12 mo → consider early intervention). Local failure: slow rise, PSADT $\geq$ 1 yr; distant: early nadir, PSADT 3-6 mo.
- **Bounce** (rise >0.2 ng/mL then durable fall): ~1/3 after LDR (2/3 in young men), up to 12-18 mo, often within first 3 yr; predicts *better* outcome. Rising PSA after BT: within 30 mo → watch; persists >30 mo → biopsy; approaching 10 ng/mL → systemic staging. Testosterone recovery after ADT also causes bounces.
- **Biopsy**: optimal at **30-36 mo** (post-mitotic cell death). Do not grade tumours with marked radiation effect; marked effect behaves as negative. Basal cell marker (high-molecular-weight cytokeratin) separates radiation atypia from cancer. mpMRI-targeted biopsy; recurrence arises at the original dominant nodule.

10 · TOXICITY AND MANAGEMENT

Toxicity	Features	Management / prevention
Acute GU	Dysuria, frequency, weak stream peak ~2 mo, settle by 12 mo; BT: IPSS rises, falls with isotope half-life (¹⁰³ Pd worse at 1 mo, resolves faster)	α -blockers; catheter for retention
Late GU	Grade $\geq$ 2 ~20% (IMRT 86.4 Gy, permanent implants); stricture; incontinence uncommon (~3%), higher after prior TURP	IGRT halves late GU toxicity; peripheral loading
Acute GI	Urgency, frequency, self-limiting	Symptomatic
Late GI (proctitis)	Bleeding 6-18 mo after RT; IMRT 81 Gy grade $\geq$ 2 ~5%; BT 1-2%; recto-urethral fistula rare (colostomy ~0.3%)	Steroid suppositories/sucralfate enemas → argon plasma coagulation or hyperbaric oxygen → diverting colostomy. Prevent: IMRT, IGRT, spacer, rectal V100 limits.
Erectile dysfunction (ED)	Vascular; gradual over 24 mo; driven mostly by ADT; penile bulb dose; potency best with BT alone (~76% at 5 yr)	Phosphodiesterase-5 inhibitors effective in ~3/4
Second cancers	↑ Small absolute excess of bladder and rectal cancers years after pelvic RT	Investigate haematuria or rectal bleeding fully (cystoscopy, colonoscopy)

- BT + EBRT is more toxic than either alone; counsel.

11 · SALVAGE AFTER LOCAL FAILURE

- Prerequisite: **biopsy-proven, local-only** recurrence. Options: salvage RP (incontinence ~63%, rectal injury), BT (LDR or HDR — preferred, concentrates dose inside the gland), SBRT re-irradiation, cryotherapy, high-intensity focused ultrasound.

12 • LANDMARK TRIALS

Trial	Design	Result	Take-home
RTOG 0126	70.2 vs 79.2 Gy, low-intermediate	8-yr biochemical failure-free 65% → 80%; OS same	Dose escalation ↑ control, not OS
CHHiP	74/37 vs 60/20 vs 57/19	60/20 noninferior	60 Gy/20 standard
HYPO-RT-PC	42.7/7 vs 78/39	Noninferior; equal late toxicity	Ultrahypofractionation valid
PACE-B	36.25/5 vs conventional/moderate	Similar 2-yr toxicity	SBRT safe
ASCENDE-RT	EBRT vs LDR boost (+12 mo ADT)	Biochemical failure HR 2.04 for EBRT boost; more GU toxicity with LDR	BT boost for high risk
TROG 96.01	RT ± 3 or 6 mo ADT	6 mo ↑ OS; 3 mo no mortality gain	≥6 mo short-course
RTOG 94-08	66.6 Gy ± 4 mo ADT, T1b-T2b	10-yr OS 57% → 62%; intermediate only	ADT for intermediate, not low
EORTC 22863	RT ± 3 yr goserelin, T3-T4	10-yr OS 39.8% → 58.1%	Long ADT in high risk
EORTC 22961	6 vs 36 mo ADT	Short course not noninferior	Long ADT for high risk
PCS IV	18 vs 36 mo	No survival difference	18 mo acceptable
SPCG-7; NCIC PR.3/MRC PR07	Lifelong ADT ± RT	RT ↓ cancer death, ↑ OS	Never ADT alone if fit
RTOG 94-13; GETUG-01	Pelvic vs prostate RT	No durable benefit	Field height matters
SPPORT	Salvage bed RT ± ADT ± nodes	Pelvic RT + ADT best	Treat nodes in salvage

13 • MCQ ONE-LINERS

1. α/β of prostate cancer ≈ 1.5 Gy — lower than rectum (~ 3 Gy).
2. Low α/β is the radiobiological rationale for hypofractionation.
3. Moderate hypofractionation standard: 60 Gy in 20 fractions (CHHiP, PROFIT).
4. CHHiP 57 Gy/19 was not shown noninferior.
5. HYPO-RT-PC: 42.7 Gy in 7 fractions; PACE-B: 36.25 Gy in 5.
6. SBRT dose ceiling: no benefit above 40 Gy in 5 fractions.
7. Dose escalation improved biochemical control in every trial but never OS.
8. IMRT reduces GI but not GU toxicity versus 3D-CRT.
9. Rectal spacer sits in Denonvilliers' fascia; halves-to-thirds late rectal toxicity.
10. Half-life: ^{125}I 59.6 d, ^{103}Pd 17 d, ^{131}Cs 9.7 d.
1. LDR ^{125}I monotherapy dose: 140–160 Gy; boost ~ 110 Gy.
2. Single-fraction 19 Gy HDR monotherapy fails — multifraction is standard.
3. D90 goal $>90\%$ of prescription; V100 $>85\%$ of gland.
14. BT relative contraindications: IPSS >20 , Qmax <10 mL/s, recent TURP, pubic arch interference.
15. ASCENDE-RT: EBRT boost doubled biochemical failure versus LDR boost.
16. RTOG 0232: adding EBRT to BT in intermediate risk adds toxicity, not control.
17. ADT radiosensitises by reducing Ku70-dependent DNA repair.
18. Unfavourable intermediate risk: 4–6 months ADT with RT.
19. High risk: 18–36 months ADT with EBRT (PCS IV: 18 = 36).
20. HDR boost made extra ADT redundant in TROG RADAR.
21. Nodal trials likely negative because L5/S1 border misses common iliac nodes.
22. Phoenix definition: PSA nadir + 2 ng/mL.
23. Benign PSA bounce: $\sim 1/3$ after LDR; associated with better outcome.
24. Optimal post-RT biopsy timing: 30–36 months.
25. PSADT <3 months after RT predicts prostate cancer death.

- | | |
|---|---|
| <p>!6. Radiation proctitis bleeding: 6-18 months; argon plasma coagulation effective.</p> <p>!7. SPPORT: pelvic nodal RT + short-term ADT best for salvage.</p> | <p>28. Early salvage preferred over adjuvant RT (RADICALS-RT, RAVES, GETUG-AFU 17).</p> |
|---|---|

1 • DEFINITION AND RATIONALE

- **Focal therapy (FT)**: guided ablation of a cancer lesion that is image-defined and biopsy-confirmed (or biopsy-confirmed only), with a **safety margin**. Aim (in men selected on prostate-specific antigen (PSA), imaging and biopsy): treat only the foci that threaten survival, preserving the neurovascular bundles, sphincter and rectum → sexual and urinary function.
- Model: **breast conservation** (lumpectomy replacing Halsted mastectomy). Evolution: subtotal ablation → hemiablation ("male lumpectomy") → image-guided lesion ablation.
- Enablers: multiparametric magnetic resonance imaging (mpMRI), MRI-fusion targeted biopsy, and recognition that **Gleason pattern 4/5** is the lethal subgroup (carries far more cancer hallmarks — angiogenesis, apoptosis resistance, invasion — than pattern 3).

Index lesion hypothesis

- Prostate cancer is multifocal in **50-91%** of prostatectomy specimens. **Index lesion (IL)** = largest/highest-grade focus; hypothesis: the IL drives the natural history and metastasis.
- **For**: most extraprostatic extension (EPE) arises from the IL; IL holds **80-90%** of tumour volume; satellites are mostly **<0.5 mL** (Stamey's "insignificant" cancer) — discounting them makes 52-93% of glands functionally unifocal; metastases are monoclonal.
- **Against**: aggressive features lie in a satellite in 12-38%; IL Gleason matches overall Gleason in only ~two-thirds; a lethal clone has been traced to a small pattern 3 focus.

KEY POINT "Multifocal, therefore unsuitable for FT" is a trap: what matters is one significant lesion plus insignificant satellites, not the count of foci. The hypothesis's exceptions are why FT is always paired with surveillance.

- **FT + active surveillance (AS)** are two halves of one strategy: ablate grade 4 disease (Prostate Cancer Intervention Versus Observation Trial (PIVOT) benefit was in intermediate risk) and watch the low-grade remainder, retreating new significant foci → prostate cancer as a chronic disease.

2 • PATIENT SELECTION

Selection is the fundamental tenet: the aggressive lesion must be identified and its extent delineated for complete ablation. Recurrence is likelier than after radical therapy, so the patient must be **compliant with close surveillance**.

Lesion localisation

- **Transrectal ultrasonography (TRUS) biopsy**: random, poor anterior sampling; lateralisation has high negative predictive value (NPV) but poor specificity. Against **transperineal mapping biopsy (TMB, 5-mm grid)**, TRUS misses almost half the disease relevant to hemiablation; >half of "TRUS-unilateral" men are disqualified.
- **mpMRI** preferentially detects high-grade lesions (diffusion-weighted imaging, DWI; dynamic contrast-enhanced, DCE), finds anterior tumours (15-33%, missed by TRUS, ideal for FT) and locates the IL. 3 tesla > 1.5 tesla; reader experience matters.
- **PROMIS** (mpMRI vs TMB) for clinically significant prostate cancer (csPCa): sensitivity **93%**, specificity 41%, NPV 89% → lesions need histological confirmation and MRI-negative areas still need systematic biopsy.
- mpMRI **underestimates tumour volume** (5-30%) → margin needed: **5 mm** ablates >95%, 6 mm 98.6% of csPCa → often hemi- or zonal ablation.
- **~90%** of outfield csPCa lies within **1 cm** of the target (the "penumbra") → biopsy in and around the lesion.

- Targeted biopsy: in-bore MRI (most accurate, costly), software MRI-ultrasound fusion, cognitive fusion (as good only in expert hands). Prostate-specific membrane antigen (PSMA) PET: more specific; mpMRI more sensitive.

KEY POINT Imaging finds the lesion, only biopsy grades it, and only systematic biopsy vouches for the rest of the gland — every consensus has kept all three.

Consensus criteria (2016-2019)

Variable	Consensus
Work-up	mpMRI + targeted biopsy + systematic biopsy of the "normal" gland
Risk group	Intermediate ; Gleason 3+4 = "sweet spot" ; 4+3 maximum; no cribriform or ductal histology
Size	≤1.5 mL (≤3 mL if not crossing midline); 20–25% of gland if crossing
Residual untreated disease	Gleason 3+3 only (≤1 mm core length in 2016 — tightened from 5 mm)
Life expectancy	Per guidelines; no fixed limit
Erectile dysfunction	Not a contraindication (FT also spares urinary/rectal toxicity)
Not suitable	High risk (no long-term locoregional data); low risk usually → AS

- Goal evolution: eradicate all cancer (2009) → eradicate csPCa (2013–16) → treat and retreat to **delay or avoid radical therapy** (2019).
- Counsel: investigational, ideally within a trial; retreatment possible; later salvage may be harder.

3 · ABLATION PATTERNS AND ROUTES

- FT = **lesion (targeted) ablation, quadrant ablation, hemiablation**. "Hockey-stick" (subtotal) ablation is partial-gland ablation, not FT.
- Transrectal → posterior zone; transperineal → anterior zone; transurethral → anterior gland. "À la carte" choice by lesion and patient.

4 · ENERGY SOURCES COMPARED

Modality	Mechanism / route	Pros	Cons
Cryotherapy	Heat extraction (Joule-Thomson, 17-gauge probes): ice crystals, vascular stasis, apoptosis; transperineal	Largest literature; ice edge visible on TRUS; thermocouples; urethral warmer; Cryo On-Line Data (COLD) registry	Retention (catheter few weeks); rectourethral fistula (RUF) ≤0.8%
High-intensity focused ultrasound (HIFU)	Focused beams ≥55 °C → coagulative necrosis; transrectal	Posterior lesions; short focal time; HEAT registry (1379 men, incl. high risk)	Anterior lesions (energy dissipation, oedema); gland height, size, calcification (transurethral resection of the prostate may help)
MRI-guided transurethral ultrasound ablation (TULSA)	Transurethral transducer + MR thermometry	Anterior gland; spatial control within 1.3 mm	In-bore MRI; csPCa on 12-month biopsy ~1/3
Focal laser ablation (FLA)	Interstitial fibre ≥60 °C; MR thermometry; transperineal/transrectal	Most predictable ablation zone; safe near vital structures	Extra fibres for big lesions; narrow margin → highest infield failure (~40% at 6 years)

Modality	Mechanism / route	Pros	Cons
Irreversible electroporation (IRE)	Non-thermal: direct-current pulses >1600 V/cm make membrane pores; probes 10-20 mm apart	Claimed to spare connective tissue; best function in matched comparison with robot-assisted radical prostatectomy (RARP)	Deep anaesthesia (muscle contraction); histology shows neurovascular bundle and urethral injury
Vascular-targeted photodynamic therapy (VTP)	Intravenous padeliporfin (WST11, TOOKAD) 4 mg/kg + 753-nm light → reactive oxygen species → vascular occlusion, necrosis	Only randomised evidence (PCM301)	Gleason 6 only; perineal pain ~15%; light precautions
Radiofrequency ablation	~100 °C molecular friction	Fat limits conduction	Early data; infield csPCa 20-30%
Focal brachytherapy	Low-dose-rate seeds or high-dose-rate (HDR) catheters; sharp dose fall-off	Mild early toxicity	HDR: stricture ≤8.2%, erectile dysfunction in half; biopsy only at 24 months; late toxicity

- Experimental: focal intensity-modulated radiotherapy and stereotactic body radiotherapy (SBRT), gold nanoshells (AuroLase), PRX302 (PSA-activated pore toxin), intratumoral paclitaxel (NanoPac).

KEY POINT Every energy fails at the edge of the ablation zone, where temperature or field strength becomes sublethal. Margins, thermometry and adjuvants (cryo-enhancers, SBRT boost) all aim to make that edge lethal. Hence laser's precise but narrow zone gives the most infield recurrence.

5 • OUTCOMES

Modality	Infield positive biopsy	Outfield	Continence	Erections
Cryotherapy	7-39% overall (~¼ csPCa)		95-100%	47-100%
HIFU	17-54% (csPCa 0-22%)	7-43% (csPCa 2-8%)	90-95%	50-78%
Laser	11-50%	17-60%	95-100%	Minimal change
IRE	csPCa 1-10%	csPCa 5-16%	~All	Some decline
VTP	19-37%	17-19%	Preserved	Nil-mild fall

- Continence is preserved by every modality (~90-100%); infield failure differs, but heterogeneous definitions forbid ranking.
- Cryotherapy 5-year biochemical progression-free survival (BPFS) ~62-70%. **COLD registry:** partial vs whole-gland cryo — same BPFS, partial ~2× better erectile recovery.
- HEAT (HIFU): 7-year failure-free survival **69%** (68% intermediate, 65% high risk).
- IRE vs matched RARP: better pad-free continence and erections, but some residual csPCa needing salvage.

6 • COMPLICATIONS

- Clavien-Dindo ≥3 ≤4% (cryotherapy). Commonest: **urinary retention** from oedema (0-9%; prevent with catheter for a few weeks); urinary tract infection; haematuria, haemospermia; urethral sloughing/debris; stricture (~4-5% HIFU, IRE).
- **Rectourethral fistula:** rare (≤2%) — the most feared; some (after laser) healed with urinary catheterisation for 4-6 weeks.
- Erectile decline is mild in most series; worst after focal HDR brachytherapy.

7 · FOLLOW-UP AND DEFINITION OF FAILURE

- **PSA falls only 50-80%** (residual epithelium) — no "zero" target; Phoenix/American Society for Radiation Oncology (ASTRO) definitions are not validated after FT. Proposed: nadir + **1.0** ng/mL at 12 months or nadir + 1.5 at 24-36 months predicts csPCa.
- Digital rectal examination unreliable (scar).

Success (ICUD-SIU 2015)	Infield (treated)	Outfield (untreated)
Short term	Eradication of treated focus	No csPCa — any outfield cancer = selection failure
Intermediate/long term	No csPCa	No csPCa (new cancer may be de novo)

ICUD-SIU = International Consultation on Urological Diseases–Société Internationale d'Urologie.

ALGORITHM

mpMRI + targeted biopsy (4-6 cores) of treated zone at **3-6 months** → mpMRI at 1-2 and 5 years (targeted biopsy for new lesion) → systematic biopsy at 1-2 and 5 years for the untreated gland

- **Prostate Imaging after Focal Ablation (PI-FAB):** read **DCE first** (scar obscures T2 and DWI), then DWI, apparent diffusion coefficient (ADC) map, T2. PI-FAB 1 (DCE–, DWI–) → monitor; 2 (DCE+, DWI–) → correlate with PSA, biopsy if rising; 3 (both +) → biopsy. Enhancement >3 mm resembling the original tumour → upgrade/biopsy.
- Contrast-enhanced ultrasound matched mpMRI for early success after IRE; grey-scale TRUS cannot see the ablation zone.

8 · SALVAGE

After focal therapy fails

- Persistent infield or de novo csPCa → **focal retreatment**, or conversion to whole-gland radical prostatectomy (RP) or radiotherapy. Patients must accept possible repeat treatment upfront.
- Salvage RP after ablation is technically harder (periprostatic fibrosis) and outcomes are worse than primary therapy. † Salvage RP after FT is feasible with acceptable cancer control, but continence and potency are worse than after primary RP.

Focal salvage for radiorecurrent disease

	Cryotherapy	HIFU	IRE
Cancer control	5-yr BPFS ~50%	3-yr BPFS 48%	PFS 60%, MFS 91% (48 months)
Continence	87-100%	64-87.5%	93%
Signature complication	RUF ~1.4%	Bladder-neck stricture, fistula ~2%; grade ≥3 ~¼	Urethral sludge ~9%
Trial	FORECAST (cryo vs HIFU, ongoing)		SAFE pilot

PFS = progression-free survival; MFS = metastasis-free survival.

- Focal and whole-gland salvage give similar cancer control; focal is better functionally. Expect roughly half the primary-FT results.

9 · LANDMARK TRIALS

Trial	Design	Result	Take-home
PCM301 (Azzouzi 2017)	415 low-risk men; padeliporfin VTP vs AS	Progression 28% vs 58%; more negative biopsies at 2 yr; conversion to radical therapy 4 yr 24% vs 53% ; continence preserved	First level 1 evidence for FT — but Gleason 6 only, surrogate end points (not survival), possible reclassification

Trial	Design	Result	Take-home
PROMIS	mpMRI vs 5-mm TMB, 11 centres	csPCa sensitivity 93%, specificity 41%	MRI lesions need biopsy; systematic biopsy still needed
HEAT registry	Prospective focal HIFU, 1379 men	7-yr failure-free survival 69%	Largest FT dataset; includes high risk

KEY POINT Open problems: can insignificant satellites be safely left; cost of repeated treatment; adjuvants. Lumpectomy works because the breast gets adjuvant radiotherapy — prostate FT relies on surveillance and retreatment instead.

10 • MCQ ONE-LINERS

1. FT rests on the index lesion hypothesis.
2. Stamey insignificant cancer: volume <0.5 mL.
3. Index lesion holds ~80–90% of tumour volume.
4. FT "sweet spot": completely ablatable Gleason 3+4.
5. Maximum grade for FT by 2016 consensus: 4+3; exclude cribriform/ductal.
6. Erectile dysfunction is not a contraindication to FT.
7. PROMIS: mpMRI sensitivity 93%, specificity 41% for csPCa.
8. ~90% of outfield csPCa lies within 1 cm of the MRI target.
9. A 5-mm margin ablates >95% of csPCa.
0. Hockey-stick ablation is partial-gland ablation, not FT.
1. Transrectal HIFU suits posterior lesions; anterior lesions → transperineal or transurethral.
2. HIFU lethal temperature $\geq 55^{\circ}\text{C}$; laser $\geq 60^{\circ}\text{C}$.
3. Cryotherapy uses Joule-Thomson effect; urethral warmer reduces urethral injury.
4. Commonest complication of focal cryotherapy: urinary retention.
5. IRE is non-thermal; field strength >1600 V/cm.
16. IRE needs deep anaesthesia to prevent muscle contraction.
17. VTP drug: padeliporfin (WST11), activated by 753-nm light.
18. PCM301: VTP vs AS in low risk; only randomised FT trial.
19. PCM301 end points were progression and conversion, not survival.
20. Laser: most predictable zone but highest infield failure.
21. Focal HDR brachytherapy: urethral stricture up to 8.2%.
22. PSA falls only 50–80% after FT.
23. Outfield cancer early after FT = selection failure.
24. First follow-up biopsy of treated zone at 3–6 months.
25. PI-FAB reads DCE first.
26. COLD registry: partial cryo ~2× better erectile recovery than whole-gland.
27. Most feared salvage complication: rectourethral fistula.
28. FORECAST compares focal salvage cryotherapy and HIFU.

164 Treatment of Locally Advanced Prostate Cancer

Reading time ≈ 32 min (slow pace)

1 · DEFINITION AND SIGNIFICANCE

- **Classic (digital rectal examination, DRE):** extracapsular extension (ECE) cT3a, seminal vesicle invasion (SVI) cT3b, adjacent organs cT4.
- **Working definition:** "high-risk" disease including regional nodal spread without distant metastasis — **T3-4, N0 or N1, M0**. In the prostate-specific antigen (PSA) era most cancers are non-palpable (cT1c), so PSA, Gleason grade and imaging, not T stage, now identify risk.
- ≥10% of new diagnoses; with metastatic cases they cause a disproportionate share of deaths. Any single modality carries a high recurrence risk → **multimodal therapy**.
- National Comprehensive Cancer Network (NCCN) high risk: cT3a, or PSA >20 ng/mL, or grade group (GG) 4-5. † Very high risk: cT3b-T4, primary Gleason pattern 5, ≥2 high-risk features, or >4 cores GG 4-5.
- Stage migration: high risk now mostly means **high grade**, not palpable T3 (clinical T3-T4 fell from ~12% to 3.5% in CaPSURE).

2 · NATURAL HISTORY

- **Grade drives cancer death** under conservative management: Gleason 8-10 → 15-year cancer mortality 60-87% (pre-PSA) and still >20% in contemporary cohorts.
- Swedish cohort (15 yr): T1-T2 and T3-T4 progress at equal rates (~48%) but disease-specific survival is 81% vs 57% — locally advanced disease converts progression into death.
- Fit men in their 70s with high-grade disease are often undertreated; withholding therapy from them is an error.

3 · STAGING AND RISK ASSESSMENT

- Combine PSA, T stage, grade and extent of cancer on biopsy. Clinical staging and imaging both **understage**.
- **Tools:** Partin tables predict pathological stage (localised disease only); Kattan nomograms predict progression (the radical prostatectomy [RP] version does not apply to cT3b or node-positive disease); D'Amico; Cancer of the Prostate Risk Assessment (CAPRA, 0-10: each 2-point rise doubles recurrence risk); clinical-genomic groups (22-gene Decipher) reclassify ~two-thirds.
- Staging triggers (NCCN): cT3-T4, PSA >20 ng/mL, GG 4-5 or primary Gleason pattern 5 → bone scan + computed tomography (CT) or magnetic resonance imaging (MRI); prostate-specific membrane antigen positron emission tomography (PSMA PET) may substitute.

Modality	Strength	Weakness
Transrectal ultrasound (TRUS)	Guides seminal vesicle/capsular biopsy to confirm cT3	No better than DRE; understages
Multiparametric MRI (mpMRI)	T2 vs T3; high specificity for pT3 in higher-risk men (>3 positive cores, abnormal DRE, PSA >10)	Variable ECE accuracy; poor specificity at neurovascular bundle
Conventional (CT/MRI + bone scan)	Widely available	Less accurate
PSMA PET	Nodal specificity ≥95%; proPSMA : accuracy 92% vs 65% for conventional imaging	Nodal sensitivity only ~40-66%

4 · RADICAL PROSTATECTOMY (RP) IN HIGH-RISK DISEASE

Rationale and selection

- RP for high-risk disease is rising again (CAPRA 6–10). It removes the primary (avoids later local morbidity), gives accurate pathological staging and lets adjuvant therapy be targeted to those who need it.
- Clinical overstaging of cT3 in 7–27%**, and over a third of high-risk specimens are specimen-confined → do not exclude men from RP on clinical stage alone.
- NCCN: RP + **extended pelvic lymph node dissection (ePLND)** only for high-risk tumours that can be **completely excised**; otherwise androgen deprivation therapy (ADT; AD in trial reports) + radiation therapy (RT). Survival is better when RP with node dissection is completed. Minimally invasive RP gives comparable outcomes.
- Outcomes, cT3: about half recurrence-free at 8–10 yr on surgery alone; disease-specific survival (DSS) **~80–90% at 10 yr** although biochemical disease-free survival is often only ~50%.

KEY POINT Operative principle: a **positive surgical margin** carries the highest relative risk of progression (4.37) — higher than ECE, SVI or nodes — so remove all prostatic tissue regardless of stage.

Pathology after RP	5-yr PSA-free survival	Relative risk of progression
Organ-confined	95%	Reference
ECE	76%	2.2–2.7
SVI	37%	2.6
Node-positive	18%	3.3
Positive margin	—	4.37 (highest)

- pT3bN0 at 10 yr: biochemical recurrence-free ~25%, metastasis-free 70%, DSS 80%.
- Node-positive at RP:** ~half alive at 10 yr; single positive node → nearly half never progress, DSS >80%. Poor predictors: Gleason >8, >3 positive nodes. PSMA PET may select limited, excisable nodal disease.

Neoadjuvant androgen deprivation (NAD) before RP

- PSA falls ~90% and gland volume 30–50%; **clinical downstaging frequent (32–90%), pathological uncommon (8–31%)**; only ~20% of cT3 become organ-confined.
- Reduces positive margins in cT1–T2 but not cT3; does **not** reduce SVI or nodal positivity, PSA recurrence, DSS or overall survival (OS). 8 vs 3 months: fewer positive margins (23% → 12%) but no PSA benefit.
- Neoadjuvant chemohormonal therapy: **CALGB 90203** (AD + docetaxel before RP) missed its primary endpoint but improved metastasis-free survival (MFS) and OS. Pathological complete response (pT0) is essentially never achieved (0–4%).

Adjuvant and salvage RT after RP

Trial	Population / comparison	Result
EORTC 22911 (European Organisation for Research and Treatment of Cancer)	pT3 and/or margin+; immediate RT vs observation	Biochemical disease-free survival 74% vs 53%; benefit only with positive margins
SWOG 8794 (Southwest Oncology Group)	Same	MFS (hazard ratio [HR] 0.71) and OS (HR 0.72) improved — the only OS-level evidence
ARO 96-02	pT3N0, undetectable PSA	Biochemical HR 0.53
RADICALS-RT, GETUG-AFU 17, RAVES + meta-analysis	Adjuvant vs early salvage RT (64–66 Gy)	Early salvage preferred ; 5-yr event-free 87–94% in both arms

- Benefit concentrates where residual disease is **local** (positive margin) and fades where failure is systemic (SVI, PSA never undetectable).
- **Trap:** very high-risk men (Gleason 8-10, pT3b) were only ~20% of the timing trials — adjuvant RT may still benefit them, and extensive or multiple positive margins justify it.
- Dose: **≥64 Gy** (dose response as in primary RT). **GETUG-AFU 16: 3 months** of AD with salvage RT improves 10-yr progression-free survival (PFS) 64% vs 49% (HR 0.54). Pelvic nodal salvage RT (SPPORT) and technique → *Ch 162*.

Adjuvant androgen deprivation and management of pN1

- **pT3N0:** adjuvant AD reduces progression but gives **no DSS or OS gain**.
- **ECOG 7887** (Eastern Cooperative Oncology Group; nodes at RP): immediate vs delayed AD → improved OS, DSS and progression-free survival.
- **AD + RT beats AD alone** in pN1: matched pairs 10-yr DSS 86% vs 70%, OS 74% vs 55%; observation vs AD vs AD + RT → AD + RT OS HR ~0.4-0.46, greatest in higher risk. AD alone ↑ other-cause mortality (HR 3.05).
- Adjuvant docetaxel (VA CSP #553, SPCG-12) or mitoxantrone (SWOG S9921) after RP: no benefit.

ALGORITHM

pN1 after RP: favourable (single or few nodes, Gleason ≤8) → observation with PSA monitoring is reasonable → adverse features (>3 nodes, Gleason >8) → AD + RT (best OS) → avoid AD alone where fit.

5 · RT + ADT

- Monotherapy external beam RT (EBRT) or brachytherapy is inadequate; the standard is **dose-escalated RT (>72, ideally ~78-81 Gy) + ADT**, often with pelvic RT. Technique, fractionation and brachytherapy → *Ch 162*.
- **Why ADT before RT works but not before RP:** with RT it shrinks the target and is synergistic with radiation (↓ DNA repair); before RP it only downstages clinically. Canadian three-arm trial: positive 24-month biopsy 65% (RT alone) → 28% (+3 mo NAD) → 5% (+ neoadjuvant, concurrent and adjuvant AD), supporting prolonged AD.

Trial	Question	Verdict
RTOG 86-10 (Radiation Therapy Oncology Group)	4 mo neoadjuvant/concurrent AD, bulky T2-T4	↓ metastases, disease-specific mortality 36% → 23%; OS unchanged
TROG 96.01	RT ± 3 or 6 mo NAD	6 mo ↓ all-cause mortality (HR 0.63); 3 mo did not
RTOG 85-31	Indefinite adjuvant goserelin	Better control, MFS; OS gain only Gleason 8-10
EORTC 22863	3 yr goserelin from start of RT, mainly T3-T4	OS 62% → 78-79% at 5 yr; local control 97% vs 79%
RTOG 92-02	4 vs 28 mo AD	Better all endpoints except OS; Gleason 8-10 OS 81% vs 71%
18 vs 36 mo (PCS IV)	Duration, high risk	OS comparable; 18 mo better quality of life
RTOG 94-13	Pelvic vs prostate RT × neoadjuvant vs adjuvant 4 mo AD	Early PFS gain lost; whole pelvis + adjuvant AD had worst OS

- **Duration:** intermediate risk 4-6 mo; high risk prolonged — NCCN **18-36 mo** with EBRT, 12-36 mo with brachytherapy boost.
- **Brachytherapy boost:** ASCENDE-RT (12 mo AD + pelvic RT 46 Gy) — low-dose-rate boost halved biochemical recurrence vs EBRT boost (HR 2.04; 9-yr 85% vs 65%); OS equal. Gleason 9-10: EBRT + boost ↓ cancer-specific mortality (HR ~0.4) vs RP or EBRT alone.

- **Trap:** benefits of AD + RT over RT alone often exclude OS — OS gain is seen with long AD in high-risk cohorts (EORTC 22863; Gleason 8-10 subsets).

Adding RT to ADT

- **SPCG-7/SFUO-3** (78% cT3; total AD → continuous flutamide ± RT): cancer death relative risk (RR) **0.44**, overall mortality RR 0.68; PSA recurrence 75% vs 26%.
- **NCIC CTG PR.3/MRC PR07** (lifelong AD ± 65-69 Gy, pelvic nodes treated): DSS HR 0.54, OS HR 0.77; 7-yr OS 66% → 74%.
- RT adds an absolute **8-10% OS** at 7-10 yr → ADT alone is inadequate for fit men; doses were below modern standard.

Systemic intensification

Trial	Added agent	Verdict
RTOG 0521	Docetaxel to long-term AD + RT (>half Gleason 9-10)	OS HR 0.69
GETUG 12	Docetaxel + estramustine to AD	Relapse-free 62% vs 50%
RTOG 9902	Paclitaxel + estramustine + etoposide	No benefit; thromboembolism
STAMPEDE (high-risk M0/N1)	Abiraterone + prednisolone 2 yr (± enzalutamide) to 3 yr AD + RT	MFS HR 0.53, OS HR 0.60

- STAMPEDE high-risk M0 eligibility: **pelvic nodes (N1), or two of Gleason ≥8, cT3-4, PSA ≥40 ng/mL**. RT was mandatory for N0 and encouraged for N1.
- **NCCN 2023: RT + ADT + 2 yr abiraterone for very high-risk disease.** Unresolved whether chemotherapy benefit needs high disease burden.

6 · ADT ALONE AND ITS TIMING

- **MRC** immediate vs deferred AD: fewer cancer deaths, mainly in M0 disease.
- Immediate vs deferred orchiectomy in men declining local therapy: OS equal; many deferred men never needed AD.
- **Bicalutamide 150 mg** (Early Prostate Cancer programme): better PFS in locally advanced disease, but **increased death in localised disease** (HR 1.23) — never use at low risk.
- Intermittent AD improves potency and quality of life; role in this setting investigational.
- AD alone does not prevent local progression: over half need TURP (transurethral resection of the prostate) for obstruction; ureteric obstruction and haematuria follow.

7 · OTHER OPTIONS AND COMPARISONS

- **Cryoablation** vs RT (cT2c-T3, randomised): equal DSS/OS but worse biochemical control (failure 87% vs 53%). High-intensity focused ultrasound: confounded series; NAD improves results.
- **RP vs RT + ADT:** no completed randomised comparison; **SPCG-15** (cT3N0M0) is testing it. Retrospective data conflict.
- Leaving the prostate untreated has its own cost: after RT alone for cT3, 10-yr local recurrence 24-74%, TURP for obstruction ~1/3, hydronephrosis 20%.
- Toxicity: AD + RT adds late gastrointestinal (GI)/genitourinary (GU) morbidity; adjuvant RT after RP worsens bowel and urinary function but not erections. RTOG 85-31 and 86-10: no rise in cardiovascular deaths.

ALGORITHM

Locally advanced / high-risk M0: stage (PSMA PET or CT + bone scan) → fit, resectable → RP + ePLND → adverse pathology: early salvage RT (adjuvant if pT3b/Gleason 8-10/margins+); pN1: observe or AD + RT. Alternative → dose-escalated RT (± brachytherapy boost, ± pelvic nodes) + 18-36 mo ADT → very high risk or N1: + 2 yr abiraterone → unfit for local therapy: ADT alone.

KEY POINT Viva: "Why not ADT alone for T3 disease?" — SPCG-7 and PR.3 show RT added to ADT halves cancer death and improves OS; AD alone neither cures nor controls the primary, and it carries its own toxicity.

8 • LANDMARK TRIALS

Trial	Design	Result	Take-home
SPCG-7	ADT ± RT, mostly T3	Cancer death RR 0.44	Add RT to ADT
NCIC PR.3/MRC PR07	Lifelong ADT ± RT	OS HR 0.77	Add RT to ADT
EORTC 22863	RT ± 3 yr goserelin	5-yr OS 62% → 79%	Long ADT in high risk
RTOG 92-02	4 vs 28 mo ADT	OS gain Gleason 8-10	Prolong ADT for high grade
STAMPEDE (abiraterone)	ADT + RT ± 2 yr abiraterone	MFS HR 0.53, OS HR 0.60	Standard for very high risk/N1
SWOG 8794	Adjuvant RT vs observation	MFS, OS improved	Postoperative RT works
RADICALS-RT / RAVES / GETUG-AFU 17	Adjuvant vs early salvage	No difference	Early salvage for most
ECOG 7887	pN+ immediate vs delayed AD	OS improved	Treat pN+ early
proPSMA	PSMA PET vs conventional	Accuracy 92% vs 65%	PSMA PET for staging

9 • MCQ ONE-LINERS

- Locally advanced = T3-4, N0/N1, M0.
- Clinical overstaging of cT3 occurs in 7-27%.
- Strongest pathological predictor of progression after RP: positive surgical margin.
- Kattan RP nomogram does not apply to cT3b or node-positive disease.
- CAPRA: each 2-point rise doubles recurrence risk.
- PSMA PET nodal specificity ≥95%, sensitivity only ~40-66%.
- proPSMA: PSMA PET accuracy 92% vs 65% for conventional imaging.
- NAD before RP: clinical but rarely pathological downstaging; no survival gain.
- NAD does not reduce SVI or nodal positivity.
- EORTC 22911: adjuvant RT benefit only with positive margins.
- SWOG 8794: only adjuvant RT trial showing OS benefit.
- Postoperative RT dose ≥64 Gy.
- GETUG-AFU 16: 3 months AD with salvage RT improves PFS.
- Adjuvant AD in pT3N0: no survival benefit.
- pN1: AD + RT beats AD alone; AD alone raises other-cause mortality.
- RTOG 86-10: 4 months AD reduced disease-specific mortality, not OS.
- TROG 96.01: 6 months, not 3, reduced all-cause mortality.
- EORTC 22863: 3 years goserelin improved OS.
- RTOG 94-13: whole pelvis + adjuvant AD arm had worst OS.
- NCCN ADT with EBRT for high risk: 18-36 months.
- SPCG-7: adding RT to ADT more than halves cancer death.
- RTOG 0521: docetaxel added to RT + ADT, OS HR 0.69.
- STAMPEDE high-risk M0: N1, or two of Gleason ≥8, cT3-4, PSA ≥40.
- Bicalutamide 150 mg increased mortality in localised disease.
- Cryoablation vs RT: equal survival, worse biochemical control.
- SPCG-15 compares RP with RT + ADT in cT3N0.

165 Management of Recurrent and Newly Metastatic Prostate Cancer

Reading time ≈ 45 min (slow pace)

1 · BIOCHEMICAL RECURRENCE: CONCEPT AND DEFINITIONS

Biochemical recurrence (BCR) is a rise in prostate-specific antigen (PSA) after curative-intent therapy that precedes clinical recurrence (disease visible on imaging). The governing asymmetry: not every detectable PSA leads to clinical recurrence (residual benign tissue, non-prostatic PSA, indolent residual cancer), but clinical recurrence is almost always preceded by a rising PSA. Hence the search for a threshold early enough to allow effective salvage yet specific enough to avoid overtreatment.

- After radical prostatectomy (RP): $\frac{1}{4}$ – $\frac{2}{5}$ develop BCR within 10 years; only about half of intermediate/high-risk men with BCR ever fail clinically.
- Memorial Sloan Kettering (MSK): PSA **≥ 0.4 ng/mL followed by a further rise** best predicted metastasis and secondary therapy.
- **Ultrasensitive PSA** (to 0.01 ng/mL) is prognostic, not diagnostic; useful mainly in pT3/high-risk disease. About half of early (1–3 month) detectable values fall on retesting → repeat, do not treat.
- Context matters: 0.1 ng/mL 6 months after grade group (GG) 5 $\neq$ 0.1 ng/mL 8 years after GG 1.

Setting	Definition	Logic
Post-RP — American Urological Association (AUA)	PSA ≥ 0.2 ng/mL	No gland → any PSA suspect
Post-RP — National Comprehensive Cancer Network (NCCN)	Failure to reach undetectable, two consecutive rises, or PSA ≥ 0.1	Any of three triggers
Post-RP — European Association of Urology (EAU)	No threshold; risk groups (below)	Kinetics + grade
After external beam radiotherapy (EBRT) ± androgen deprivation therapy (ADT)	Phoenix: nadir + 2 ng/mL	Remnant gland keeps making PSA; validated against metastasis and death
After low-dose-rate (LDR) brachytherapy	≥ 0.2 may suit better; < 0.2 at 4 years ≈ cure	$\geq 2\times$ EBRT dose → very low nadir; Phoenix may flatter brachytherapy

EAU BCR risk group	After RP	After radiotherapy (RT)
Low risk (needs BOTH)	PSA doubling time (PSADT) ≥ 1 year AND GG < 4	Interval to failure ≥ 18 months AND GG < 4
High risk (EITHER)	PSADT ≤ 1 year OR GG 4–5	Interval ≤ 18 months OR GG 4–5

KEY POINT PSA bounce after RT: transient rise of 0.1–0.5 ng/mL at a mean ≈ **9 months** (post-radiation prostatitis, delayed cell death), then a fall. It is **favourable** (better BCR-free survival) → monitor closely; never salvage on a bounce.

2 · RISK PREDICTION AND NATURAL HISTORY

- Predictors after RP: pre-treatment PSA, stage, GG, margin status, PSADT. **Positive margin** is the only modifiable factor (pT3a margin-negative did better than pT2 margin-positive); margin length (per mm) and Gleason ≥ 4 at the margin independently predict BCR → meticulous dissection.
- Genomic classifiers (Decipher, Genomic Prostate Score, Prolaris) predict metastasis but are unproven for selecting early salvage.
- **Nomograms**: biopsy-based before treatment, pathology-based after. MSK pre-op (PSA, stage, Gleason, biopsy cores; concordance 0.76–0.79) and post-op (adds extracapsular extension, seminal vesicle, nodes, margins, adjuvant RT; 10-year BCR-free survival, concordance 0.79–0.81) — only

MSK tools reach 15-year cancer-specific survival. **CAPRA** (Cancer of the Prostate Risk Assessment) = pre-op; **CAPRA-S** = post-op ("surgery") [MCQ]. **Salvage RT nomogram** (PSA, pathological Gleason, PSADT, margins, nodes, ADT): concordance 0.69; in GG 3-5, a positive margin >3 mm roughly halves survival.

- **Pound series** (untreated post-RP BCR): median **8 years** BCR → metastasis, then **5 years** metastasis → death; predictors: time to BCR <2 years, PSADT <10 months, Gleason 8-10. Most men with BCR do not die of prostate cancer.
- After Phoenix failure: median ≈5.4 years to metastasis, ≈10.5 years to cancer death. Predictors: short PSADT, stage, Gleason, short interval to BCR.
- PSA follow-up is **lifelong** (¼ recur after 5 years).

3 · IMAGING AT RECURRENCE

Purpose: **exclude metastasis before further local therapy**, because irradiating or operating on the pelvis in metastatic disease is risk without benefit. Then localise local recurrence.

- Bone scan and computed tomography (CT): rarely positive below PSA **10 ng/mL**.
- Positron emission tomography (PET)/CT tracers: choline; ¹⁸F-fluciclovine (**EMPIRE-1**: fluciclovine-guided salvage improved 3-year control); **prostate-specific membrane antigen (PSMA)** PET (⁶⁸Ga-PSMA-11, ¹⁸F-piflufolastat) — highest detection: ≈½ positive at PSA **<0.5**, ≈100% above 2. Detection rises with PSA for every tracer.
- ⁶⁸Ga-PSMA PET finds disease outside the planned salvage field in ≈1 in 5 at low PSA.
- **Multiparametric magnetic resonance imaging (mpMRI)** beats PET for **local** recurrence, especially at low PSA; after RT, T2 is poor, so dynamic contrast-enhanced and diffusion-weighted sequences carry the diagnosis.
- **Biopsy after RT** is mandatory before local salvage (MRI targets → seminal vesicles → systematic).

KEY POINT Two curves run in opposite directions: salvage RT works best at the lowest PSA, but imaging detection and certainty of true recurrence improve only as PSA rises. Risk features (grade, margins, kinetics) let us act early without overtreating.

4 · SALVAGE AFTER RADICAL PROSTATECTOMY

Salvage radiotherapy (SRT) to the prostate bed is the only curative-intent option for post-RP BCR. About ⅓-½ remain BCR-free at 4-5 years. Cancer-specific survival benefit was confined to men treated within 2 years of BCR with PSADT <6 months (highest metastatic risk).

- **Timing**: lower PSA = better; each 0.1 ng/mL costs ≈2.6% recurrence-free survival (≈10% per 0.1 with ≥2 adverse features). **Early salvage at PSA 0.1-0.2** equals adjuvant RT and spares ≥60% from RT (ARTISTIC).
- **Dose**: **≥64-65 Gy** to the fossa (guidelines); 70 Gy adds nothing (SAKK 09/10). Pelvic nodes 40-45 Gy. Intensity-modulated RT (IMRT) standard.
- **Field + hormones**: SPPORT shows ADT and pelvic nodal RT are additive.
- No visible lesion on mpMRI predicts SRT failure (occult metastasis). Continue PSA surveillance after SRT.

5 · SALVAGE AFTER RADIOTHERAPY

Prerequisites for every option: biopsy-proven local recurrence, imaging-confirmed non-metastatic disease, excellent performance status, life expectancy ≥5-10 years and shared decision — morbidity in irradiated tissue is high and control lower than primary therapy. Every modality gives ≈**50-70%** control, so selection (local-only disease, low PSA, GG <4) predicts success more than modality choice.

Option	Selection / features	Signature morbidity
Salvage RP (open or robotic)	EAU: life expectancy ≥ 10 years, PSA ≤ 10 , biopsy GG ≤ 3 , no nodes/metastases on imaging, initial cT1-T2; gives pathological staging. Poor if seminal vesicle or nodal involvement	Incontinence, bladder neck contracture, rectal injury, near-universal impotence (robotic: fewer)
Salvage brachytherapy (LDR $\approx$ high-dose-rate)	Only after EBRT; spares irradiated neighbours	Grade ≥ 3 urinary toxicity ≈ 1 in 6
Salvage stereotactic body RT (SBRT)	After EBRT	Low toxicity, modest control
Salvage cryotherapy	PSA < 10 (ideally < 4), PSADT ≥ 16 months; focal or whole-gland; post-cryo nadir < 0.2 predicts success	Urethrorectal fistula (now 1–3%), erectile dysfunction; incontinence now single-digit thanks to urethral warming
Salvage high-intensity focused ultrasound (HIFU)	≥ 2 years after EBRT	Fistula, osteitis pubis, stricture, incontinence

Fistula is the ablative signature, incontinence the surgical one. Recurrence after **focal therapy** is worked up like radiation failure (PET $\rightarrow$ MRI $\rightarrow$ targeted biopsy $\rightarrow$ any salvage).

6 · NON-CURATIVE APPROACH TO BCR: OBSERVATION VS ADT

Observation preserves quality of life; ADT delays metastasis but costs toxicity and drives castration resistance. Most men with BCR never die of it.

- AUA: observation (or trial) over hormones after maximal local therapy; ADT only for higher metastatic risk — PSADT ≤ 12 months or high grade; intermittent preferred. EAU: observe EAU low-risk BCR, limited life expectancy or men declining local therapy. Surveillance PET/CT, especially if PSADT < 12 months.
- Primary ADT for localised disease (no local therapy) is **not** recommended; bicalutamide 150 mg in low-risk localised disease worsened overall survival (OS) (Early Prostate Cancer programme).
- **Messing (ECOG) trial**: immediate ADT after RP for node-positive (pN+) disease improved OS and progression-free survival (PFS) — but it was small, pre-dated modern imaging, had bulky nodes (up to 20), a fifth had persistent PSA, and the delayed arm was treated only at clinical progression. Low-volume pN+ with undetectable PSA: **20–40%** cured by surgery alone.
- MRC (locally advanced/asymptomatic metastatic): immediate ADT reduced progression-related symptoms; OS gain faded with follow-up.
- Early ADT helps only in proportion to risk: TOAD (rising PSA) OS hazard ratio (HR) 0.55; EORTC 30891 benefit concentrated in PSA > 50 or PSADT < 12 months.
- **EMBARK** — high-risk BCR (PSADT ≤ 9 months, PSA ≥ 1 after RP or ≥ 2 above nadir after RT): leuprolide vs leuprolide + enzalutamide vs enzalutamide; metastasis-free survival (MFS) 71.4% / 87.3% / 80.0% (combination HR 0.42) $\rightarrow$ enzalutamide approved for **high-risk non-metastatic hormone-sensitive prostate cancer (nmHSPC)**.

Intermittent vs continuous ADT

- **NCIC PR.7** (rising PSA after RT): intermittent **non-inferior** for OS (8.8 vs 9.1 years), better hot flashes, libido, urinary symptoms $\rightarrow$ acceptable.
- **SWOG 9346** (new metastatic): OS 5.1 vs 5.8 years, HR 1.10 (90% confidence interval 0.99–1.23) — **inconclusive** (upper limit exceeded the 1.20 margin) $\rightarrow$ intermittent may be worse when metastatic.
- Modern metastatic care = continuous combination therapy; intermittent reserved for men unsuited to combinations.

7 · ANDROGEN DEPRIVATION THERAPY: BIOLOGY AND AGENTS

The androgen receptor (AR) is the oncogenic driver. Castration removes the ligand, not the receptor: castration-resistant prostate cancer (CRPC) keeps AR signalling alive by AR amplification (20-30%), ligand-binding-domain mutations (antagonist becomes agonist), splice variants lacking the ligand-binding domain (**AR-V7** — poor prognosis), intratumoural steroidogenesis and bypass (glucocorticoid receptor). Testosterone → dihydrotestosterone (DHT, $\approx 13\times$ more avid) via 5 α -reductase.

Castrate testosterone: ≤ 50 ng/dL.

Method	Key facts
Bilateral orchiectomy (simple or subcapsular)	$\approx 95\%$ of testosterone is testicular; castrate within hours (median ≈ 15 ng/dL). Fewer cardiac events, fractures and peripheral arterial disease than agonists, equal survival → offer for lifelong ADT.
Luteinising hormone-releasing hormone (LHRH) agonists: leuprolide, goserelin, triptorelin, histrelin	Initial luteinising hormone (LH) surge (up to $10\times$, 10-20 days) → receptor desensitisation → $\downarrow$ LH → $\downarrow$ testosterone. Follicle-stimulating hormone (FSH) rises. Survival = orchiectomy
LHRH antagonists: degarelix, relugolix (oral)	Immediate blockade, no flare , FSH falls. Degarelix: castrate by day 3 in $>95\%$. HERO (relugolix vs leuprolide): castrate by day 4 56% vs 0%, sustained castration 97% vs 89%, faster testosterone recovery after stopping
Steroidal antiandrogen: cyproterone acetate	AR blockade + progestational central suppression (testosterone $\downarrow 70-80\%$). Inferior to goserelin; severe cardiovascular events (up to 10%), gynaecomastia, rare fulminant hepatotoxicity → no longer used for cancer; 50-100 mg/day still used for hot flushes
Non-steroidal antiandrogens: flutamide (diarrhoea), nilutamide (delayed dark adaptation, pneumonitis), bicalutamide	Alone raise LH and testosterone → gynaecomastia/mastodynia; all hepatotoxic. Role: combined androgen blockade (CAB) and flare prevention . Monotherapy not approved
AR-targeting agents (ARTAs): enzalutamide 160 mg, apalutamide 240 mg, darolutamide 600 mg twice daily	Block binding, nuclear translocation, DNA binding. Seizures → enzalutamide; rash, hypothyroidism → apalutamide; darolutamide fewest effects
Androgen synthesis inhibitors: abiraterone (selective CYP17), ketoconazole (non-selective)	Abiraterone $\downarrow$ cortisol → $\uparrow$ adrenocorticotrophic hormone (ACTH) → mineralocorticoid excess (hypokalaemia, hypertension, oedema) → give with prednisone. Ketoconazole: castrate within 4 h — emergency option

KEY POINT Flare matters almost only with spinal or weight-bearing bone metastases: give an antiandrogen with (or before) the agonist for **21-28 days**; simultaneous start is as good as a week's lead-in. For **impending cord compression** or severe bone pain choose an **antagonist or orchiectomy** (or ketoconazole).

Antiandrogen withdrawal: stopping bicalutamide on CAB at progression gives $>50\%$ PSA fall in 15-30% (mutant AR now activated by the drug); 0-10% with ARTAs; no OS gain.

8 · ADT SIDE EFFECTS AND THEIR MANAGEMENT

Effect	Key facts	Management
Osteoporosis / fracture	Loss of peripheral oestrogen; fracture risk $\uparrow 20-45\%$, rises with duration	Baseline hip dual-energy x-ray absorptiometry (DEXA) + FRAX; all: calcium 1000-1200 mg + vitamin D 400-1000 international units (IU)/day, weight-bearing exercise, stop smoking. Drug if T-score -1 to -2.5 with 10-year hip fracture $\geq 3\%$ or major fracture $\geq 20\%$ (or osteoporosis): denosumab 60 mg SC 6-monthly, zoledronic acid 4-5 mg IV yearly or alendronate 70 mg weekly; repeat DEXA at 1 year

Effect	Key facts	Management
Hot flushes (commonest, 50–80%)	Hypothalamic oestrogen loss	Treat if bothersome: oestrogens most effective (thrombosis, gynaecomastia); megestrol, cyproterone, medroxyprogesterone; venlafaxine (taper), gabapentin. Clonidine ineffective
Sexual dysfunction	Libido loss worst; penile shortening	Phosphodiesterase-5 inhibitors usually fail; vacuum, injections, prosthesis; couple therapy
Metabolic	Fat ↑, sarcopenia; atypical metabolic syndrome (subcutaneous fat, raised high-density lipoprotein); diabetes ↑ (agonist HR 1.44)	Exercise; monitor glucose and lipids
Cardiovascular	Agonists ↑ coronary disease, myocardial infarction, sudden death, thromboembolism	Risk-factor control; orchiectomy/antagonist favourable
Anaemia	Normocytic normochromic, very common (most on CAB lose ≥10% haemoglobin, within 6 months), even without marrow disease; pre-ADT haemoglobin <12 g/dL predicts shorter survival	Check blood count at follow-up; usually asymptomatic; reversible within ≈1 year of stopping
Cognitive function	Memory and visuospatial decline (verbal skills, mood spared); evidence conflicting; ADT–dementia link methodologically weak	Monitor; refer if decline
Gynaecomastia	≈½ with bicalutamide 150 mg or enzalutamide; ≤5% with castration	Prophylactic breast RT 10–15 Gy (useless once established), tamoxifen; established → liposuction/mastectomy

Follow-up on ADT

Review 3–6-monthly (PSA, blood count, chemistry). Faster fall to undetectable predicts survival.

Rising PSA → check testosterone first: inadequate suppression is a drug problem (switch agent or orchiectomy); rising PSA with castrate testosterone = **CRPC** → *Ch 166* → restage (soft tissue + bone). Germline *HSD3B1* (1245A>C) variant predicts ADT resistance **[MCQ]**.

9 • METASTATIC HORMONE-SENSITIVE PROSTATE CANCER (MHSPC)

- **Synchronous** (de novo at diagnosis) vs **metachronous** (after local therapy). Metastasis arising on ADT = CRPC.
- Prognosis: metachronous low volume best (median OS ≈92 vs ≈43 months synchronous/high volume); nodes only > bone > visceral > bone + visceral. **SWOG 9346 PSA nadir at 7 months:** ≤0.2 → median OS **75** months; 0.2–4 → 44; >4 → 13.
- Trials used conventional imaging; PET-only metastases are largely unstudied.
- **Germline and somatic testing for all metastatic men:** >10% carry actionable DNA-repair mutations (germline *BRCA1/2*, *ATM*, *PALB2*, *CHEK2*, *HOXB13*, mismatch-repair genes; somatic also *CDK12*, microsatellite instability-high [MSI-H]).

Definition	Criteria	Use
CHAARTED high volume	≥4 bone metastases with ≥1 outside the axial skeleton , OR any visceral metastasis (no grade criterion). Low volume = not high volume	Docetaxel benefit subgroup; RT to primary in low volume

Definition	Criteria	Use
LATITUDE high risk	≥2 of: Gleason ≥8, ≥3 bone lesions, visceral metastasis	Entry for abiraterone

Treatment principles

- **ADT is the floor, never the ceiling:** backbone of orchiectomy or LHRH agonist/antagonist **plus** intensification (docetaxel, abiraterone or an ARTA) unless contraindicated. Antiandrogen monotherapy and long-term CAB not recommended.
- Docetaxel **75 mg/m² 3-weekly × 6**; meta-analysis HR 0.77; neutropenia; benefit greatest in high volume.
- Abiraterone 1000 mg + prednisone 5 mg; enzalutamide or apalutamide — benefit regardless of volume or prior docetaxel.
- **Triplet** (ADT + docetaxel + abiraterone or darolutamide) for chemotherapy-fit men, especially high-volume and synchronous disease.
- Observation only if asymptomatic with life expectancy <5 years, or during metastasis-directed therapy.
- Follow-up 3-6-monthly: PSA, bloods, testosterone with any PSA rise; bone and cardiovascular health.

Local therapy to the primary

The untreated primary may keep seeding metastases. **STAMPEDE arm H:** prostate RT added to ADT — no OS gain overall (HR 0.92) but benefit in pre-specified **low-volume** disease (HR 0.68; 3-year OS 81% vs 73%) → **RT to the prostate for low-volume synchronous mHSPC**. **HORRAD** (ADT ± prostate RT): no OS difference (HR 0.9); suggested benefit with <5 bone lesions (HR 0.68, not significant) — hypothesis-generating; STAMPEDE then confirmed the low-volume signal. PEACE-1 also tested prostate RT. Caveat: men had ADT alone; benefit on top of modern combinations unknown.

Cytoreductive prostatectomy: feasible with low morbidity and good continence, but no randomised survival evidence → only within trials or to palliate local symptoms.

Metastasis-directed therapy (MDT)

Oligometastatic disease (**≤3 sites** on PET) can be treated by SBRT or surgery (salvage lymph node dissection) to defer systemic therapy. **STOMP** (choline PET; surgery/SBRT vs observation): ADT-free survival 21 vs 13 months. **ORIOLE** (SBRT vs observation): progression at 6 months 19% vs 61%. No OS data → for motivated men wishing to avoid lifelong ADT.

10 · LANDMARK TRIALS

Trial	Design	Result	Take-home
RADICALS, RAVES, GETUG-AFU 17 (ARTISTIC)	Adjuvant vs early salvage RT (PSA 0.1–0.2), high-risk pathology	Equivalent control; ≥60% spared RT	Early salvage preferred
Radiation Therapy Oncology Group (RTOG) 0534 (SPPORT)	Bed vs bed + 4–6 months ADT vs pelvis + ADT	5-year freedom from progression 70.9 / 81.3 / 87.4%	Add ADT and pelvic nodes (endpoint not OS)
RTOG 9601	SRT ± 2 years bicalutamide 150 mg	12-year OS 76.3 vs 71.3%; fewer cancer deaths	OS benefit
GETUG-AFU 16	SRT ± 6 months goserelin	10-year BCR-free 64 vs 49% (HR 0.54); MFS HR 0.73; OS unchanged	Short ADT: no OS gain
EMBARK	High-risk BCR; leuprolide ± enzalutamide vs enzalutamide	MFS 87.3% (combination) / 80.0% (enzalutamide) / 71.4% (leuprolide); combination HR 0.42, enzalutamide alone HR 0.63	Enzalutamide for high-risk nmHSPC

Trial	Design	Result	Take-home
CHAARTED (ECOG 3805)	ADT ± docetaxel	OS 57.6 vs 44.0 months (HR 0.61); high volume 49.2 vs 32.2 months	Benefit in high volume only
GETUG-AFU 15	ADT ± docetaxel (up to 9 cycles)	OS HR 0.88 (0.68–1.14), not significant; better PFS	The negative docetaxel trial
STAMPEDE (docetaxel; arm G abiraterone)	Multi-arm; metastatic + high-risk non-metastatic	Docetaxel OS HR 0.78 (metastatic 0.76); abiraterone (arm G) OS HR 0.63, 3-year OS 83 vs 76%, failure-free survival HR 0.29	Both work
LATITUDE	High-risk mHSPC; ADT ± abiraterone	OS HR 0.62; radiographic progression-free survival (rPFS) 33.0 vs 14.8 months; OS not reached vs 34.7 months	Abiraterone in high risk
ARCHES	ADT ± enzalutamide	rPFS HR 0.39	Volume-independent
ENZAMET	ADT + enzalutamide vs ADT + older antiandrogen	OS HR 0.67	Control was not placebo
TITAN	ADT ± apalutamide	rPFS HR 0.48; OS HR 0.67	Volume-independent
PEACE-1	Synchronous; 2×2 ± abiraterone ± prostate RT on standard care	All: rPFS HR 0.54, OS HR 0.82; docetaxel-treated: rPFS HR 0.50, OS HR 0.75	Triplet with abiraterone
ARASENS	ADT + docetaxel ± darolutamide	OS HR 0.68	Darolutamide in mHSPC only with docetaxel

11 • MCQ ONE-LINERS

1. AUA BCR after RP: PSA ≥ 0.2 ng/mL; NCCN uses ≥ 0.1 .
2. Phoenix definition after EBRT: PSA nadir + 2 ng/mL.
3. PSA < 0.2 ng/mL 4 years after LDR brachytherapy ≈ cure.
4. EAU low-risk BCR after RP: PSADT > 1 year AND GG < 4 .
5. EAU low-risk BCR after RT: interval to failure > 18 months AND GG < 4 .
6. Only modifiable predictor of BCR after RP: positive surgical margin.
7. Pound: median 8 years BCR → metastasis, 5 years metastasis → death.
8. PSMA PET detects ≈ half of recurrences at PSA < 0.5 ng/mL.
9. Minimum salvage RT dose to prostate bed: 64–65 Gy.
0. RTOG 9601: 2 years bicalutamide 150 mg with SRT improved OS.
1. Salvage cryotherapy selection: PSA < 10 , PSADT ≥ 16 months.
2. EAU salvage RP: PSA < 10 , GG ≤ 3 , life expectancy ≥ 10 years, initial cT1–T2.
3. EMBARK high-risk BCR: PSADT ≤ 9 months.
4. Castrate testosterone level: < 50 ng/dL.
5. LHRH agonist raises FSH; antagonist lowers FSH and causes no flare.
6. Flare prophylaxis: antiandrogen for 21–28 days.
7. Impending cord compression in hormone-naïve man: antagonist or orchiectomy.
8. Apalutamide: rash and hypothyroidism; enzalutamide: seizures.
9. Abiraterone needs prednisone to suppress ACTH-driven mineralocorticoid excess.
0. Commonest ADT side effect: hot flushes; most effective treatment: oestrogens.
1. SWOG 9346: intermittent ADT non-inferiority unproven in metastatic disease.
2. CHAARTED high volume: ≥ 4 bone metastases (≥ 1 appendicular) or visceral.
3. LATITUDE high risk: ≥ 2 of Gleason ≥ 8 , ≥ 3 bone lesions, visceral.
4. Negative docetaxel mHSPC trial: GETUG-AFU 15.
5. Messing trial: immediate ADT for pN+ after RP improved OS, but nodes were bulky and imaging dated.
6. HORRAD: prostate RT in mHSPC — no overall OS benefit; low-volume signal later confirmed by STAMPEDE.

- | | |
|--|---|
| <p>!7. Cyproterone acetate: steroidal antiandrogen; now used 50–100 mg/day for hot flushes.</p> <p>!8. SWOG 9346: PSA nadir ≤ 0.2 ng/mL at 7 months → median OS 75 months.</p> | <p>29. Darolutamide in mHSPC is approved only with docetaxel (ARASENS).</p> <p>30. STOMP and ORIOLE: MDT delays ADT/progression in oligometastases; no OS data.</p> |
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166 Management of Castration-Resistant Prostate Cancer

Reading time ≈ 38 min (slow pace)

1 · DEFINITION AND BIOLOGY

Castration-resistant prostate cancer (CRPC) = disease progression (biochemical, radiographic or clinical) despite castrate serum testosterone **≤50 ng/dL** on androgen deprivation therapy (ADT). Almost every man on ADT eventually progresses. Historically, with ADT alone in metastatic disease, progression-free survival (PFS) was 12–20 months and overall survival (OS) 24–36 months.

- **Castration resistance is not androgen independence:** the androgen receptor (AR) remains the driver through AR overexpression/amplification, gene and enhancer amplification, ligand-binding-domain mutations (activated by other steroids), splice variants lacking the ligand-binding domain (e.g. **AR-V7**), adrenal androgens (up to 10%) and intratumoural androgen synthesis (CYP17 overexpressed). This is the rationale for next-generation AR-pathway inhibitors (ARPIs).
- Once AR dependence is lost, therapy moves to non-AR targets: microtubules (taxanes), bone (radium-223), prostate-specific membrane antigen (PSMA), DNA repair (poly-ADP-ribose polymerase inhibitors, PARPIs) and immunity.
- **Lineage plasticity** (neuroendocrine/anaplastic change) produces AR-indifferent disease with low PSA (prostate-specific antigen).

2 · CLINICAL STATES AND PROGRESSION CRITERIA

Tumour-node-metastasis (TNM) staging is purely anatomic and does not guide advanced disease. The **clinical states model** (Prostate Cancer Clinical Trials Working Group 3, **PCWG3**, 2016) classifies men by response to prior therapy: localised → rising PSA (non-castrate) → non-castrate metastases or **non-metastatic CRPC (nmCRPC)** → **metastatic CRPC (mCRPC)** first line → second line → later lines.

- Baseline evaluation: extent of disease; mode of progression (PSA only, new bone, nodal or visceral lesions); symptoms (bone pain); response to prior hormonal therapy. Prognostic factors: performance status, pain, haemoglobin, PSA, alkaline phosphatase, lactate dehydrogenase (LDH), extent of bone disease, visceral disease.
- † PCWG3 PSA progression: rise ≥25% and ≥2 ng/mL above nadir, confirmed ≥3 weeks later.
- **PSA doubling time (PSADT)** predicts the pace: PSADT **≤3 months** → rapid course → consider aggressive therapy.
- Radiographic progression without PSA progression is now common because ARPIs are used earlier (ARCHES: up to ⅔ of radiographic progressors) → PSA alone is insufficient; image periodically.
- Check testosterone if non-compliance is suspected or the prior regimen may not sustain castration.

Modality	Pitfall	PCWG3 rule
PSA	Chemotherapy may cause a rise before a fall; little change with sipuleucel-T and radium-223; flare with radium-223 and ¹⁷⁷ Lu-PSMA	Confirm rise; declines on abiraterone/enzalutamide predict outcome
Bone scan	New lesions early may be flare of healing disease; rarely shows response	Rescan at 12–16 weeks as new baseline; progression = ≥2 new lesions ("2 + 2" rule)
Computed tomography (CT)	Soft tissue is a small fraction of (mostly bone) burden	RECIST (Response Evaluation Criteria in Solid Tumours); PFS preferred over response rate
PSMA positron emission tomography (PET)	Apparent progression at 4 weeks resolving by 3 months in ≈13%	Response criteria not yet defined

KEY POINT Why the rules exist: every marker (PSA, bone scan, PSMA PET) can flare early, so progression is called on confirmed, combined evidence — otherwise effective therapy is abandoned too soon.

General principles

- **Continue ADT indefinitely** (luteinising hormone-releasing hormone [LHRH] agonist/antagonist) during every later therapy: stopping may worsen survival; exogenous testosterone can cause a severe flare. All CRPC trials mandated it.
- **Antiandrogen withdrawal:** if progressing on combined androgen blockade, stop the first-generation antiandrogen (e.g. bicalutamide) and follow PSA for 4–8 weeks — a short-lived PSA fall in a minority.
- Limited, asymptomatic disease suits sequential hormonal therapy; **symptomatic bone pain or visceral-predominant disease → upfront chemotherapy.**
- Rapid progression with relatively low PSA → biopsy a metastasis to exclude neuroendocrine change.

3 · GERMLINE AND SOMATIC TESTING

- **Germline testing** for: all metastatic prostate cancer; family history of *BRCA1/2*-, Lynch- or *HOXB13*-associated cancers; Ashkenazi ancestry.
- **Somatic** next-generation sequencing: tumour tissue is standard (usually the archival primary, as bone biopsy is difficult). **Liquid biopsy** (circulating tumour DNA) is an alternative, but **clonal haematopoiesis of indeterminate potential (CHIP)** can give false DNA-repair calls → such men do not respond to PARPIs.
- Somatic homologous recombination repair (HRR) mutations in up to **20–25%** of mCRPC; mismatch repair deficiency (dMMR)/microsatellite instability-high (MSI-H) in **2–5%** → if found, genetic counselling for Lynch syndrome.

4 · NON-METASTATIC CRPC (NMCRPC)

nmCRPC = rising PSA on castration with no metastases on **conventional** imaging (CT, bone scan). It arose from stage migration (ADT started at first PSA rise) and is shrinking as PSMA PET reclassifies many men as metastatic. PSA kinetics (PSADT, velocity) are the strongest predictors of metastasis; baseline PSA >10 ng/mL also predicts. Median time to bone metastasis in unselected men ≈30 months.

Standard: an ARPI (apalutamide, enzalutamide or darolutamide) + continued ADT when PSADT ≤10 months — all three improve metastasis-free survival (MFS) and OS. Abiraterone is active (IMAAGEN) but **not approved** for nmCRPC.

5 · MCRPC: ANDROGEN-SIGNALLING INHIBITORS

Agent	Mechanism	Dose	Signature toxicity / notes
Abiraterone	Selective CYP17 inhibitor (17 α -hydroxylase + 17,20-lyase) → blocks adrenal and intratumoural androgen synthesis	1000 mg daily fasting + prednisone (10 mg/day in mCRPC trials)	Mineralocorticoid excess (hypokalaemia, hypertension, oedema), hepatotoxicity; caution in cardiovascular disease; correct potassium first. Food ↑ absorption up to 10× (250 mg with low-fat meal is an alternative)
Enzalutamide	AR antagonist, no agonist activity; blocks nuclear translocation and DNA binding	160 mg daily, no steroid	Seizures (≈1%) , fatigue, falls; CYP3A4/2C8 interactions
Apalutamide	As enzalutamide; does not cross blood–brain barrier	240 mg daily	Rash, hypothyroidism , falls/fractures

Agent	Mechanism	Dose	Signature toxicity / notes
Darolutamide	Structurally distinct; does not cross blood-brain barrier	600 mg twice daily	Least central nervous system toxicity

KEY POINT Why prednisone with abiraterone? CYP17 blockade lowers cortisol → adrenocorticotrophic hormone (ACTH) rises → excess mineralocorticoid precursors. Low-dose prednisone suppresses ACTH. Enzalutamide needs no steroid (prednisone may even activate a promiscuous AR). Class effects of ARPIs: fatigue, falls and fractures — caution in men >75 years.

6 • CYTOTOXIC CHEMOTHERAPY

Only taxanes extend life in mCRPC. **Docetaxel** binds β -tubulin → microtubule stabilisation → G2/M arrest and BCL2 phosphorylation; it also impairs microtubule transport of AR into the nucleus.

Cabazitaxel has poor affinity for the P-glycoprotein efflux pump → active in docetaxel-resistant disease (little cross-resistance).

Agent	Dose / line	Signature toxicity
Docetaxel	75 mg/m² 3-weekly + prednisone 5 mg twice daily; first-line chemotherapy	Neutropenia, fatigue, oedema, neuropathy , hyperlacrimation, nail dystrophy
Cabazitaxel	20 mg/m² (25 if low neutropenic risk) 3-weekly + prednisone + granulocyte colony-stimulating factor (G-CSF); post-docetaxel	Neutropenia (grade 3-4 82%), diarrhoea ; little neuropathy
Mitoxantrone	12 mg/m ² + prednisone; palliation only, taxane-refractory or frail	No survival benefit
Platinum (with etoposide or taxane)	Neuroendocrine/aggressive variant; DNA-repair-deficient (e.g. germline <i>BRCA2</i>)	—

Estramustine was abandoned (thromboembolism, no added activity). Adding anti-angiogenic or bone-targeted drugs to docetaxel never improved OS.

7 • RADIOPHARMACEUTICALS

Feature	Radium-223	¹⁷⁷ Lu-PSMA-617 (lutetium vipivotide tetraxetan)
Emitter / targeting	Alpha (path <100 μm → spares marrow); calcium mimetic, bone-seeking	Beta; PSMA-directed — needs PSMA PET-positive disease
Eligible	Symptomatic bone metastases, no visceral or bulky nodal disease	PSMA-avid metastases at any site, after ARPI and taxane
Schedule	50 kBq/kg IV 4-weekly × 6	6-weekly, up to 6 cycles
Toxicity	Diarrhoea, anaemia; fewer adverse events than placebo	Fatigue, dry mouth , nausea, anaemia
Caution	Never with abiraterone/prednisone (more deaths and fractures); give a bone agent with any ARPI	—

Older beta emitters (strontium-89, samarium-153) only palliate pain.

8 • POLY(ADP-RIBOSE) POLYMERASE (PARP) INHIBITORS

PARP repairs single-strand breaks. Inhibition (catalytic block plus **PARP trapping** on DNA) causes replication-fork collapse and double-strand breaks; cells lacking homologous recombination (e.g. *BRCA2* loss) cannot repair them and die — **synthetic lethality**. PARPIs are ineffective in unselected

men. AR maintains HRR gene expression and PARP1 promotes AR signalling, so **ARPI + PARPI** is synergistic.

- Benefit is greatest with *BRCA* (especially *BRCA2*), weak or absent with *ATM* (ATM-deficient tumours may suit ATR inhibitors instead).
- Approvals: olaparib monotherapy — broad HRR panel; rucaparib — *BRCA1/2* only; olaparib + abiraterone and niraparib + abiraterone — *BRCA*-mutated; talazoparib + enzalutamide — HRR-mutated.
- Class toxicity: nausea, fatigue, **anaemia** (watch for sudden falls); rarely myelodysplastic syndrome, pulmonary embolism.

9 • IMMUNOTHERAPY

- **Sipuleucel-T**: autologous dendritic-cell vaccine (apheresis; cells loaded with prostatic acid phosphatase-granulocyte-macrophage colony-stimulating factor (GM-CSF) fusion protein). First FDA (US Food and Drug Administration)-approved therapeutic cancer vaccine. For **asymptomatic/minimally symptomatic mCRPC without visceral metastases or opioid-requiring pain**. Prolongs OS **without** PSA or PFS change (delayed immune effect). Toxicity: infusion fever, rigors.
- Checkpoint inhibitors (ipilimumab, pembrolizumab ± enzalutamide, docetaxel or olaparib) fail in unselected mCRPC — prostate cancer is immunologically "cold".
- **Pembrolizumab** (anti-programmed cell death protein 1 [PD-1], 200 mg IV 3-weekly): tumour-agnostic approval for **MSI-H/dMMR** or tumour mutational burden **≥10 mutations/megabase**, after ≥1 prior line. Immune toxicity: colitis, hepatitis, endocrinopathy, pneumonitis.

10 • NEUROENDOCRINE/AGGRESSIVE-VARIANT CRPC

Feature	Usual mCRPC	Neuroendocrine variant
AR dependence / PSA	Retained; PSA tracks disease	Lost; PSA low or undetectable despite progression
Pattern	Osteoblastic bone ± nodes; visceral <20%	Bulky pelvic/retroperitoneal masses, liver, osteolytic lesions with hypercalcaemia (parathyroid hormone-related protein), brain metastases
Molecular	AR alterations	≥2 of <i>PTEN</i> loss, <i>TP53</i> mutation, <i>RB1</i> loss; low PSMA
Treatment	ARPI, taxane, radioligand, PARPI	Platinum + etoposide (or taxane + carboplatin) ± radiotherapy (RT)
Survival	Years	<12 months

Biopsy (small cell/poorly differentiated with neuroendocrine immunostaining) is strongly encouraged; chromogranin A is not needed for diagnosis.

11 • SEQUENCING AND CROSS-RESISTANCE

Pivotal mCRPC trials enrolled men progressing on ADT alone; today most reach CRPC already exposed to docetaxel and/or an ARPI given for hormone-sensitive disease. **Cross-resistance**: progression on one ARPI predicts resistance to the next — sequential ARPI helps only **15-30%**. The logic is therefore to **switch mechanism** (AR → taxane → radioligand/PARPI), not intensify the same one (CARD).

Prior (hormone-sensitive) therapy	Reasonable next option at mCRPC
Docetaxel, no ARPI	ARPI
ARPI, no chemotherapy	Docetaxel, radium-223 or sipuleucel-T; PARPI if HRR-mutated
Docetaxel + ARPI	Cabazitaxel, ¹⁷⁷ Lu-PSMA, radium-223; PARPI if HRR-mutated
MSI-H/dMMR any time after ≥1 line	Pembrolizumab

- Practical selectors: comorbidity (seizures, cardiovascular, liver disease, diabetes), drug interactions (enzalutamide induces CYP3A4/2C9/2C19 — anticoagulants, antiepileptics), cost and pill burden.
- **AR-V7** splice variant lacks the ligand-binding domain → ligand-independent AR activity; predicts poor survival. ↑ AR-V7 in circulating tumour cells predicts resistance to abiraterone/enzalutamide, not to taxanes.

12 • COMPLICATIONS AND PALLIATIVE CARE

Spinal cord compression — an emergency

Common because of vertebral spread; mostly from vertebral bodies, sometimes paravertebral soft tissue (neurological deficit with normal bone scan). Early treatment preserves walking and sphincter function. **Every focal back pain in a man with bone metastases is cord compression until proven otherwise.**

ALGORITHM

Back pain ± motor/sensory deficit or bladder/bowel change → **high-dose IV dexamethasone** (load 10 mg IV, then 4-10 mg 6-12-hourly; 16-100 mg/day) → **urgent whole-spine magnetic resonance imaging (MRI)** → definitive RT ± decompressive surgery → taper steroids over 2-3 weeks

- **Patchell (2005)**: decompressive surgery + RT beats RT alone. Operate if progression during RT, unstable pathological fracture, or recurrence after RT (weigh against prognosis).
- Hormone-naïve man: castrate rapidly with antagonist or orchiectomy → *Ch 165*.

Pain

- Focal bone pain → analgesia + **external-beam RT**; X-ray/CT painful weight-bearing sites to exclude lytic lesions or impending fracture (surgical stabilisation).
- Diffuse bone pain → analgesia, corticosteroids, bone agents, wide-field RT, radiopharmaceuticals, chemotherapy.
- Neuropathic pain/plexopathy → tricyclics (amitriptyline), gabapentinoids, nerve block — not opioids alone.
- Circumoral numbness → skull-base metastasis → irradiate.

Urinary obstruction and marrow failure

- Obstructive uropathy (primary or pelvic/retroperitoneal nodes) → infection, pain, renal impairment precluding renally cleared chemotherapy → **relieve before chemotherapy** with internal ureteric stent or percutaneous nephrostomy. Not all unilateral cases need drainage if the other kidney functions; nephrostomies burden home care; stents need regular change.
- Anaemia (chronic disease, marrow replacement, treatment): erythropoiesis-stimulating agents not recommended (may increase mortality).
- Hospice/end-of-life care when options are exhausted and performance declines — start it in time.

13 • BONE-TARGETED THERAPY

Two indications: (1) ADT-related osteoporosis (oral bisphosphonate or denosumab per FRAX risk); (2) **skeletal-related event (SRE) prevention** in castration-resistant bone metastases — fewer fractures, less pain, RT and admissions, but **no antitumour or survival effect**. Hypercalcaemia is rare (suggests neuroendocrine disease).

Feature	Zoledronic acid	Denosumab
Mechanism	IV bisphosphonate; inhibits osteoclasts	Fully human anti-RANKL (receptor activator of nuclear factor-κB ligand) antibody
Dose	4 mg IV 4-weekly , adjusted for creatinine clearance	120 mg subcutaneous (SC) 4-weekly ; no renal adjustment

Feature	Zoledronic acid	Denosumab
SRE prevention	Better than placebo	Better than zoledronic acid (time to first SRE 20.7 vs 17.1 months)
Toxicity	Flu-like fever, myalgia, creatinine rise, hypocalcaemia, osteonecrosis of the jaw (ONJ)	Hypocalcaemia (grade ≥ 3 5%), hypophosphataemia, ONJ 2-4%

KEY POINT ONJ occurs mostly with dental extraction or poor dentition → dental assessment first; avoid in these men. Give calcium 1000 mg + vitamin D 800 international units (IU) daily with both agents to prevent hypocalcaemia. Bone agents are strongly recommended alongside radium-223.

14 • LANDMARK TRIALS

Trial	Design	Result	Take-home
SPARTAN	nmCRPC, PSADT ≤ 10 mo; apalutamide vs placebo	MFS 40.5 vs 16.2 mo (HR 0.28); OS HR 0.78	First nmCRPC drug
PROSPER	nmCRPC; enzalutamide vs placebo	MFS improved; death $\downarrow$ 27%	Enzalutamide for nmCRPC
ARAMIS	nmCRPC; darolutamide vs placebo	MFS 40.4 vs 18.4 mo; OS HR 0.69	Least toxic nmCRPC option
TAX 327	Docetaxel 3-weekly vs weekly vs mitoxantrone	OS 18.9 vs 17.3 vs 16.4 mo (HR 0.76)	First chemotherapy to prolong survival
TROPIC	Post-docetaxel; cabazitaxel vs mitoxantrone	OS 15.1 vs 12.7 mo (HR 0.70)	Second-line chemotherapy
PROSELICA / FIRSTANA	Cabazitaxel 20 vs 25 / vs docetaxel first line	20 non-inferior / not superior	20 mg/m ² ; docetaxel stays first
CARD	After docetaxel + 1 ARPI: cabazitaxel vs 2nd ARPI	OS 13.6 vs 11.0 mo (HR 0.64)	Switch mechanism
COU-AA-301	Post-docetaxel; abiraterone vs placebo	OS 14.8 vs 10.9 mo (HR 0.65)	Abiraterone post-chemo
COU-AA-302	Chemo-naive, asymptomatic	rPFS (radiographic PFS) 16.5 vs 8.3 mo	Abiraterone pre-chemo
AFFIRM	Post-docetaxel; enzalutamide vs placebo	OS 18.4 vs 13.6 mo (HR 0.63)	Enzalutamide post-chemo
PREVAIL	Chemo-naive; enzalutamide vs placebo	Death $\downarrow$ 29%; radiographic progression $\downarrow$ 81%	Enzalutamide pre-chemo
ALSYMPCA	Symptomatic bone-only; radium-223 vs placebo	OS 14.9 vs 11.3 mo (HR 0.70); SRE delayed	Alpha emitter prolongs life
VISION	Post-ARPI + taxane, PSMA PET+; ¹⁷⁷ Lu-PSMA + standard care	OS 15.3 vs 11.3 mo (HR 0.62); rPFS HR 0.40	Approved 2022
TheraP	Phase II; ¹⁷⁷ Lu-PSMA vs cabazitaxel	PSA response 65 vs 37%; fewer grade 3-4 events; OS equal	Lutetium $\geq$ cabazitaxel
IMPACT	Sipuleucel-T vs placebo	OS 25.8 vs 21.7 mo (HR 0.78); no PSA/PFS change	Vaccine approved 2010
PROfound	Post-ARPI; olaparib vs 2nd ARPI	BRCA/ATM cohort rPFS 7.4 vs 3.6 mo (HR 0.34)	Olaparib, HRR panel
TRITON2/3	Rucaparib after ARPI	BRCA rPFS HR 0.50; no ATM benefit	Rucaparib for BRCA only

Trial	Design	Result	Take-home
PROpel / MAGNITUDE / TALAPRO-2	First-line PARPI + abiraterone (olaparib, niraparib) / + enzalutamide (talazoparib)	rPFS gains, largest in <i>BRCA</i> ; PROpel OS 42.1 vs 34.7 mo (<i>P</i> = 0.054)	Combinations for HRR/ <i>BRCA</i>
Denosumab vs zoledronic acid	Bone-metastatic CRPC	First SRE 20.7 vs 17.1 mo; OS equal	Denosumab better for SRE

15 • MCQ ONE-LINERS

- CRPC requires castrate testosterone <50 ng/dL with progression.
- PCWG3 bone progression: ≥ 2 new lesions, confirmed on a scan after 12–16 weeks.
- LHRH suppression is continued throughout CRPC therapy.
- nmCRPC treatment threshold in SPARTAN, PROSPER, ARAMIS: PSADT ≤ 10 months.
- Abiraterone is active in nmCRPC but not approved there.
- Abiraterone must be taken fasting; food raises absorption up to tenfold.
- Enzalutamide: seizures; apalutamide: rash and hypothyroidism.
- Darolutamide and apalutamide do not cross the blood-brain barrier.
- TAX 327: 3-weekly docetaxel 75 mg/m² beat mitoxantrone.
- Cabazitaxel escapes P-glycoprotein efflux; toxicity neutropenia and diarrhoea.
- Mitoxantrone palliates but does not prolong survival.
- Radium-223: alpha emitter, path <100 μ m, bone-only symptomatic disease.
- Radium-223 must not be combined with abiraterone/prednisone.
- ¹⁷⁷Lu-PSMA-617 typical toxicity: dry mouth.
- Sipuleucel-T antigen: prostatic acid phosphatase fused to GM-CSF.
- Sipuleucel-T improves OS without lowering PSA.
- Pembrolizumab indication: MSI-H/dMMR or tumour mutational burden ≥ 10 /megabase.
- dMMR prevalence in mCRPC: 2–5%.
- PARPI benefit greatest in *BRCA2*; rucaparib not approved for ATM.
- CHIP can cause false-positive DNA-repair calls on liquid biopsy.
- Neuroendocrine CRPC: low PSA, lytic lesions, visceral disease — treat with platinum-etoposide.
- Small cell prostate cancer signature: ≥ 2 of PTEN, TP53, RB1 loss.
- Sequential ARPI benefit only 15–30% (cross-resistance).
- CARD: cabazitaxel beats a second ARPI after docetaxel and one ARPI.
- First step in cord compression: high-dose dexamethasone, then MRI.
- Patchell: decompressive surgery + RT superior to RT alone.
- Denosumab dose: 120 mg SC 4-weekly, no renal adjustment.
- Zoledronic acid: 4 mg IV 4-weekly, renal dose adjustment.
- Bone agents reduce SREs but do not improve survival.
- ONJ risk: dental extraction, poor dentition — dental check first.



Putting it together: the prostate cancer pathway

Reading time ≈ 9 min (slow pace) · Integrates Ch 153–166 · use these tables as the skeleton of long-essay answers

A · FROM RAISED PSA TO DIAGNOSIS

ALGORITHM

- 1. Raised prostate-specific antigen (PSA) or abnormal digital rectal examination (DRE)** → repeat PSA; exclude infection, instrumentation, 5 α -reductase inhibitors (double the PSA). Consider PSA density and biomarkers (→ Ch 154).
- 2. Multiparametric magnetic resonance imaging (mpMRI) before biopsy** → Prostate Imaging Reporting and Data System (PI-RADS) score (→ Ch 155).
- 3. Biopsy:** targeted biopsy of PI-RADS 3–5 lesions plus systematic cores; transperineal route lowers infection risk.
- 4. Grade** with the International Society of Urological Pathology (ISUP) Grade Group (GG 1–5) (→ Ch 156). **Stage and risk-group** (→ Ch 157); staging imaging (prostate-specific membrane antigen positron emission tomography [PSMA PET], or computed tomography [CT] + bone scan) from unfavourable intermediate risk upwards.

B · LOCALISED DISEASE: RISK GROUP → TREATMENT

Risk groups follow the National Comprehensive Cancer Network (NCCN). Favourable groups need ALL criteria; adverse groups need ANY one. Treatment choice also weighs life expectancy, comorbidity and the patient's preference.

Risk group	Defined by	Preferred management
Very low	T1c, GG1, PSA <10, PSA density <0.15, <3 positive cores (≤50% each)	Active surveillance (AS)
Low	T1–T2a, GG1, PSA <10	AS preferred; all treatments are options
Favourable intermediate	One intermediate factor (T2b–c, GG2, PSA 10–20); GG1–2; <50% cores positive	Radical prostatectomy (RP) or radiotherapy (RT); AS for selected low-volume GG2; focal therapy in trials
Unfavourable intermediate	2–3 intermediate factors, or GG3, or ≥50% cores positive	RP, or RT + 4–6 months androgen deprivation therapy (ADT)
High / very high	T3a, GG4–5 or PSA >20 / T3b–T4, primary pattern 5, ≥2 high-risk features	RP + extended pelvic lymph node dissection (multimodal), or RT + long-term ADT (18–36 months); add 2 years of abiraterone with RT if N1 or ≥2 of Gleason ≥8, cT3–4, PSA ≥40 (STAMPEDE)

KEY POINT Trials comparing treatment with observation (SPCG-4, PIVOT, ProtecT) show the benefit of radical treatment is confined to younger men with intermediate-risk or higher disease and long life expectancy. ProtecT showed RP and RT give equal cancer survival at 15 years, with different side-effect profiles.

C · SURGERY VS RADIOTHERAPY VS FOCAL THERAPY

	Radical prostatectomy (Ch 160–161)	Radiotherapy (Ch 162)	Focal therapy (Ch 163)
Main toxicity	Urinary incontinence, erectile dysfunction	Bowel (rectal) and urinary irritation; erectile dysfunction later; ADT side effects	Least toxicity; retreatment often needed

	Radical prostatectomy (Ch 160-161)	Radiotherapy (Ch 162)	Focal therapy (Ch 163)
Advantages	Full pathological staging; PSA should become undetectable; salvage RT remains available	No anaesthesia; suits older or comorbid men; hypofractionation shortens treatment	Preserves function; ideal for an MRI-visible index lesion
Failure defined as	PSA ≥ 0.2 ng/mL (confirmed)	Phoenix: nadir + 2 ng/mL	No agreed definition; biopsy-based
Salvage	Salvage RT to prostate bed, early at PSA 0.1–0.2 $\pm$ ADT	Salvage RP, cryotherapy, high-intensity focused ultrasound (HIFU) or brachytherapy in selected men	Repeat focal, whole-gland treatment, RP or RT

D • ADVANCED DISEASE IN FOUR STEPS

ALGORITHM

1. Biochemical recurrence (Ch 165): PSMA PET to localise. After RP: salvage RT. After RT: biopsy-proven local recurrence → local salvage. High-risk recurrence (PSA doubling time ≤ 9 months) → ADT + enzalutamide (EMBARK).

2. Metastatic hormone-sensitive (mHSPC): never ADT alone. ADT + an androgen-receptor pathway inhibitor (ARPI: abiraterone, enzalutamide, apalutamide, darolutamide); triplet with docetaxel for fit, high-volume disease. RT to the prostate for low-volume disease (STAMPEDE arm H). Offer germline and somatic genetic testing to all.

3. Non-metastatic castration-resistant (nmCRPC) with PSA doubling time ≤ 10 months: ARPI (SPARTAN, PROSPER, ARAMIS).

4. Metastatic castration-resistant (mCRPC) (Ch 166): switch mechanism. Docetaxel → cabazitaxel (CARD); ^{177}Lu -PSMA (VISION); radium-223 for symptomatic bone-only disease (ALSYMPCA); poly(ADP-ribose) polymerase (PARP) inhibitor if BRCA/homologous-recombination mutation (PROfound); pembrolizumab if microsatellite instability-high. Bone protection (denosumab or zoledronic acid) throughout.